ETSI TR 133 929-9 V19.0.0 (2026-10)



TECHNICAL REPORT

LTE;
5G;
Lawful Interception (LI) implementation guidance;
LI for PTC
(3GPP TR 33.929-9 version 19.0.0 Release 19)



Reference
DTR/TSGS-0333929-9vj00

Keywords
5G,LTE

ETSI

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Contents

Intellectual Property Rights 2
Legal Notice 2
Modal verbs terminology 2
Foreword 6
Introduction 6
1 Scope 7
2 References 7
3 Definitions of terms, symbols and abbreviations 8
3.1 Terms 8
3.2 Symbols 8
3.3 Abbreviations 8
4 Illustration of LI for PTC 11
4.1 Overview 11
4.1.1 General 11
4.1.2 Background 11
4.1.2.1 Simple views 11
4.1.2.2 PTC functions supporting half-duplex media with floor control 13
4.1.2.3 PTC concepts 15
4.1.2.3.1 PTC session handling 15
4.1.2.3.2 PTC presence and watcher functions 16
4.1.3 PTC view with LI 17
4.1.3.1 General principles 17
4.1.3.2 LI for PTC session handling 18
4.1.3.3 LI for presence watcher functions 19
4.1.4 PTC LI architecture 20
4.2 PTC topology scenarios 21
4.2.1 General 21
4.2.2 Few conventions used 21
4.2.3 PTC pre-established session 22
4.2.4 PTC 1:1 session 23
4.2.4.1 General 23
4.2.4.2 LI: PTC clients in the same CSP domain 24
4.2.4.2.1 General 24
4.2.4.2.2 PTC client A is a target 24
4.2.4.2.3 PTC client B is a target 25
4.2.4.3 LI: PTC clients in different CSP domains 26
4.2.4.3.1 General 26
4.2.4.3.2 PTC client A is the target in CSP A domain 26
4.2.4.3.3 PTC Client B is a target in CSP B domain 27
4.2.5 PTC one-to-many session 28
4.2.5.1 General 28
4.2.5.2 Ad-hoc group PTC session 29
4.2.5.2.1 General 29
4.2.5.2.2 PTC Client A is a target 29
4.2.5.2.3 PTC Client B is a target 32
4.2.5.2.4 PTC Client C is a target 33
4.2.5.3 Similarities between PTC client B and PTC client C from an LI perspective 36
4.2.5.4 Pre-arranged group PTC session 36
4.2.5.4.1 General 36
4.2.5.4.2 LI: Group management server is in different provider domain 37
4.2.5.4.3 Group management server is in same service provider domain of PTC session initiator 39
4.2.5.5 Chat-group PTC session 43
4.2.5.5.1 General 43

4.2.5.5.2 One CSP domain scenarios....45
4.2.5.5.2.1 General....45
4.2.5.5.2.2 PTC chat-group ID is the target....45
4.2.5.5.2.3 PTC clients are the targets....46
4.2.5.5.3 Different CSP domain scenarios....50
4.2.5.5.3.1 General....50
4.2.5.5.3.2 PTC chat-group ID is the target....50
4.2.5.5.3.3 PTC clients are the targets....52
4.2.6 PTC one-to-many-to-one session....54
4.2.6.1 General....54
4.2.6.2 PTC Client A is a target....55
4.2.6.3 PTC Client B is a target....56
4.2.6.4 PTC Client C is a target....58

5 Example call flows of LI for PTC....61
5.1 General....61
5.2 Pre-established session setup....62
5.3 On-demand PTC 1:1 and ad-hoc group sessions....63
5.3.1 General....63
5.3.2 PTC session initiated from a target....63
5.3.2.1 PTC session start....63
5.3.2.2 PTC session end....64
5.3.2.3 Floor control switching....65
5.3.2.4 Floor control request queued....65
5.3.2.5 Floor control request denied....66
5.3.2.6 Floor control request revoked....67
5.3.3 PTC sessions terminated to a target....68
5.3.3.1 PTC session start....68
5.3.3.2 PTC session end....69
5.3.3.3 Floor control switching....70
5.3.3.4 Floor control request queued....71
5.3.3.5 Floor control request denied....72
5.3.3.6 Floor control request revoked....73
5.4 PTC sessions (1:1 or ad-hoc group) from pre-established sessions....74
5.4.1 General....74
5.4.2 PTC session start from a target....75
5.4.2.1 Scenario 1: Pre-established to Pre-established....75
5.4.2.1.1 PTC session start....75
5.4.2.1.2 PTC session end....75
5.4.2.2 Scenario 2: Pre-established to on-demand....76
5.4.2.2.1 PTC session start....76
5.4.2.2.2 PTC session end....77
5.4.2.3 Scenario 3: On-demand to pre-established....78
5.4.2.3.1 PTC session start....78
5.4.2.3.2 PTC session end....79
5.4.3 PTC session terminated to a target....80
5.4.3.1 Scenario 1: Pre-established to Pre-established....80
5.4.3.1.1 PTC session start....80
5.4.3.1.2 PTC session end....80
5.4.3.2 Scenario 2: Pre-established to on-demand....81
5.4.3.2.1 PTC session start....81
5.4.3.2.2 PTC session end....82
5.4.3.3 Scenario 3: On-demand to pre-established....83
5.4.3.3.1 PTC session start....83
5.4.3.3.2 PTC session end....83
5.5 On-demand PTC pre-arranged group sessions....84
5.5.1 General....84
5.5.2 PTC session start....84
5.5.3 PTC session end....85
5.5.4 Floor control switching....86
5.6 Pre-arranged PTC session with pre-established sessions....86
5.6.1 General....86

5.6.2 PTC session start....87
5.6.3 PTC session end....87
5.7 On-demand chat-group PTC sessions with chat-group ID as the target....88
5.7.1 General....88
5.7.2 Chat-group PTC session start....88
5.7.3 Chat-group PTC session end....89
5.7.4 Floor control procedures....90
5.7.4.1 General....90
5.7.4.2 Flow-control switching....90
5.7.4.3 Floor control request queued....91
5.7.4.4 Floor control request denied....92
5.7.4.5 Floor control request revoked....93
5.8 Chat-group PTC sessions with clients having pre-established sessions....94
5.8.1 General....94
5.8.2 PTC session start....95
5.8.3 PTC session end....95
5.9 On-demand chat-group PTC sessions with PTC client as the target....96
5.9.1 General....96
5.9.2 Chat-group PTC session start....97
5.9.3 Chat-group PTC session end – target releases first....97
5.9.4 Chat-group PTC session end – target releases last....98
5.9.5 Floor control procedures....99
5.9.5.1 General....99
5.9.5.2 Flow-control switching....99
5.9.5.3 Floor control request queued....99
5.9.5.4 Floor control request denied....100
5.9.5.5 Floor control request revoked....101
5.10 Chat-group PTC sessions with clients having pre-established sessions, one of the PTC client is a target....102
5.10.1 General....102
5.10.2 PTC session start....103
5.10.3 PTC session end – target releases first....103
5.10.4 PTC session end – target releases last....104
5.11 One-to-Many-One PTC session – floor control procedures....105
5.11.1 General....105
5.11.2 PTC session initiator is the target....105
5.11.3 PTC session participant is the target....106

Annex A: Change history....108

History....109

Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The present document is part of a multi-part TR as described below:

- Part 1: LI for IMS based services. See TR 33.929-3 [10].
- Part 2: LI for IMS based STIR/SHAKEN. See TR 33.929-2 [11].
- Part 3: LI for messaging services. See TR 33.929-3 [12].
- Part 4: LI for data in 5G core. See TR 33.929-4 [13].
- Part 5: LI for ID association caching. See TR 33.929-5 [14].
- Part 6: LI for IMS based RCS. See TR 33.929-6 [15].
- Part 7: LI location acquisition capabilities. See TR 33.929-7 [16].
- Part 8: LI for MMS. See TR 33.929-8 [17].
- Part 9: LI for PTC (present document).

In Part 1, the illustrations of LI for IMS-based services focus on the LI aspects of IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (local break-out and home-routed roaming).

In Part 2, the illustrations of LI for IMS based STIR/SHAKEN focus on various STIR/SHAKEN related LI reporting scenarios.

In Part 3, the illustrations of LI for messaging services focus on SMS over NAS and SMS over IP.

In Part 4, the illustrations of LI for data in 5G core focus on the LI aspects of data interception focussing on the PDU session related events including the handover scenarios.

In Part 5, the illustrations of LI for ID association caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

In Part 6, the illustrations of LI for IMS based RCS focus on the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider.

In Part 7, the illustrations of LI location acquisition capabilities focus on the conceptual overview and the flow diagrams for location reporting, LALS and location acquisition.

In Part 8, the illustrations of LI for MMS focus on the conceptual overview and the flow diagrams for MMS.

In Part 9 (present document), the illustrations of LI for PTC focus on the conceptual overview and the flow diagrams for MCPTT and PoC, the two commonly referred to as Push to Talk over Cellular (PTC).

Introduction

Unlike the previous generation of LI technical specifications, the latest LI technical specifications (TS 33.126 [2], TS 33.127 [3], TS 33.128 [4]) contain only the normative part of the requirements. However, the implementers may need additional information such as architectural diagrams, conceptual scenarios, flow-diagrams and additional background information to better understand the requirements in those specifications. The present document collects the relevant informative annexes from the previous generation of LI technical specifications and then expands them to include the similar level of details as for latest technologies such as 5G.

1 Scope

The present document provides architectural diagrams, conceptual scenarios, flow-diagrams, examples, and other background information which can be useful to implement the LI functions defined in TS 33.126 [2], TS 33.127 [3] and TS 33.128 [4].

The present document covers the LI for PTC that include the illustrations covering the conceptual overview and the flow diagrams for MCPTT and PoC commonly referred to as Push to Talk over Cellular.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.126: "Lawful Interception requirements".

[3] 3GPP TS 33.127: "Lawful Interception (LI) architecture and functions".

[4] 3GPP TS 33.128: "Protocol and procedures for Lawful Interception (LI)".

[5] 3GPP TS 24.147: "Conferencing using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".

[6] 3GPP TS 23.501: "System architecture for the 5G System".

[7] 3GPP TS 24.174: "Support of multi-device and multi-identity in the IP Multimedia Subsystem (IMS); Stage 3".

[8] 3GPP TS 23.228: "IP Multimedia Subsystem (IMS), Stage 2".

[9] 3GPP TR 33.928: "ADMF logic for provisioning Lawful Interception (LI)".

[10] 3GPP TR 33.929-1: "Lawful Interception (LI) implementation guidance; LI for IMS-based services".

[11] 3GPP TR 33.929-2: "Lawful Interception (LI) implementation guidance; LI for IMS based STIR/SHAKEN".

[12] 3GPP TR 33.929-3: "Lawful Interception (LI) implementation guidance; LI for messaging services".

[13] 3GPP TR 33.929-4: "Lawful Interception (LI) implementation guidance; LI for data in 5G core".

[14] 3GPP TR 33.929-5: "Lawful Interception (LI) implementation guidance; LI for ID Association Caching ".

[15] 3GPP TR 33.929-6: "Lawful Interception (LI) implementation guidance; LI for IMS based RCS".

[16] 3GPP TR 33.929-7: "Lawful Interception (LI) implementation guidance; LI location acquisition capabilities".

[17] 3GPP TR 33.929-8: "Lawful Interception (LI) implementation guidance; LI for MMS".

[18] 3GPP TS 24.379: "Mission Critical Push to Talk (MCPTT) call control; protocol specification".

[19] OMA-TS-PoC-System_Description-V2_1-20110802-A: "OMA PoC System Description".

[20] 3GPP TS 23.280: "Common functional architecture to support mission critical services".

[21] 3GPP TS 23.279: "Functional architecture and information flows to support Mission Critical Push to Talk (MCPTT); Stage 2".

[22] ATIS-07000016: "LAES Capabilities for 3GPP IMS Based PoC".

[23] 3GPP TS 24.380: "Mission Critical Push To Talk (MVCPTT) media plane control".

[24] OMA-AD_PoC-V1_0-20050428-C: "Push to talk over Cellular (PoC) – Architecture".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

None.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

3GPP	3rd Generation Partnership Program
ADMF	Administration Function
AMF	Access and Mobility Management Function
APN	Access Pointe Name
AS	Application Server
BBIFF	Bearer Binding Intercept and Forward Function
BBIFF-C	Bearer Binding Intercept and Forward Function Control plane
BBIFF-U	Bearer Binding Intercept and Forward Function User plane
BGCF	Border Gateway Control Function
CC	Content of Communication
CSCF	Call Session Control Function
CSP	Communication Service Provider
CUPS	Control and User Plane Separation
DNN	Data Network Name
E-CSCF	Emergency Call Session Control Function
eCNAM	Enhanced Calling NAMe
EPC	Evolved Packet Core
H-SMF	Home SMF
H-UPF	Home UPF
HPLMN	Home PLMN
HSS	Home Subscriber Server
IBCF	Interrogating Border Control Function
I-CSCF	Interrogating CSCF

ID	Identity
IMPI	IP Multimedia Private Identity
IMPU	IP Multimedia Public Identity
IMS	IP Multimedia Subsystem
IMS-AGW	IMS Media Gateway
IRI	Intercept Related Information
LALS	Lawful Access Location Services
LBO	Local Break Out
LEA	Law Enforcement Agency
LEMF	Law Enforcement Monitoring Facility
LI	Lawful Interception
LICF	Lawful Interception Control Function
LI_HI1	LI_Handover Interface 1
LI_HI2	LI_Handover Interface 2
LI_HI3	LI_Handover Interface 3
LIPF	LI Provisioning Function
LI_T1	Lawful Interception Triggering Interface 1
LI_T3	Lawful Interception Triggering Interface 3
LI_X1	Lawful Interception Internal Interface 1
LI_X2	Lawful Interception Internal Interface 2
LI_X3	Lawful Interception Internal Interface 3
LMISF	LI Mirror IMS State Function
LMISF-CC	LMISF for the handling of CC
LMISF-IRI	LMISF for the handling of IRI
LTE	Long Term Evolution
MCPTT	Mission Critical Push To Talk
MDF	Mediation and Delivery Function
MDF2	Mediation and Delivery Function 2
MDF3	Mediation and Delivery Function 3
MGCF	Media Gateway Control Function
MMS	Multimedia Message Service
MRFC	Media Resource Function Controller
MRFP	Multimedia Resource Function Processor
N9HR	N9 Home Routed
NAS	Non-Access Stratum
NR	New Radio
PASSporT	Personal ASSertion Token
P-CSCF	Proxy Call Session Control Function
PDN	Packet Data Network
PEI	Permanent Equipment Identifier
PGW	PDN Gateway
PGW-C	PDN Gateway Control Plane
PGW-U	PDN Gateway User Plane
PLMN	Public Land Mobile Network
PoC	Push to talk over Cellular
POI	Point Of Interception
PTC	Push to Talk over Cellular
RCD	Rich Call Data
RCS	Rich Communication Suite
S8HR	S8 Home Routed
S-CSCF	Serving Call Session Control Function
SDP	Session Description Protocol
SGW	Serving Gateway
SGW-C	Serving Gateway Control Plane
SGW-U	Serving Gateway User Plane
SHAKEN	Signature-based Handling of Asserted information using toKENs
SIP	Session Initiation Protocol
SMF	Session Management Function
SMS	Short Messaging Service
STIR	Secure Telephony Identity Revisited
SUPI	Subscriber Permanent Identifier
TF	Triggering Function

TrGW	Transit Gateway
UDM	Unified Data Management
UE	User Equipment
UPF	User Plane Function
V-SMF	Visted SMF
V-UPF	Visited UPF
VPLMN	Visited PLMN
xCC	LI_X3 Communications Content
xIRI	LI_X2 Intercept Related Information

4 Illustration of LI for PTC

4.1 Overview

4.1.1 General

As defined in 3GPP TS 33.127 [3] and 3GPP TS 33.128 [4], the term PTC is used in the present document to refer collectively:

- Push to talk over Cellular (PoC) as defined in OMA PoC System Description document [19].
- Mission Critical Push To Talk (MCPTT) as defined in 3GPP TS 23.279 [21] and 3GPP TS 24.379 [18].

NOTE: The term PTC stands also for Push to Talk over Cellular. ATIS has published an LI standard titled LAES for IMS based Push to talk over Cellular (PoC), ATIS-07000016 [22].

The present document assumes that PTC is a service offered over the cellular networks by the CSPs. At the service level, the PTC service may also be offered by a third party service provider.

4.1.2 Background

4.1.2.1 Simple views

In a general sense, Push to talk is a conversation in half-duplex mode – sort of walkie-Talkie. One user speaks and the rest listens. The user who speaks is said to have the Floor. After speaking, the user can relinquish the Floor and at that time, the Floor is considered to be open. When the Floor is open (i.e. no one has the Floor), another user on the PTC session can acquire the Floor to speak. The users on a PTC session are referred to as PTC Clients.

Figure 4.1.2.1-1 illustrates simple view of PTC concept when two PTC Clients are on a PTC session.

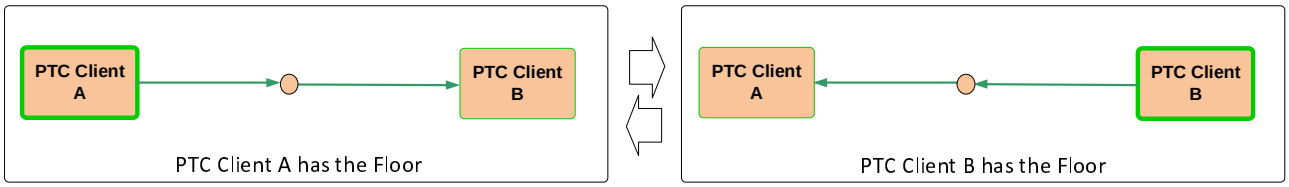


Figure 4.1.2.1-1: Two PTC Clients on a PTC session

The PTC session with two PTC Clients, as shown in figure 4.1.2-1, is referred One-One PTC session. At any given time, either PTC Client A or the PTC Client B can have the Floor and hence be talking and the other PTC Client would be listening. It is also possible (not shown in figure 4.1.2.1-1) that the Floor can be open, meaning neither PTC Client A nor PTC Client B is talking at a given time.

The PTC session can also be One-to-Many PTC session or One-to-Many-to-One PTC session.

Figure 4.1.2.1-2 illustrates a simple view of One-to-Many PTC session where one speaks and many listens.

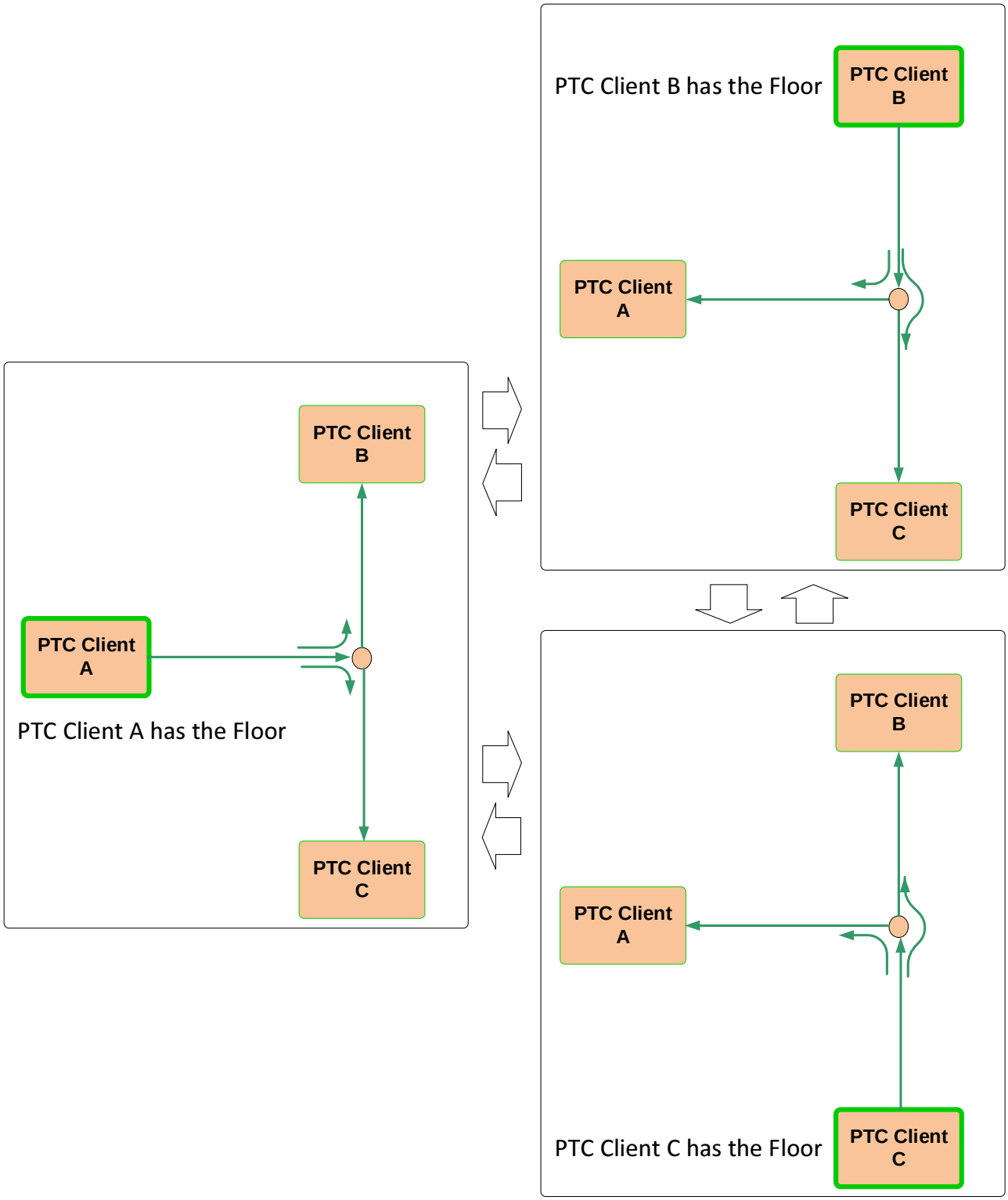


Figure 4.1.2.1-2: Simple view of One-to-Many PTC session

In the illustration shown in figure 4.1.2.1-2, three PTC Clients are on a One-to-Many PTC session. At any given time, any one of the three PTC Clients can have the Floor and hence, be speaking while other two PTC Clients are listening.

Figure 4.1.2.1-3 illustrates simple view of One-to-Many-to-One PTC session where one speaks with many listening or one speaks and only one listens.

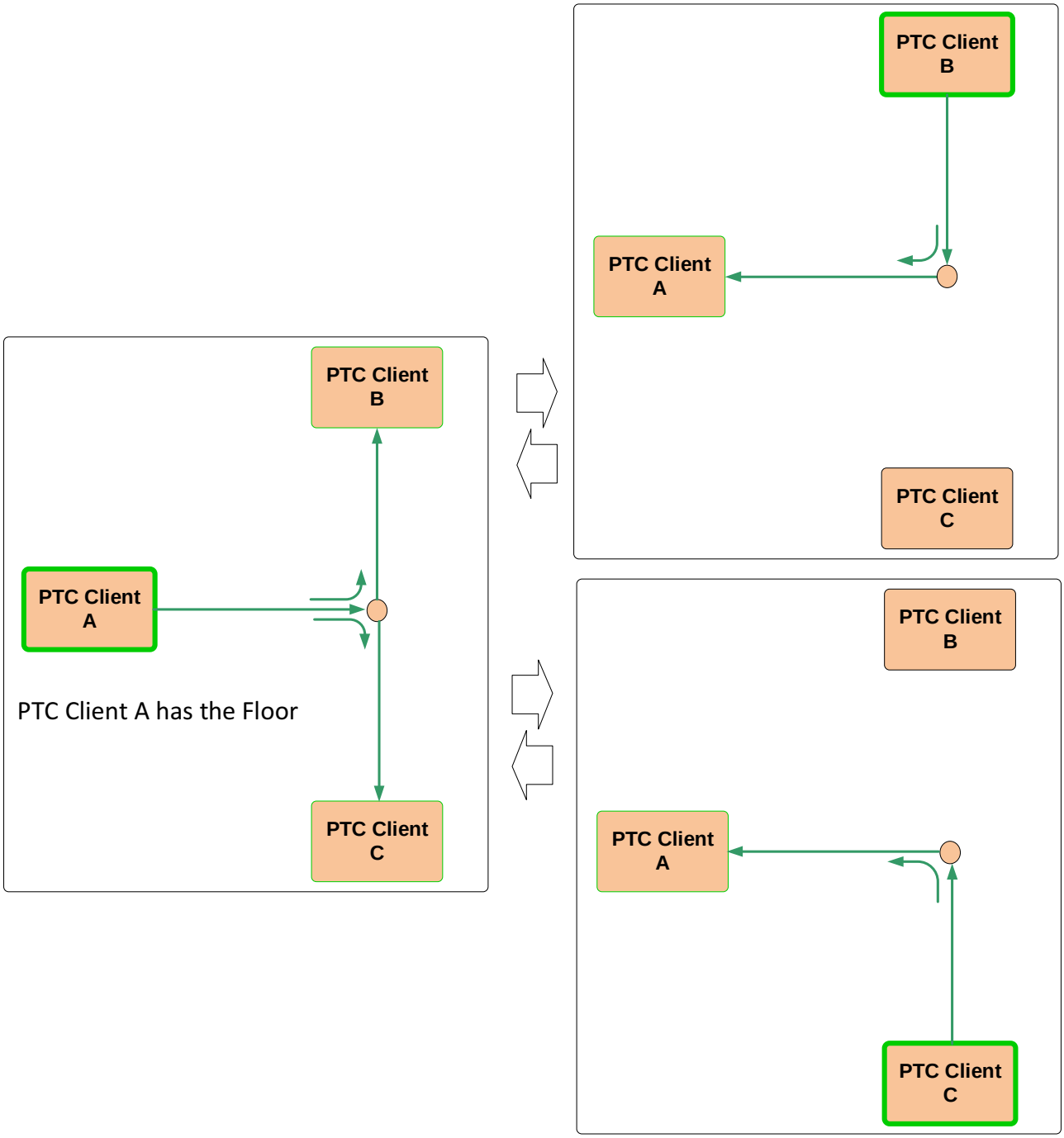


Figure 4.1.2.1-3: Simple view of One-to-Many-to-One PTC session

In the illustration shown in figure 4.1.2.1-3, the speech from PTC Client A is heard by the PTC Client B and PTC Client B (One-to-Many), while the speech from PTC Client B and PTC Client C can be heard by only PTC Client A (Many-to-One).

In general, the PTC session can be initiated by a PTC Client inviting other PTC Clients or all PTC Clients can join a PTC chat-group session. In the former case, the members of the group to whom the invitation sent can be pre-arranged (by the CSP) or determined by session initiating client on an ad-hoc basis. The one-to-many-one PTC group session can happen when the PTC group members are determined based on the pre-arranged list.

4.1.2.2 PTC functions supporting half-duplex media with floor control

With PTC, as illustrated with the simple views in clause 4.1.2.1, the media flows in a half-duplex mode from one PTC Client to another PTC Client. The PTC Client, currently speaking, is said to have the Floor which can be relinquished

by that PTC Client and can be acquired by a differ PTC Client using the Floor Control procedures. The network function that handles the PTC service is referred to as PTC Server. All signalling and Floor Control messages and the half-duplex media pass through the PTC Server. As said in clause 4.1.1, the PTC service may be provided by the CSP or by a Third Party Service Provider.

When the PTC service is provided by the CSP, the PTC Server is said to be within the CSP domain. When a PTC service is provided by a Third Party Service Provider, the PTC Server is within that Third Party Service Provider domain.

The 3GPP TS 23.280 [20] defines various deployment possibilities depending on where the PTC Server is. Figure 4.1.2.2-1 depicts those possibilities.

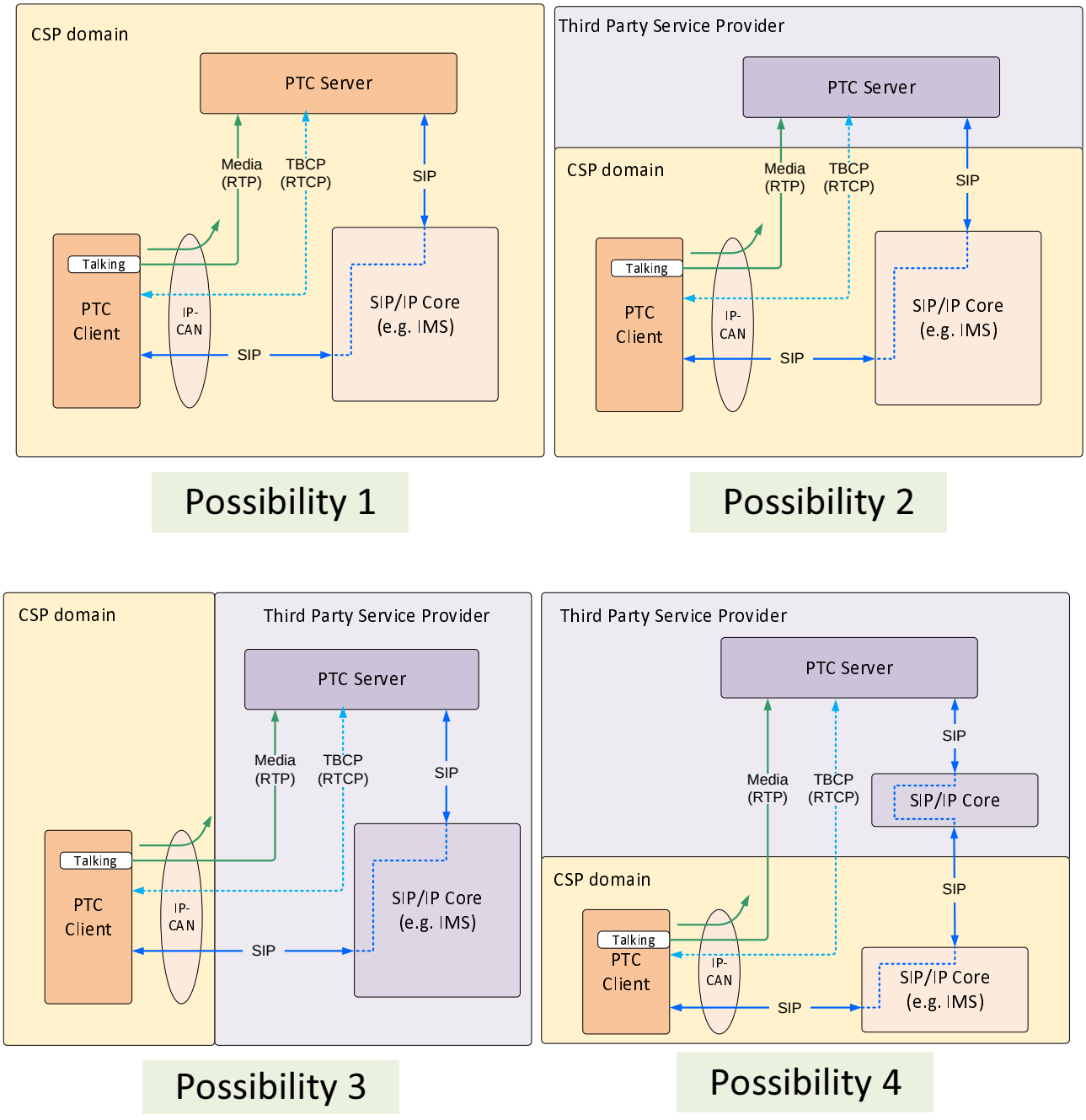


Figure 4.1.2.2-1: Deployment possibilities of PTC service (per TS 23.280 [20])

The PTC Server may be within the CSP domain (Possibility 1). The PTC Server may be within the Third Party Service Provider domain (Possibilities 2 to 4).

Session handling related signalling is provided by Session Initiation Protocol (SIP). All the SIP signalling message pass through SIP/IP core which can be IMS, specially, when the SIP/IP core is within the CSP domain. SIP/IP Core may also be in the Third Party Service Provider domain either in entirety (Possibility 3) or in part (Possibility 4). In Possibility 3, the CSP domain has just the IP-CAN which is presumed to include the RAN and the packet core (e.g. 5GC, EPC).

Real Time Protocol (RTP) is used to transfer the half-duplex media. The Floor Control procedure related signalling is provided by Talk Burst Control Protocol which is based on RTCP.

The PTC service also includes the presence and watcher functions and the details of this are illustrated in clause 4.1.2.4.

4.1.2.3 PTC concepts

4.1.2.3.1 PTC session handling

The PTC session handling functions are handled by an IMS-based Application Server referred to as PTC Server. In the illustrations shown here, the PTC Server is within the CSP domain. However, note that either one or more of the PTC Servers can be in a Third Party Service Provider as well.

Each PTC Client would be registered to a PTC Server which is referred to as a PTC Server with participating PTC function. When a PTC Client initiates a PTC session inviting the other PTC Clients, the PTC Server providing the participating PTC function for the initiating PTC Client controls the PTC session and is presumed to be providing the controlling PTC function. Figure 4.1.2.3.1-1 illustrates the view of a PTC session showing PTC Servers providing participating PTC function and controlling PTC function. In the illustration, only two PTC Clients are shown.

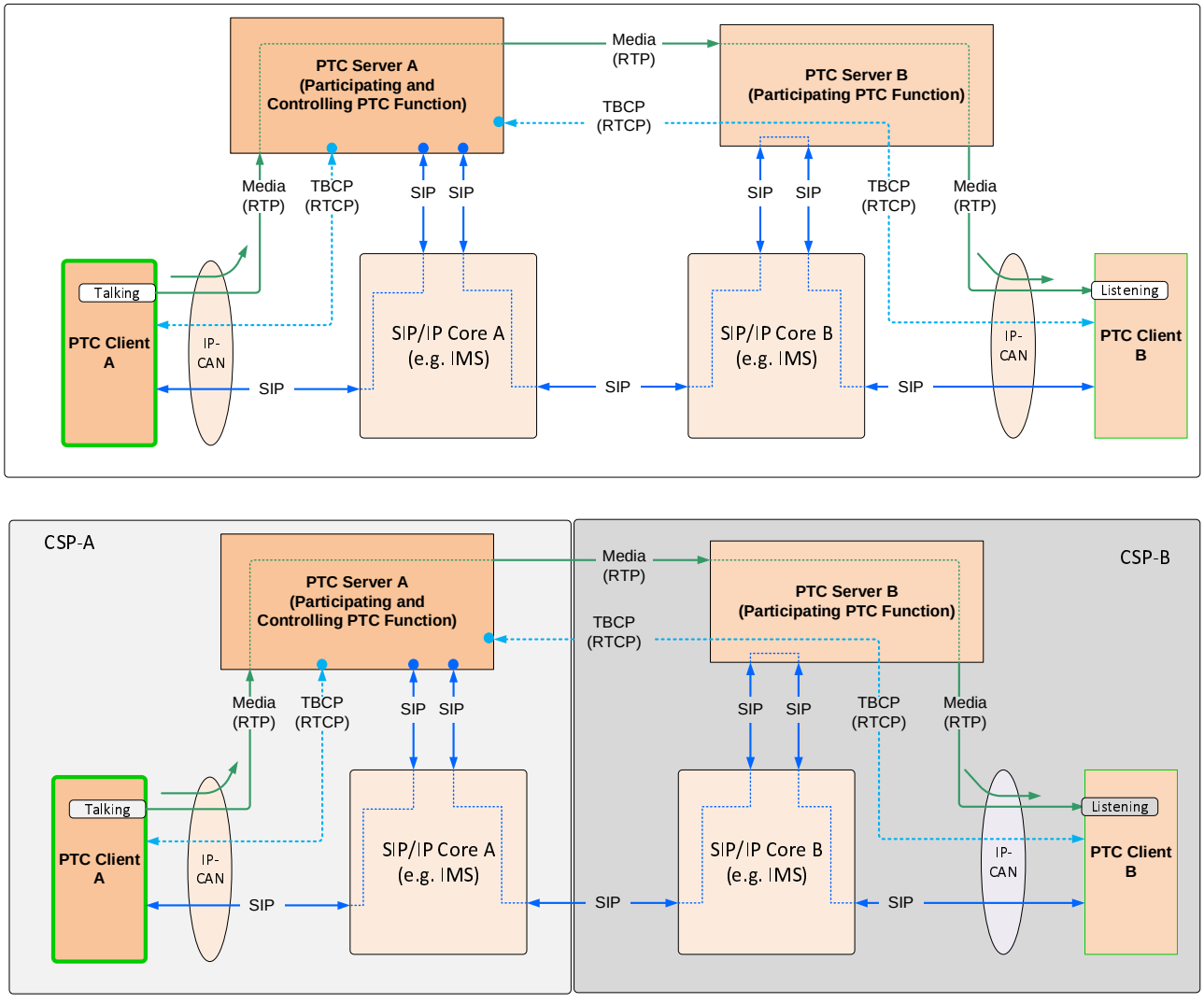


Figure 4.1.2.3.1-1: PTC Servers providing participating PTC Function and controlling PTC Function

In the illustration shown in figure 4.1.2.3.1-1, two PTC Clients (A and B) are present. The PTC Server A provides the participating PTC function for PTC Client A and the PTC Server B provides the participating PTC function for PTC Client B. The PTC Client A initiates a PTC session to PTC Client B and therefore, the PTC Server A also provides the controlling PTC function for the PTC session.

As shown in figure 4.1.2.3.1-1, the two PTC Clients can be within the same CSP domain or in two separate CSP domains. The media (which is in RTP form) is between the PTC Client and the PTC Server that provides the participating function. The mechanism used to manage the floor-control is referred to as Talk Burst Control using the Talk Burst Control Protocol (TBCP) which is based on RTCP when media is in the RTP form. In the illustration, PTC Client A has the floor. TBCP is used to relinquish or acquire the Floor.

The session related signalling is based on the SIP signalling with SIP/IP Core providing the session control. The present document assumes that SIP/IP Core is IMS based.

The PTC session can also be initiated by a PTC Client to multiple destinations that are determined either on an ad-hoc basis or based on administered pre-arranged list. In the latter case, a separate server that manages the group provides the controlling PTC function.

Multiple PTC clients can join chat-group PTC session. In this case, a separate server that manages the chat-group PTC session provides controlling PTC function.

NOTE: As per the OMA specification, when the PTC Server that provides the participating PTC function may not be on the media path when a different PTC Server provides the controlling PTC function. However, with such a deployment approach, the LI cannot be provided, specially, when the CSP serving the LI warrant has just the PTC Server that provides the participating PTC function.

Therefore, the present document assumes that the PTC Server that provides the participating PTC function is always on the media path. Therefore, the present document assumes that the media and TBCP messages always pass through the PTC Server that provide the participating PTC function.

4.1.2.3.2 PTC presence and watcher functions

In reference to the PTC, the real-time status along with the willingness to involve in a PTC session is known as presence information of a PTC client. The presence information is stored in Presence Server. The entity that provides the presence information to the Presence Server is referred to as Presence Source. The Presence Source can either be in the PTC Client's device or in the PTC Server. In the latter case, the PTC Server is providing the Presence Source functions on behalf of the PTC Client. Figure 4.1.2.3.2-1 illustrates the PTC architecture that shows Presence Source and Presence Server functions.

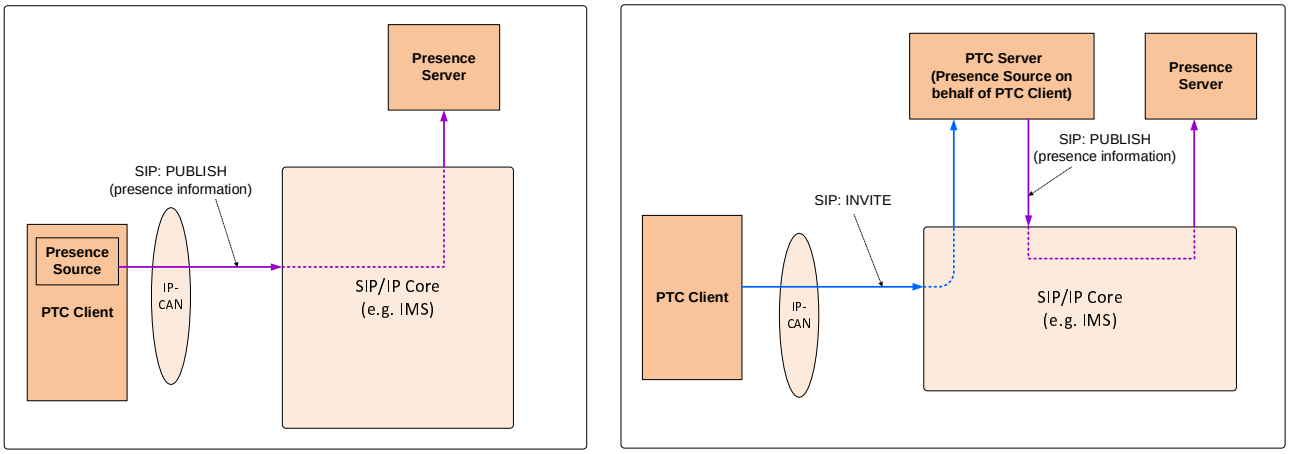


Figure 4.1.2.3.2-1: Presence source and Presence Server functions

In the left part of the diagram in figure 4.1.2.3.2-1, the Presence Source is in the PTC Client. The Presence Source sends the presence information to the Presence Server using the SIP: PUBLISH message.

In the right part of the diagram in figure 4.1.2.3.2-1, the PTC Server provides the Presence Source functions on behalf of the PTC Client. The Presence Source present in the PTC Server sends the presence information of the PTC Client to the Presence Server using the SIP: PUBLISH message when it receives the SIP: INVITE message from the PTC Client.

In reference PTC, the Watcher requests and receives the presence information of another PTC Client from the Presence Server. A Resource List Server may be used to consolidate the requests from multiple Watchers. Figure 4.1.2.3.2-2 illustrates the PTC architecture that shows Watcher functions.

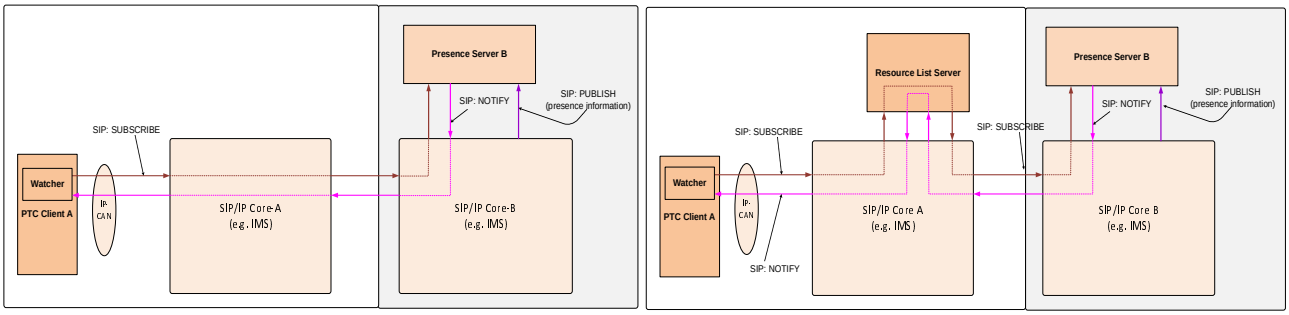


Figure 4.1.2.3.2-2: Watcher functions

In figure 4.1.2.3.2-2, the Watcher functions are in the PTC Client's device. In the right part of the diagram, a Resource List Server is used to consolidate the requests from Watcher. The Watcher requests for the presence information from the Presence Server (in the illustration, Presence Server B) using the SIP: SUBSCRIBE message. The Presence Server B provides the presence information to the Watcher using the SIP: NOTIFY message.

The Watcher functions may also be provided by a PTC Server on behalf of a PTC Client.

Figure 4.1.2.3.2-3 illustrates the case where the PTC Server provides the Watcher functions on behalf of the PTC Client. The illustration also shows the Presence Server storing the presence source information.

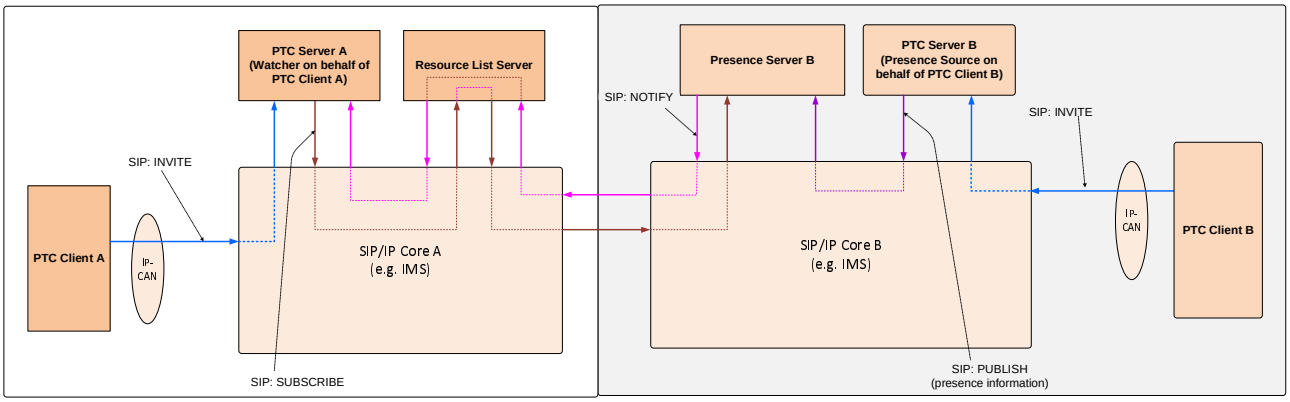


Figure 4.1.2.3.2-3: Participating PTC Server providing Watcher functions on behalf of PTC Client

In the illustration, the PTC Server A providing the Watcher Functions on behalf of PTC Client A requests for the presence information of PTC Client B using the SIP: SUBSCRIBE message when it receives the SIP: INVITE from the PTC Client A. Resource List Server is used in this illustration to consolidate the SIP: SUBSCRIBE messages. The PTC Server B providing the Presence Source functions on behalf of the PTC Client B sends the presence information of the PTC Client B to the Presence Server B using the SIP: PUBLISH message, when it receives the SIP: INVITE message from the PTC Client B.

The Presence Server B, upon receiving the SIP: SUBSCRIBE from the Watcher, sends the presence information of PTC Client B using the SIP: NOTIFY message.

4.1.3 PTC view with LI

4.1.3.1 General principles

The LI functions for PTC is provided by the PTC Server that in most situations provides the participating PTC function (i.e. PTC server to which the target has registered to), but may also be providing the controlling PTC function. When PTC chat group ID is a target, the PTC server that handles the chat group PTC session provides the LI functions.

When the PTC Server is in the Third Party Service Provider domain, the LI functions are provided by the PTC Server present in that Third Party Service Provider domain.

Figure 4.1.3.1-1 illustrates the LI for two cases.

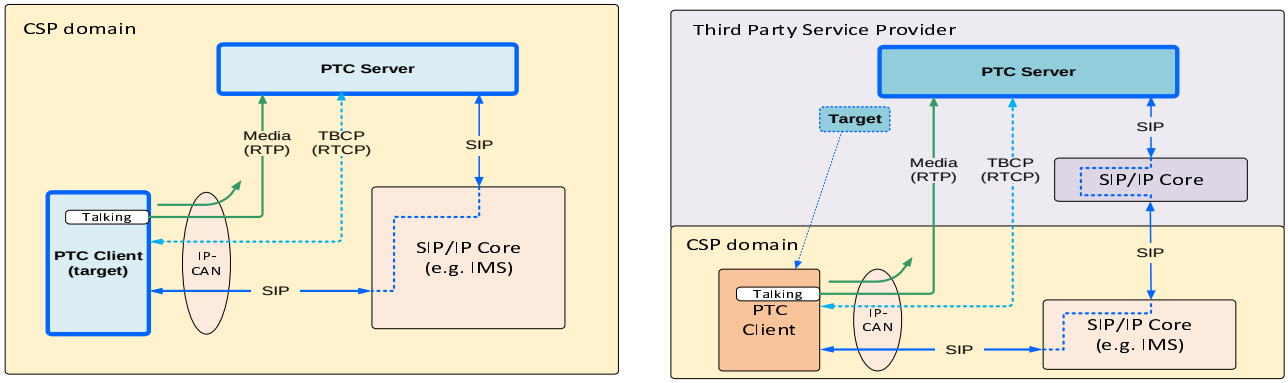


Figure 4.1.3.1-1: PTC LI general principles

In the left part of the diagram shown in figure 4.1.3.1-1, the PTC Server is in the CSP domain. Here, the PTC Client is a target in the CSP domain and the PTC Server present in the CSP domain provides the LI functions.

In the right part of the diagram shown in figure 4.1.3.1-1, the PTC Server is in the Third Party Service Provider domain (possibility 4 shown in figure 4.1.2.2-1). Here, the PTC Client is a target in the Third Party Service Provider domain and hence, the PTC Server present in the Third Party Service Provider domain provides the LI functions. For this, the Third Party Service Provider has to be served with the warrant.

The LI functions in the PTC Servers generate the xIRI based on the SIP and TBCP messages sent to or received from the target and xCC based on the media (RTP) sent to or received from the target.

Watcher and Presence

For Watcher and Presence functions, the LI functions are provided when the Participating PTC Server provides the Watcher or Presence functions on behalf of a target PTC Client.

4.1.3.2 LI for PTC session handling

A high-level view of LI aspects for PTC session handling is depicted in figure 4.1.3.2-1 below.

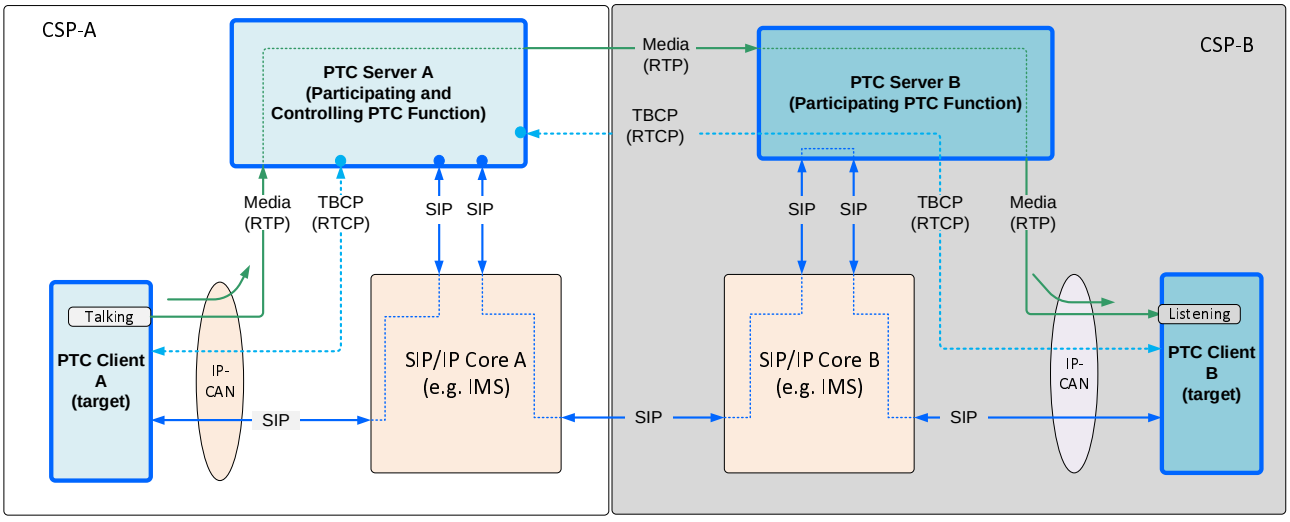


Figure 4.1.3.2-1: High level view of LI functions for PTC session handling

In the illustration shown in figure 4.1.3.2-1, the PTC Client A initiates the PTC session to PTC Client B and therefore, the PTC Server A that provides the participating PTC function for PTC Client A also provides the controlling PTC function. The PTC Server B provides the participating PTC function for PTC Client B.

The PTC Server A provides the LI functions when PTC Client A is a target and PTC Server B provides the LI functions when the PTC Client B is a target. PTC Client A is a target in CSP-A and PTC Client B is a target in CSP-B. The scope of LI provided by the LI functions in PTC Server A may be more than the scope of the LI functions provided by the LI functions in PTC Server B since the PTC Server A also provides the controlling PTC function.

The LI functions in the PTC Servers generate the xIRI based on the SIP and TBCP messages sent to or received from the target and xCC based on the media (RTP) sent to or received from the target.

Note that the two PTC Clients and hence, the two PTC Servers can be in the same CSP domain.

4.1.3.3 LI for presence watcher functions

A high-level view of LI aspects for PTC presence and watcher handling functions is depicted in figures 4.1.3.3-1 and 4.1.3.3-2.

Presence Source and Watcher in the device

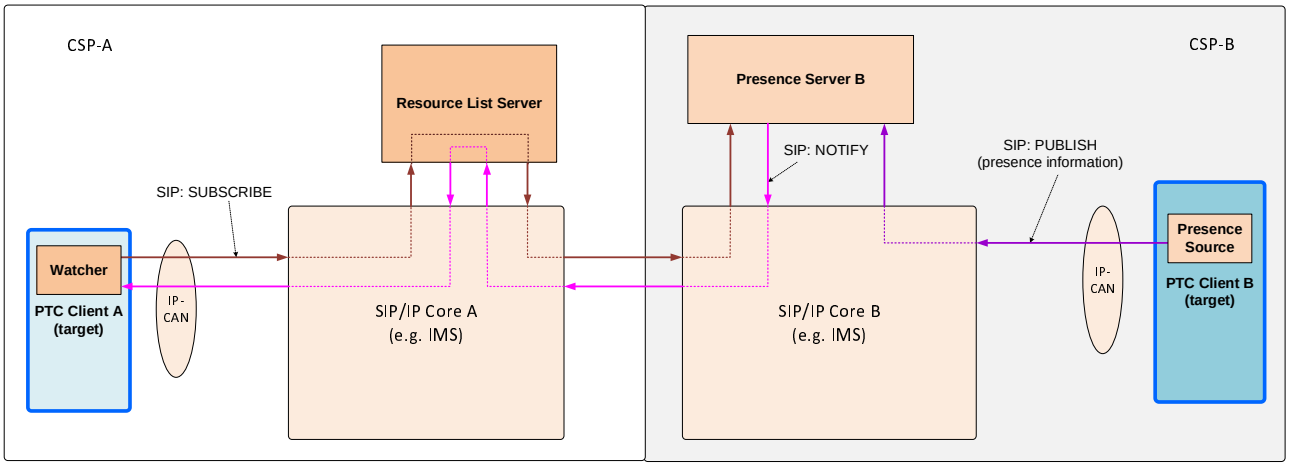


Figure 4.1.3.3-1: Presence source and Watcher in the PTC Client device

In the illustration shown in figure 4.1.3.3-1, Watcher and Presence Source functions are in the PTC Client devices. The PTC Client A provides the Watcher functions and PTC Client B is the Presence Source.

PTC Client A is a target in CSP-A and PTC Client B is a target in CSP-B. However, since a PTC Server is not involved in the handling of presence and watcher functions LI is not possible. In other words, LI does not apply when Watcher and Presence Source functions are in the PTC Client devices.

Presence Source and Watcher in the PTC Server

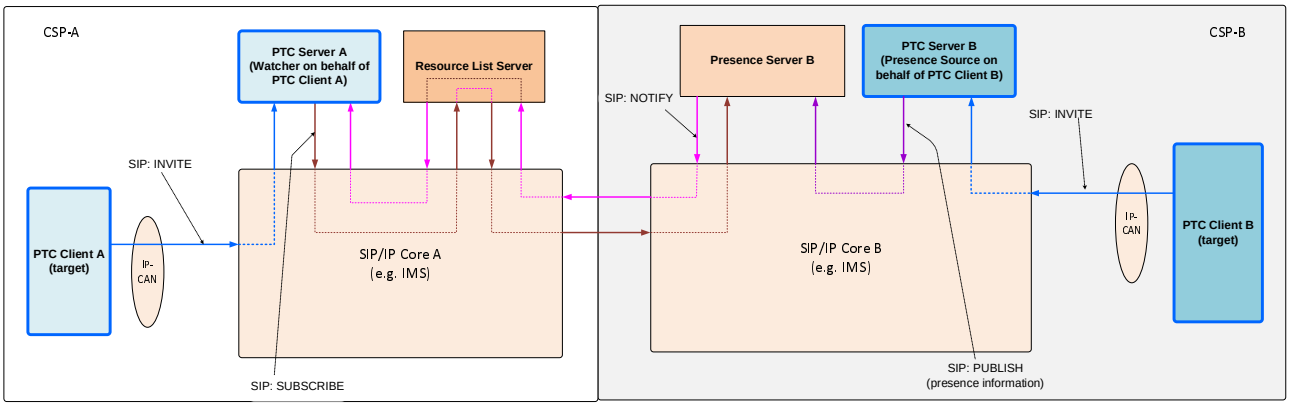


Figure 4.1.3.3-2: Presence source and Watcher in the PTC server

In the illustration shown in figure 4.1.3.3-2, the PTC Server A provides the Watcher functions on behalf of PTC Client A and PTC Server B provides the Presence Source functions on behalf of PTC Client B.

With PTC Client A as a target in CSP A, the PTC Server would provide the LI functions by intercepting (i.e. xIRIs) the SIP: SUBSCRIBE and SIP: NOTIFY messages. With PTC Client B as a target, the PTC Server B provides the LI functions by intercepting the SIP: PUBLISH message.

4.1.4 PTC LI architecture

As per the LI architecture defined in TS 33.127 [3], TS 33.128 [4], the PTC Server provides the IRI-POI and the CC-POI with the following target identifiers:

- IMPU.
- IMPI.
- MCPTT ID.
- Instance Identifier URN.
- PTC Chat Group ID.

Among the identities listed above, the first two are associated with a PTC Client. The PTC Chat Group ID is a network resource used to identify a PTC Chat Group. The TS 33.127 [3] states that the concept of target non-local ID does not apply to PTC.

The LI architecture for PTC is depicted in figure 4.1.4-1.

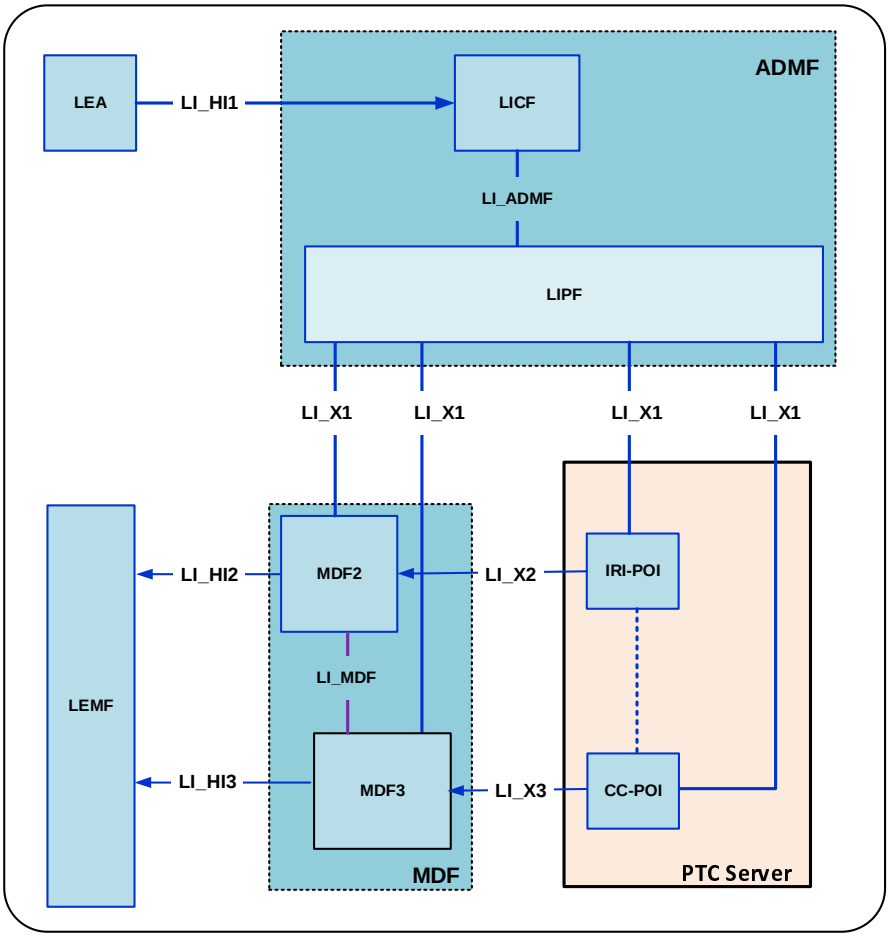


Figure 4.1.4-1: LI architecture with POIs in PTC server

The POIs in the PTC Server report the SIP messages used to setup the PTC sessions, presence and watcher functions and the TBCP messages used for the floor-control and the RTP media sent and received from the target.

When the target is a PTC Chat Group, the scope of SIP messages, TBCP and RTP media include the exchanges that happen between the PTC Server providing the controlling PTC function for the Chat Group and all PTC Clients.

The LI for Presence and Source does not apply when the target is a PTC Chat Group.

4.2 PTC topology scenarios

4.2.1 General

The subsequent clauses illustrate various topology scenarios for PTC which are built on the high-level views presented in clauses 4.1.1, 4.1.2 and 4.1.3.

Four cases are illustrated here:

- PTC pre-established session.
- PTC 1:1 session.
- PTC One-to-Many sessions which can also be referred to as part of PTC group sessions. Three cases are there:
  - PTC ad-hoc group session.
  - PTC pre-arranged group session.
  - PTC chatgroup session.
- One-to-Many-to-One session.

The focus is on the illustration of LI for PTC sessions. From a topology perspective, in some instances, an ad-hoc group and pre-arranged group may have similarity. For such instances, the two cases are combined in the illustration.

The PTC topology scenarios for presence and watcher functionalities do not have more to illustrate than what are presented in clauses 4.1.1 and 4.1.2.

4.2.2 Few conventions used

The term PTC Client is used in the diagrams and the descriptions that refer to the diagram. In a general description, the term PTC client may be used. Likewise, the term PTC Server is used in the diagrams and the descriptions that refer to the diagram. In a general description, the term PTC server may be used. Similarly, the terms Floor and Floor Control are used in the diagrams and the descriptions that refer to the diagrams. In the general descriptions the terms floor and floor-control may be used.

In all the illustrations, when the PTC client is a subscriber of a CSP domain with 5G/4G radio and packet core are assumed. However, none of the radio or packet core elements are part of these topology illustrations. The NFs within the IMS domain are shown for illustrative purpose and they all are named the same way (e.g. S-CSCF, IBCF).

Figure 4.2.2-1 shows some of the conventions used within the topology drawings (in a general way):

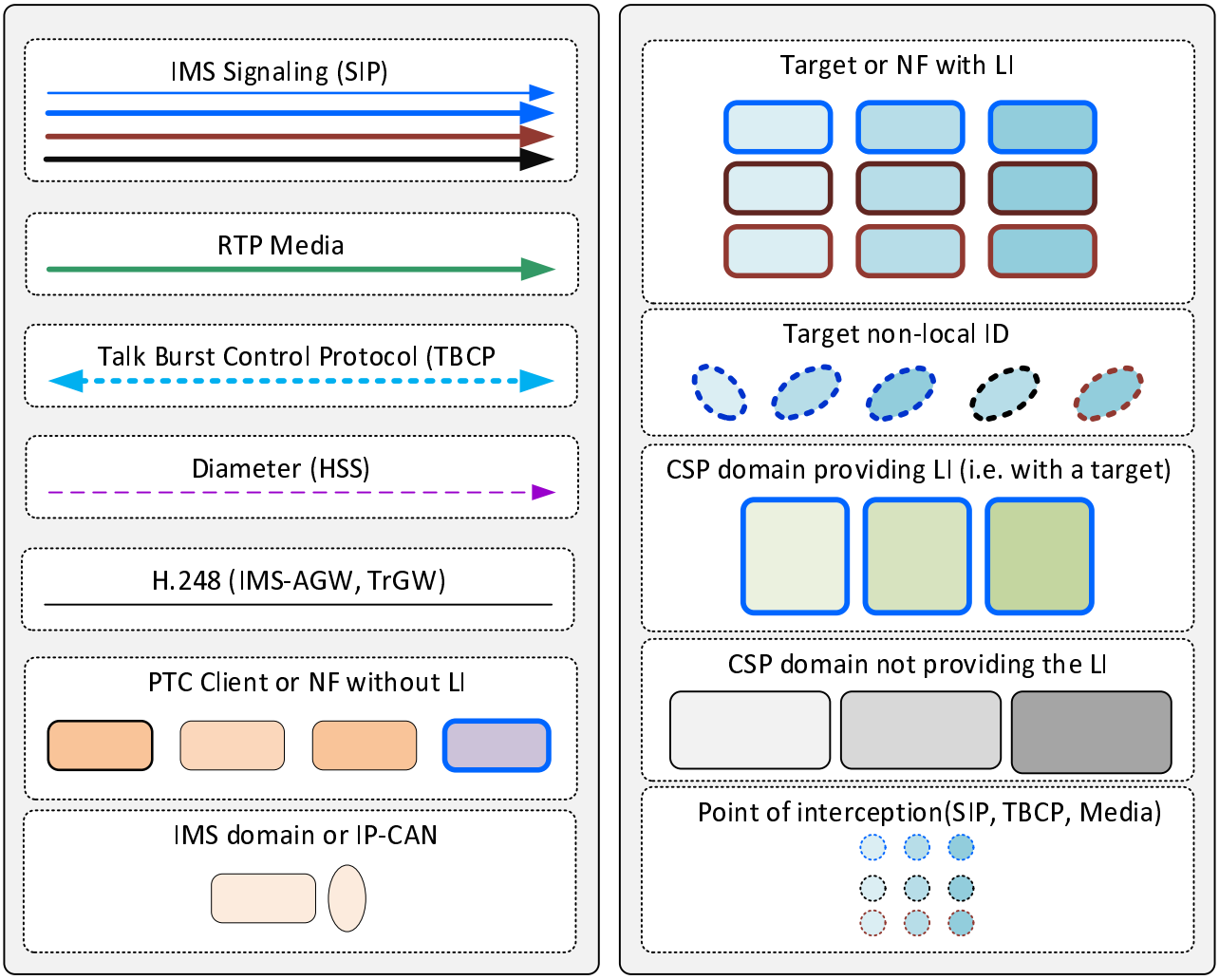


Figure 4.2.2-1: Few conventions used within the drawings

In all the illustrations, the RTP media flows in one direction only since in a PTC session, the media flow is in half-duplex mode, one is talking and others are listening. The PTC server that manages the floor control is said to provide the controlling PTC function. The TBCP handling and media flow are managed by the PTC server that provides the controlling PTC function.

In all the illustrations, the direction shown for the SIP messages is for the initial INVITE in the sense it indicates how the initial signalling has happened. However, it has to be noted that the SIP flows on either direction (e.g. INVITE Vs 200 OK). Except for the PTC chat group sessions, A indicates the initiator of a PTC session.

4.2.3 PTC pre-established session

Before any of the PTC sessions are established, a PTC client can have pre-established session setup to the respective PTC Server that provides the participating PTC function. With pre-established session setup, the PTC client would use the SIP REFER instead of a SIP INVITE to initiate a PTC session. At the terminating end too, SIP REFER is used instead of a SIP INVITE.

In the illustration, three PTC clients with each having the pre-established session with the respective PTC Server that provide the participating PTC function.

Figure 4.2.3-1 illustrates the case where all three PTC clients are targets in the respective CSP domains.

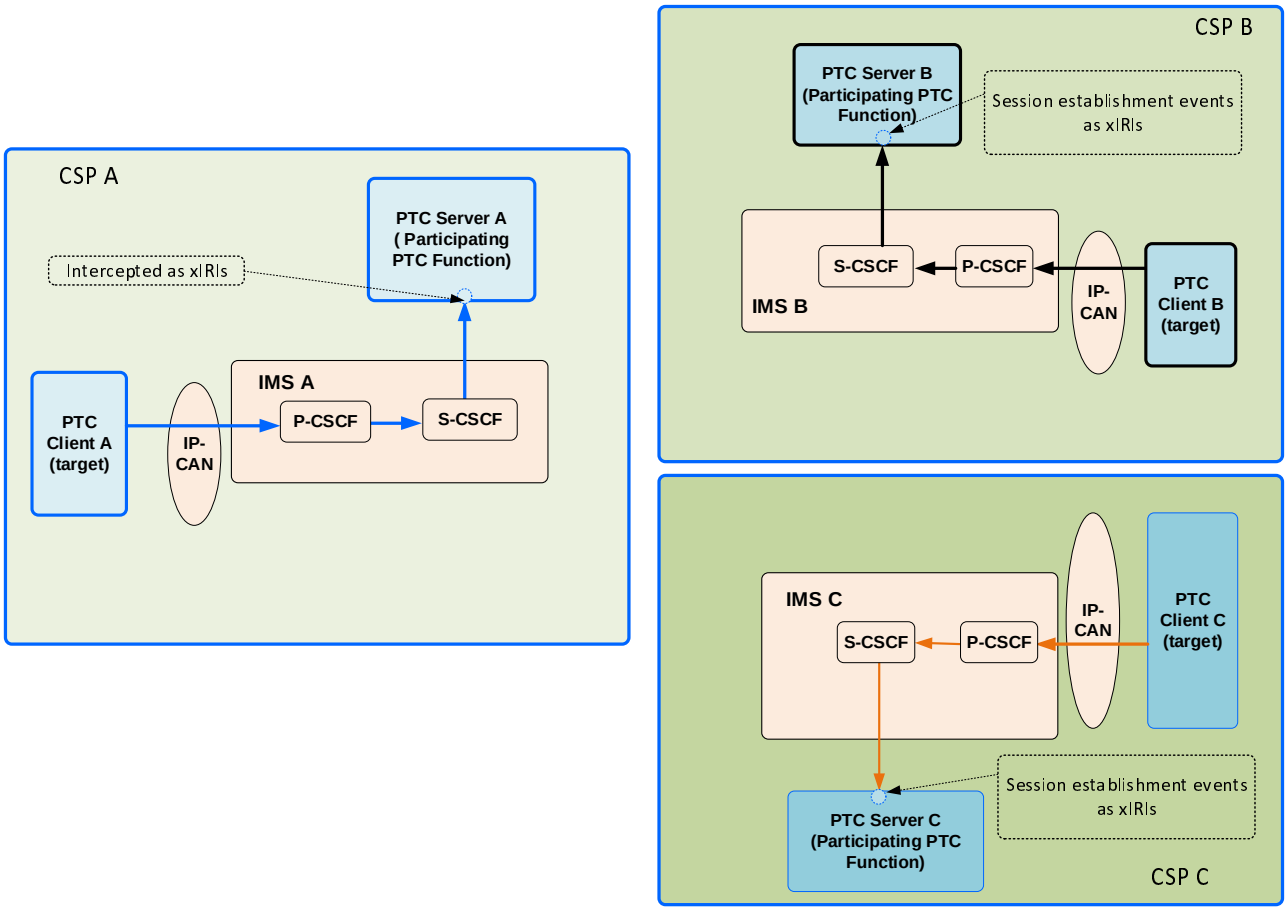


Figure 4.2.3-1: PTC pre-established sessions

When PTC Client A is a target in the CSP A domain, the PTC Server A that provides the participating PTC function for PTC Client A provides the LI functions. Likewise, when PTC Client B is a target in the CSP B domain, the PTC Server B that provides the participating PTC function for PTC Client B provides the LI functions. Similarly, when PTC Client C is a target in the CSP C domain, the PTC Server C that provides the participating PTC function for PTC Client C provides the LI functions.

Since, pre-established sessions are from PTC client to the PTC Server that provides the participating PTC function, the PTC pre-establishment events are intercepted as xIRIs in the respective PTC Server that provide the participating PTC function.

4.2.4 PTC 1:1 session

4.2.4.1 General

In a PTC 1:1 (read as one-on-one) session, two PTC clients are involved. One PTC client establishes a PTC session to another PTC client. Either of the two PTC clients can have the floor, and hence, speaking. In a PTC language, these types of session may also be referred to as PTC private sessions.

In the illustrations PTC Client A initiates the 1:1 PTC session to PTC Client B.

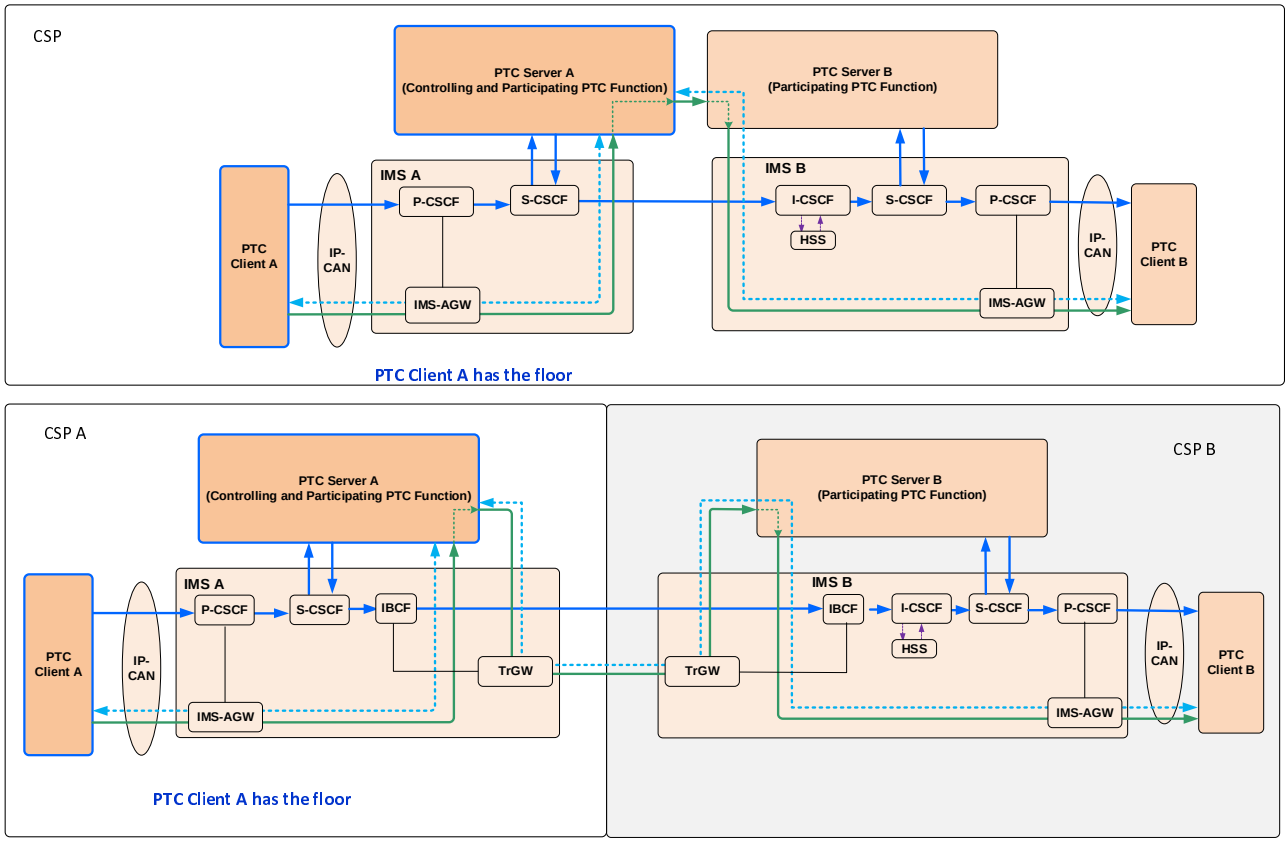


Figure 4.2.4.1-1: PTC Client A initiates a 1:1 PTC Session to PTC Client B

As shown in figure 4.2.4.1-1, the PTC Client A and PTC Client B can be in the same CSP domain or in different CSP domains.

4.2.4.2 LI: PTC clients in the same CSP domain

4.2.4.2.1 General

In the illustrations shown here, PTC Client A and PTC Client B are in the same CSP domain.

4.2.4.2.2 PTC client A is a target

Figure 4.2.4.2.2-1 illustrates a case of PTC Client A who initiated the 1:1 PTC session to PTC Client B is a target. In the illustration, the LI is applicable only for PTC.

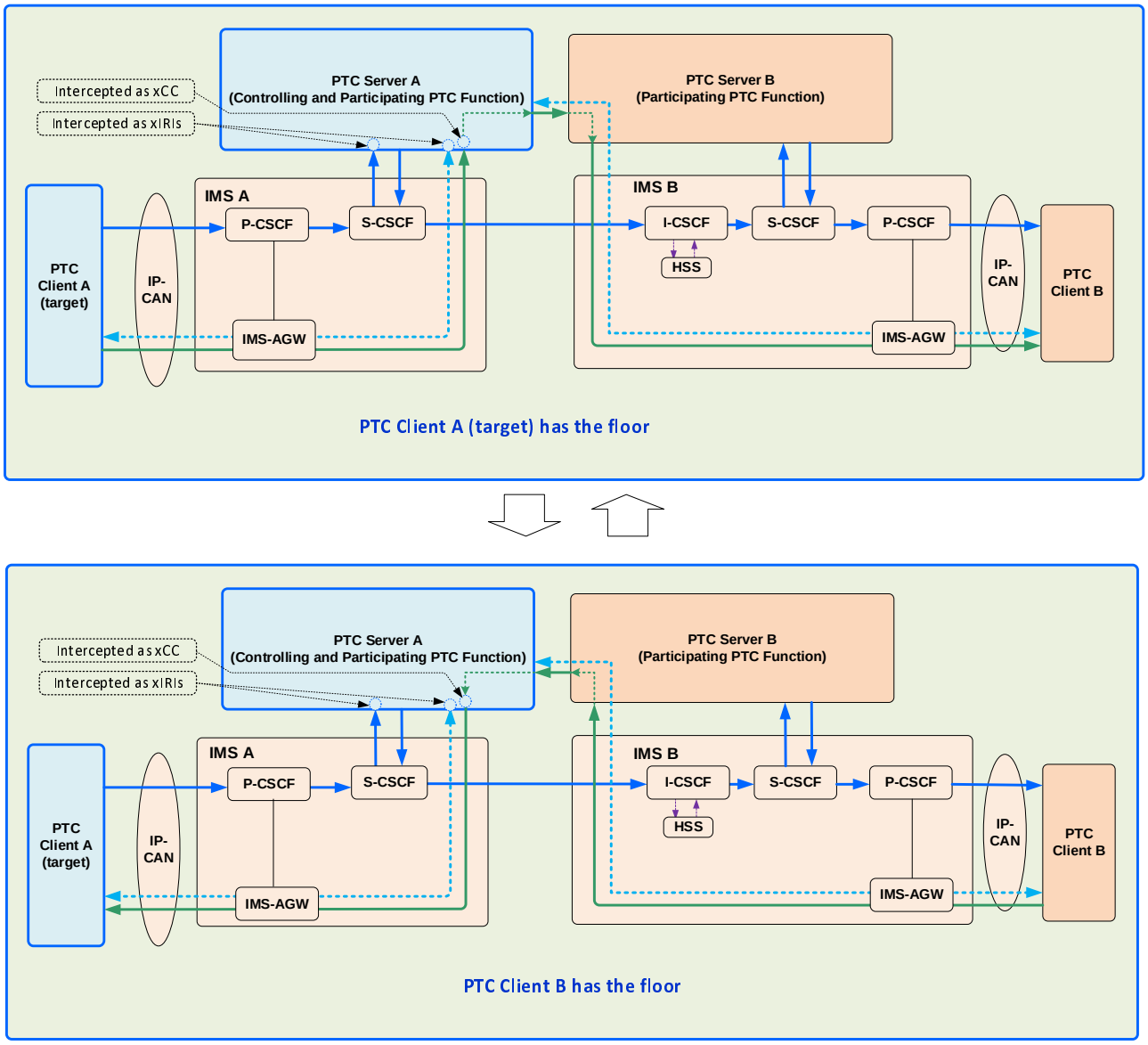


Figure 4.2.4.2.2-1: 1:1 PTC session, PTC Client A is a target

When PTC Client A is a target, the PTC Server A that provides the participating PTC function for PTC Client A provides the LI function. Since, the PTC Client A has initiated the 1:1 PTC session, the PTC Server A also provides the controlling PTC function. As shown in figure 4.2.4.2.2-1, either PTC Client A (target) or the PTC Client B can have the floor at any given time.

4.2.4.2.3 PTC client B is a target

Figure 4.2.4.2.3-1 illustrates a case of PTC Client B who receives an incoming 1:1 PTC session from PTC Client A is a target. In the illustration, the LI is applicable only for PTC.

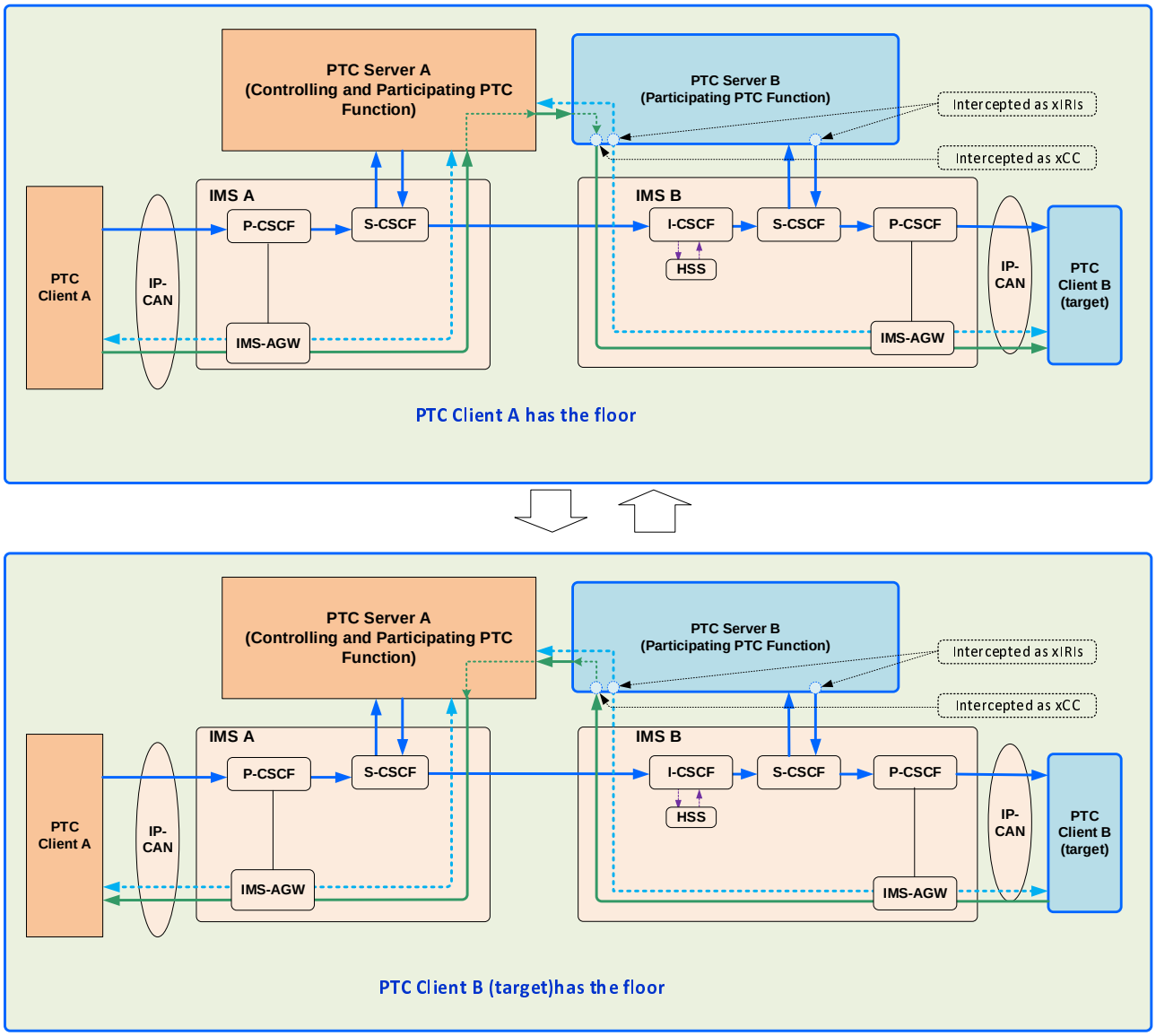


Figure 4.2.4.2.3-1: 1:1 PTC session, PTC Client B is a target

When PTC Client B is a target, the PTC Server B that provides the participating PTC function for PTC Client B provides the LI function. The scope of LI includes the interception of SIP messages, TBCP messages and the RTP media as observed in the PTC Server B. As shown in figure 4.2.4.2.3-1, either PTC Client A or the PTC Client B (target) can have the floor at any given time.

4.2.4.3 LI: PTC clients in different CSP domains

4.2.4.3.1 General

In the illustrations shown here, PTC Client A and PTC Client B are in different CSP domains.

4.2.4.3.2 PTC client A is the target in CSP A domain

Figure 4.2.4.3.2-1 illustrates a case of PTC Client A who initiated the 1:1 PTC session to PTC Client A is a target in CSP A domain. In the illustration, the LI is applicable only for PTC.

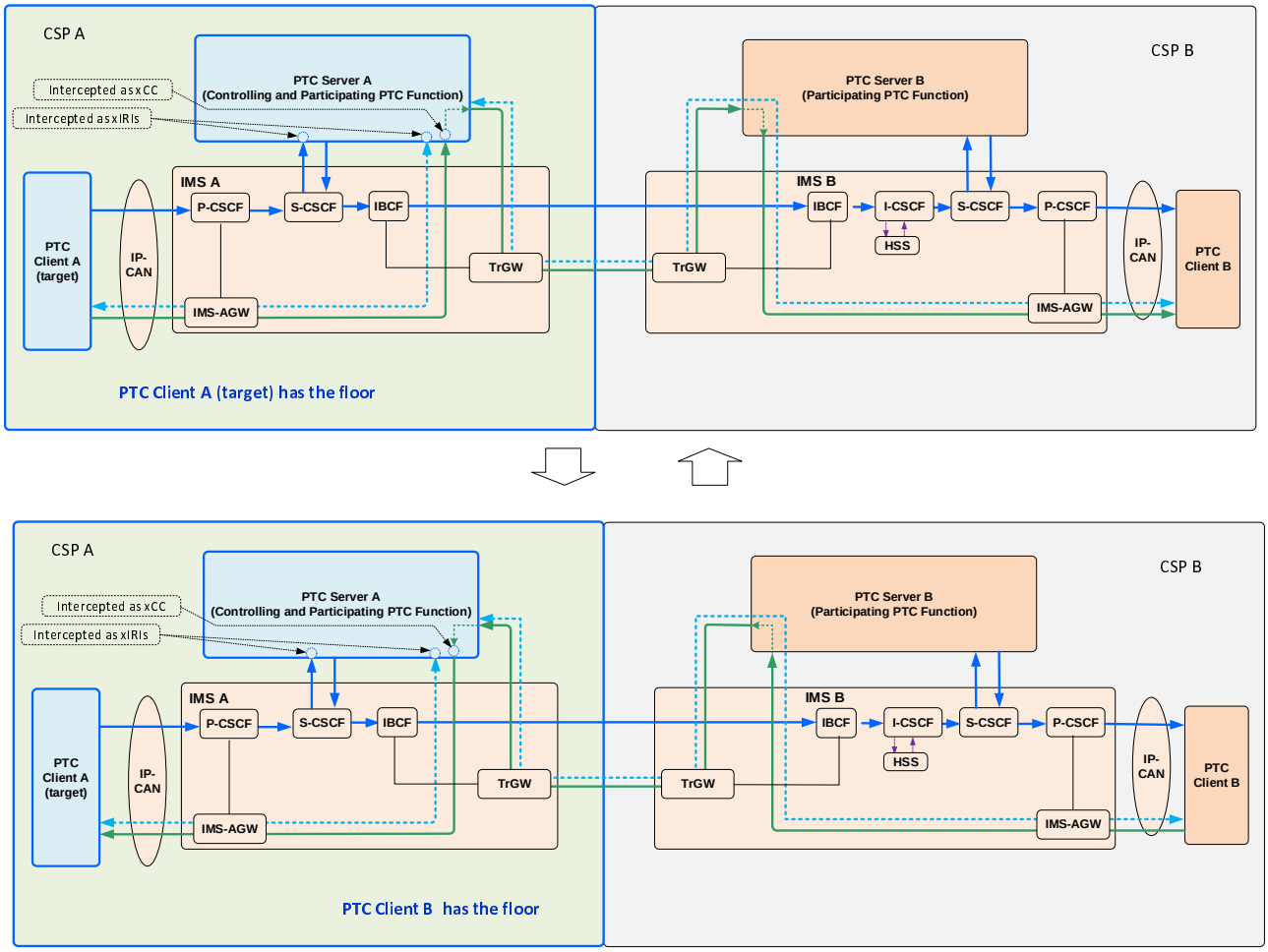


Figure 4.2.4.3.2-1: 1:1 PTC session, PTC Client A is a target in CSP A domain

When PTC Client A is a target, the PTC Server A that provides the participating PTC function for PTC Client A provides the LI function. Since, the PTC Client A has initiated the 1:1 PTC session, the PTC Server A also provides the controlling PTC function. As shown in figure 4.2.4.3.2-1, either PTC Client A (target) or the PTC Client B can have the floor at any given time.

4.2.4.3.3 PTC Client B is a target in CSP B domain

Figure 4.2.4.3.3-1 illustrates a case of PTC Client B who receives the 1:1 PTC session from PTC Client B is a target in CSP B domain. In the illustration, the LI is applicable only for PTC.

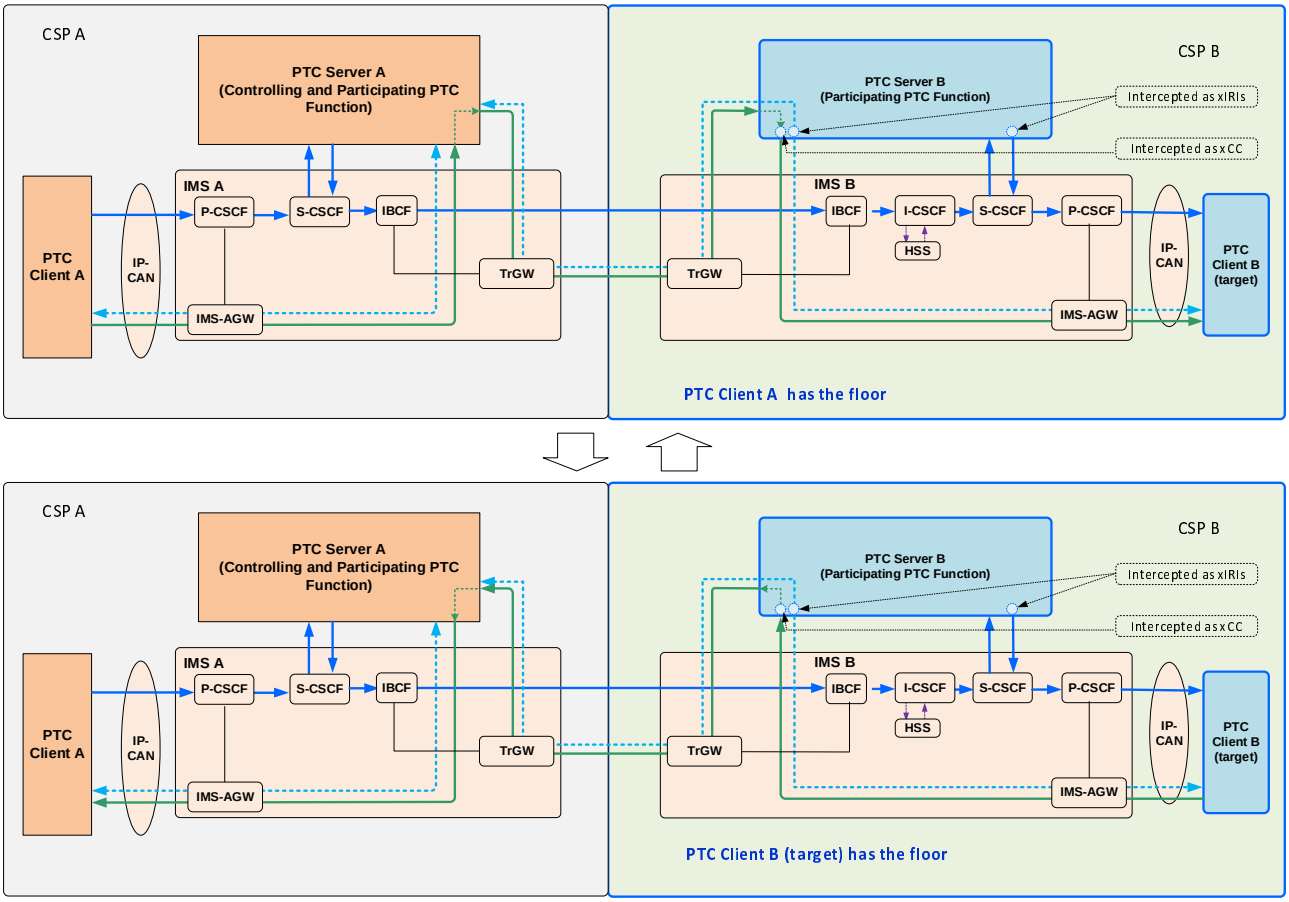


Figure 4.2.4.3.3-1: 1:1 PTC session, PTC Client B is a target in CSP B domain

When PTC Client B is a target, the PTC Server B that provides the participating PTC function for PTC Client B provides the LI function. The scope of LI includes the interception of SIP messages, TBCP messages and the RTP media as observed in the PTC Server B. As shown in figure 4.2.4.3.3-1, either PTC Client A or the PTC Client B (target) can have the floor at any given time.

4.2.5 PTC one-to-many session

4.2.5.1 General

In a PTC One-to-Many session, multiple PTC clients are involved as group in a PTC session. This is also referred to as PTC group session. While one of PTC clients in the group is speaking, the rest of the PTC clients are listening. Any PTC client in the group can acquire the floor (when it is open) to begin speaking.

The PTC session can be setup either with one PTC client initiating the PTC session with multiple destination PTC clients or the PTC clients can join a PTC chat-group session. In the first case, the multiple destination PTC clients in the group can either be pre-arranged (by the CSP) or determined on an ad-hoc basis (by the PTC client) during the session initiation. In the second case, a separate PTC Server may handle the PTC chat group session.

In an ad-hoc group PTC session, the list of multiple destination PTC clients may be provided by the PTC client who initiates the session or based on some other criteria determined by the PTC Server that provides the participating PTC function to the session-initiating PTC client.

In a pre-arranged PTC group session, the participants in the PTC group are pre-determined. Upon receiving the PTC session-initiation request from one of the PTC clients in the pre-arranged group, the PTC server that provides the participating PTC function to the session initiating PTC client would interact with the group management system to determine the PTC clients in the group which in turn would send out the PTC session invitation to all those PTC clients.

4.2.5.2 Ad-hoc group PTC session

4.2.5.2.1 General

In figure 4.2.5.2.1-1, three PTC clients are shown. PTC Client A initiates the PTC session. The PTC Client B and PTC Client C are the other two PTC clients in the group.

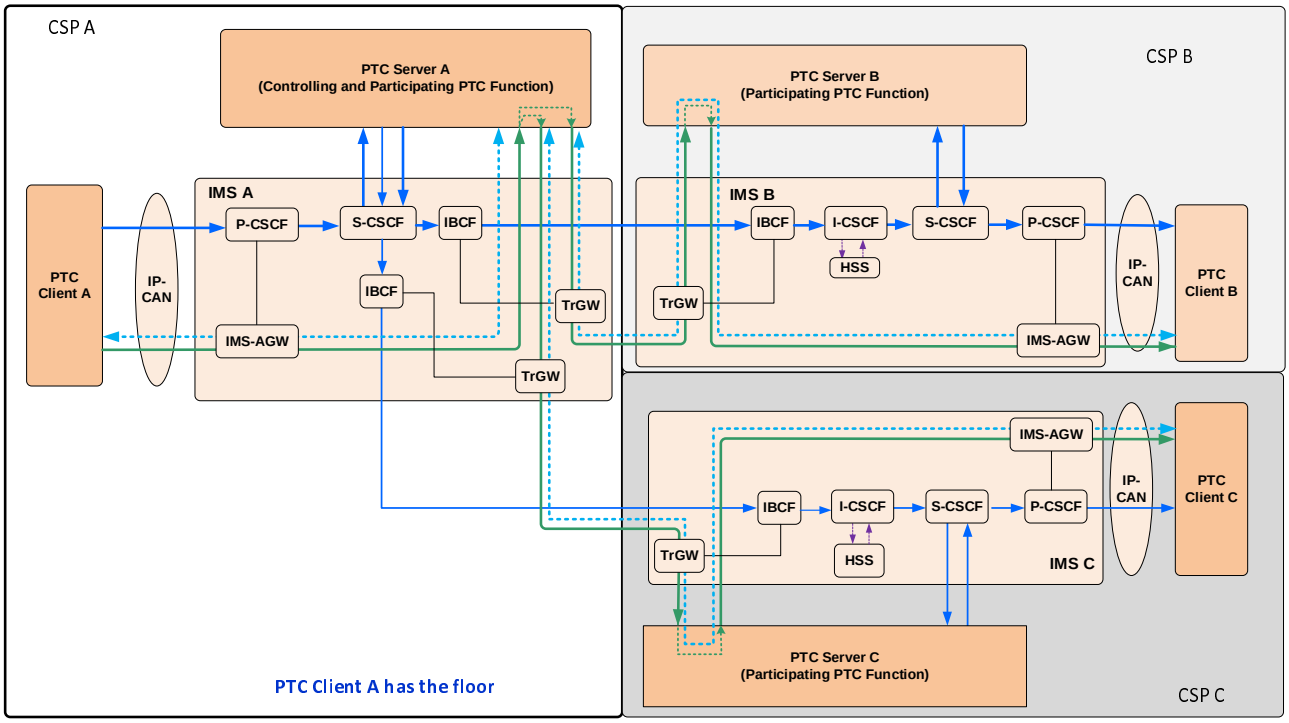


Figure 4.2.5.2.1-1: An ad-hoc or pre-arranged group PTC session – a topology view

In the illustrations, PTC Client A is in CSP A domain, PTC Client B is in CSP B domain and PTC Client C is in CSP C domain.

Either two or all three of the PTC clients may also be in the same CSP domain. When the two PTC clients are in the same CSP domain, IBCFs and TrGWs are not present between the IMS domain serving those PTC clients (see figure 4.2.4.1-1).

The topology shown in figure 4.2.5.2.1-1 (all three PTC clients in different CSP domains) is expanded with the LI aspects shown in the rest of the illustrations of this clause.

4.2.5.2.2 PTC Client A is a target

Figure 4.2.5.2.2-1 illustrates a case of PTC Client A who initiated the ad-hoc group PTC session to PTC Client B and PTC Client C is a target. In the illustration, the LI is applicable only for PTC.

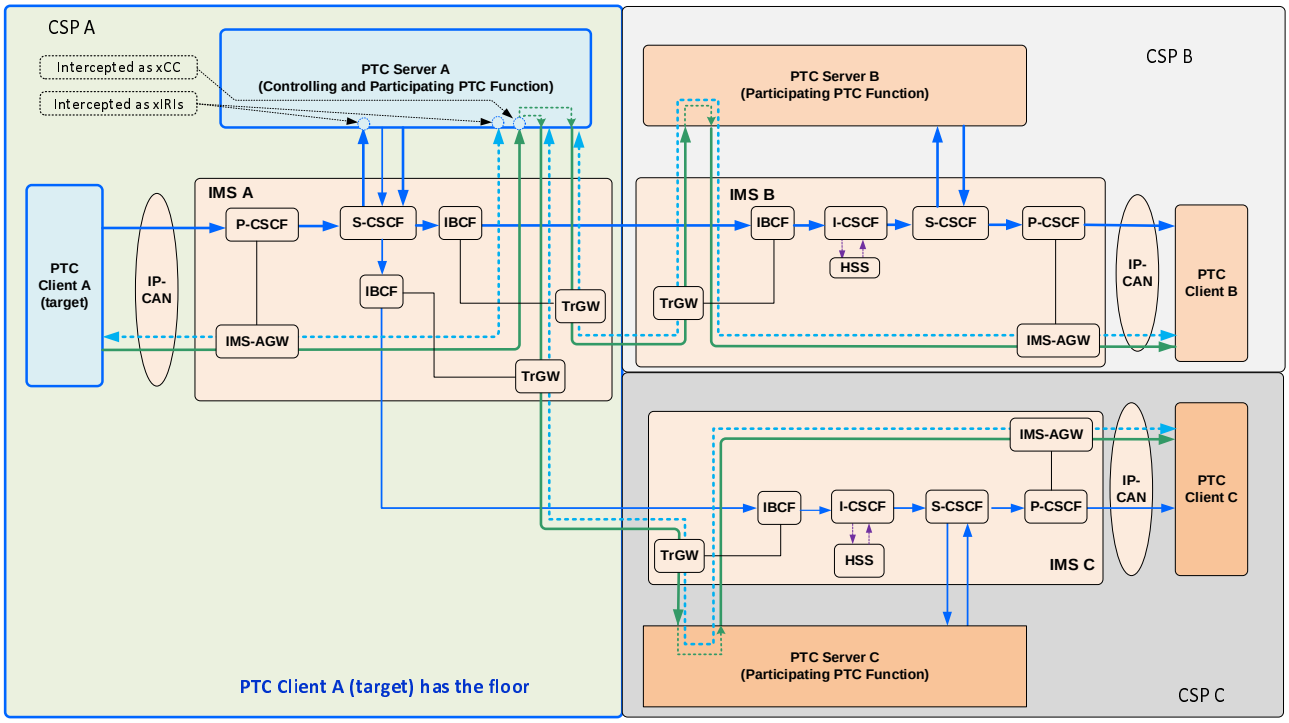


Figure 4.2.5.2.2-1: Ad-hoc group PTC session, PTC Client A is a target

When PTC Client A is a target, the PTC Server A that provides the participating PTC function for PTC Client A provides the LI functions. Since, the PTC Client A has initiated this PTC group session, the PTC Server A also provides the controlling PTC function. As shown in figure 4.2.5.2.2-1, the interception happens at the ingress end of signalling and media flow. In this illustration, PTC Client A has the floor.

Figure 4.2.5.2.2-2 illustrates the topology with different PTC clients having the floor.

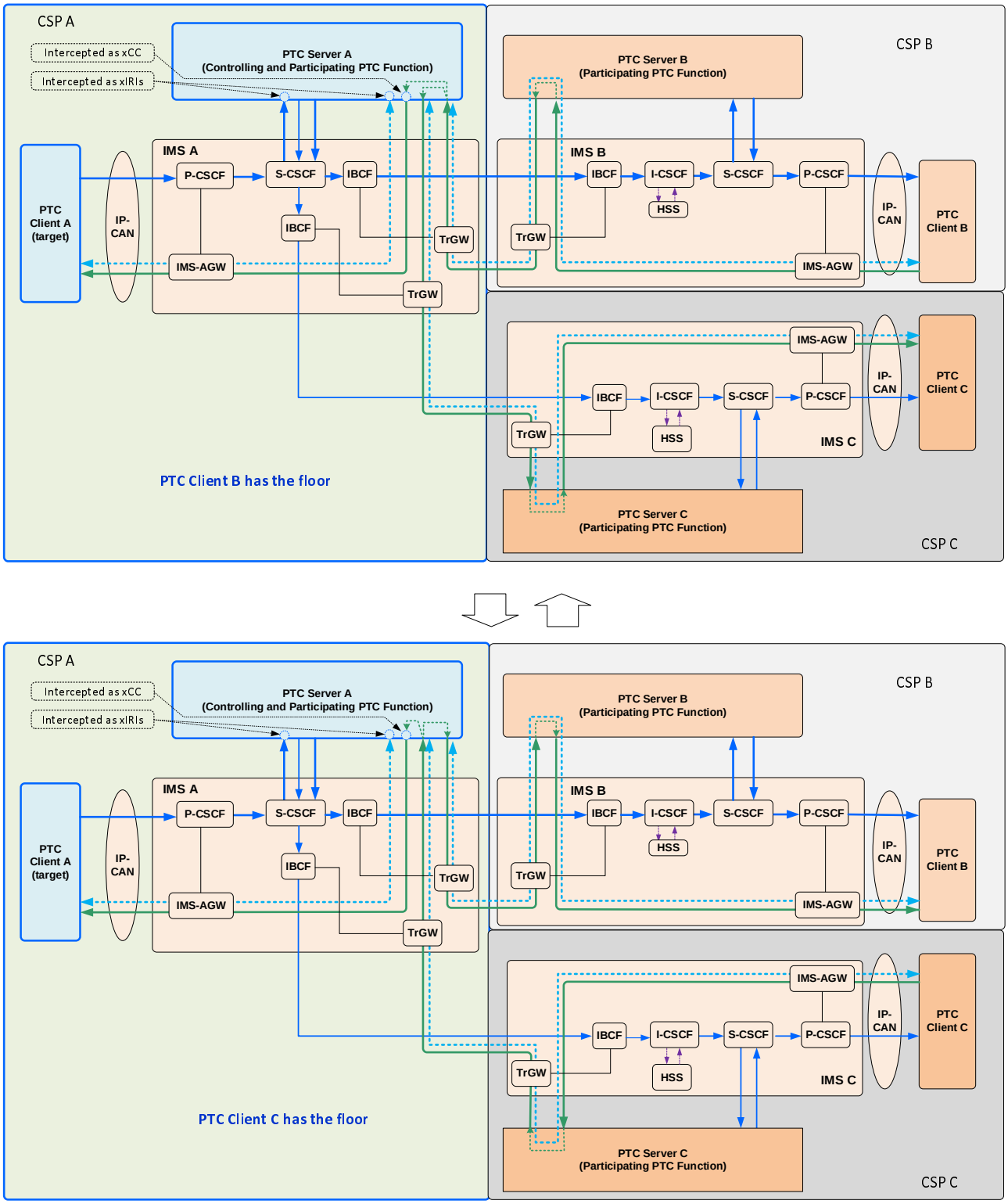


Figure 4.2.5.2.2-2: Ad-hoc group PTC session, PTC Client A is a target – other floor control options

As shown in figure 4.2.5.2.2-2, either the PTC Client B or PTC Client C can also have the floor at any given time in addition to PTC Client A (target) having the floor (figure 4.2.5.2.2-1). The difference in the LI aspects is mainly the direction of media that is being intercepted. In all cases, the interception happens at the ingress end of signalling and media flow with respect to the PTC Server A.

4.2.5.2.3 PTC Client B is a target

Figure 4.2.5.2.3-1 illustrates a case of PTC Client B one of the destination PTC clients of an ad-hoc group PTC session is the target. In the illustration, the LI is applicable only for PTC.

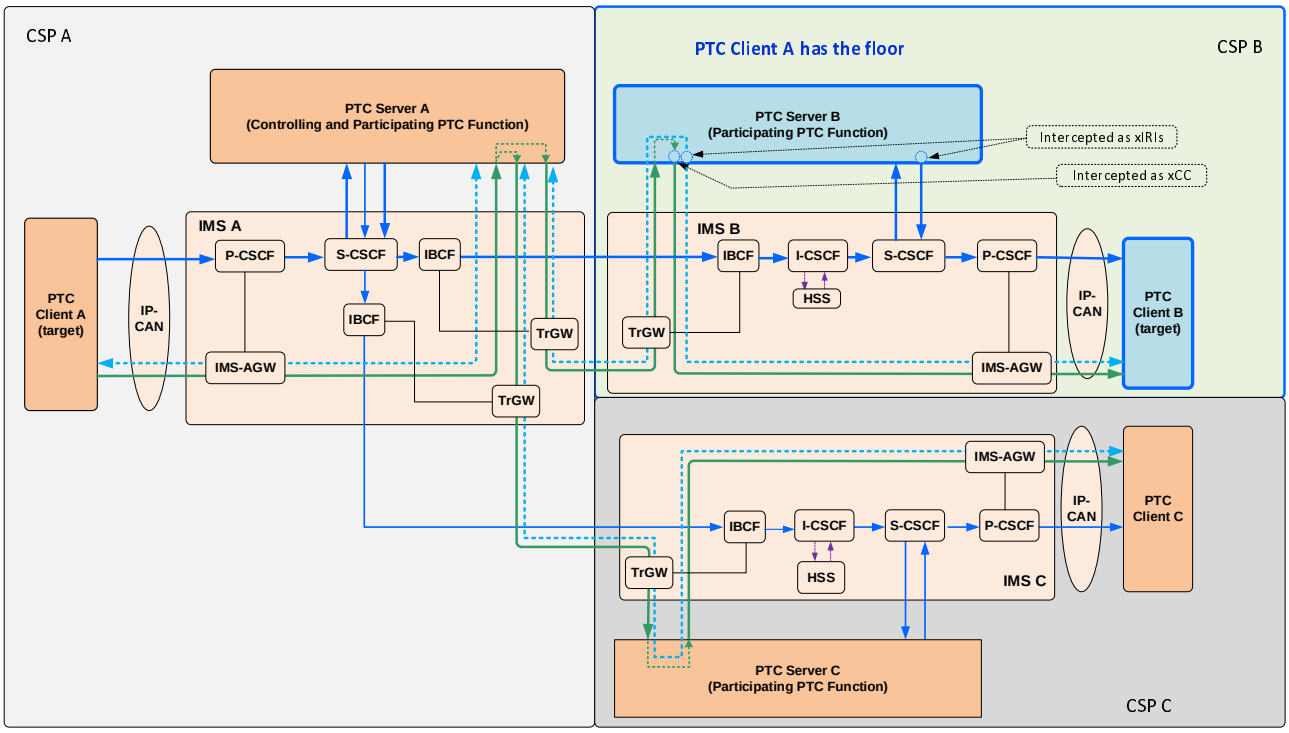


Figure 4.2.5.2.3-1: Ad-hoc group PTC session, PTC Client B is a target

When PTC Client B is a target, the PTC Server B that provides the participating PTC function for PTC Client B provides the LI functions. The scope of LI functions provided by PTC Server B is limited to the information available in PTC Server B. In this illustration, the controlling PTC function is provided by PTC Server A present in the CSP A domain. As shown in figure 4.2.5.2.3-1, the interception happens at the egress end of signalling and media flow. In this illustration, PTC Client A has the floor.

Figure 4.2.5.2.3-2 illustrates the topology with different PTC clients having the floor.

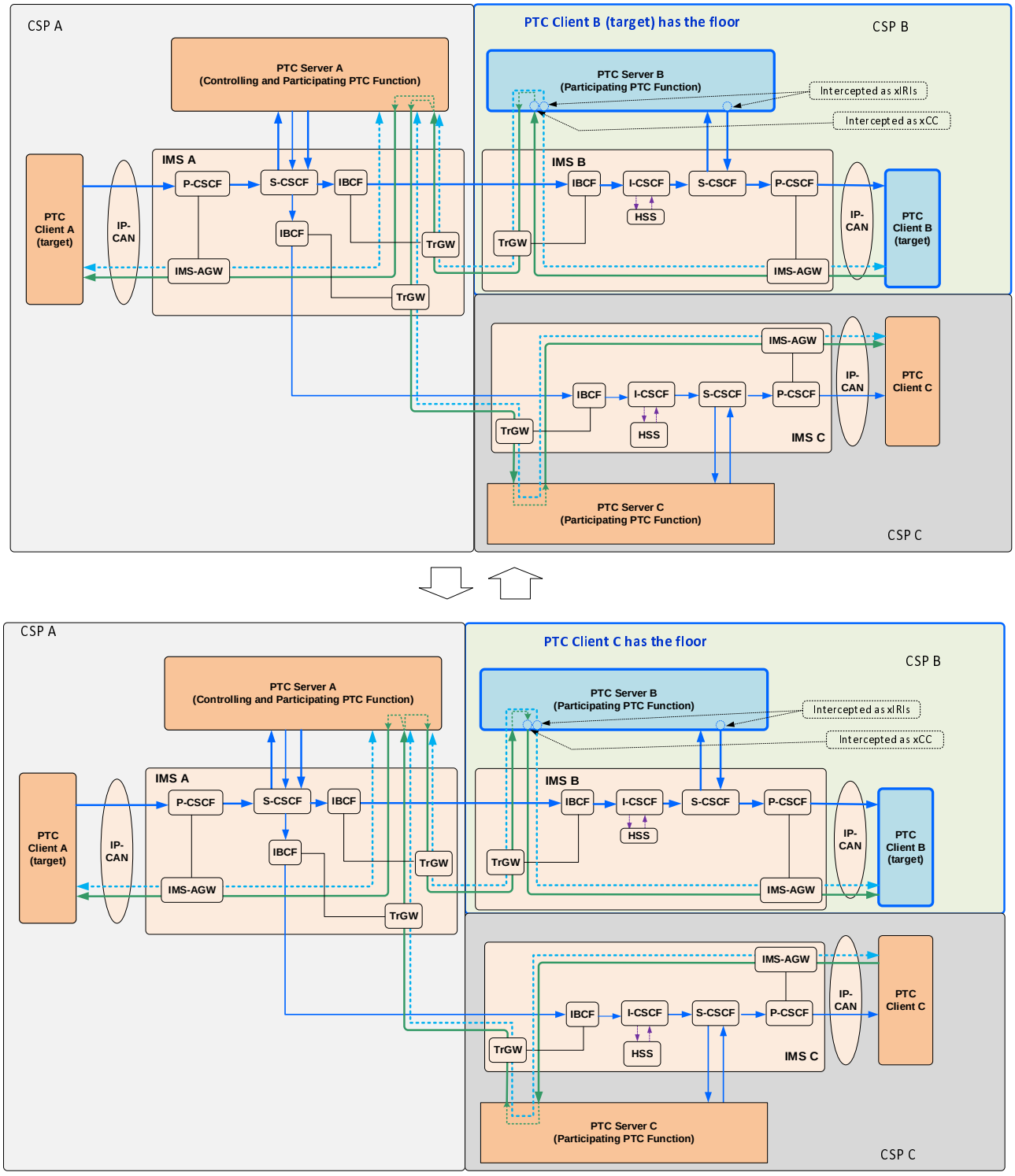


Figure 4.2.5.2.3-2: Ad-hoc group PTC session, PTC Client B is a target – floor control options

As shown in figure 4.2.5.2.3-2, either the PTC Client B (target) or PTC Client C can also have the floor at any given time in addition to PTC Client A having the floor (figure 4.2.5.2.3-1). The difference in the LI aspects is mainly the direction of media that is being intercepted. In all cases, the interception happens at the egress end of signalling and media flow with respect to the PTC Server B.

4.2.5.2.4 PTC Client C is a target

Figure 4.2.5.2.4-1 illustrates a case of PTC Client C the other destination PTC client of an ad-hoc group PTC session is the target. In the illustration, the LI is applicable only for PTC.

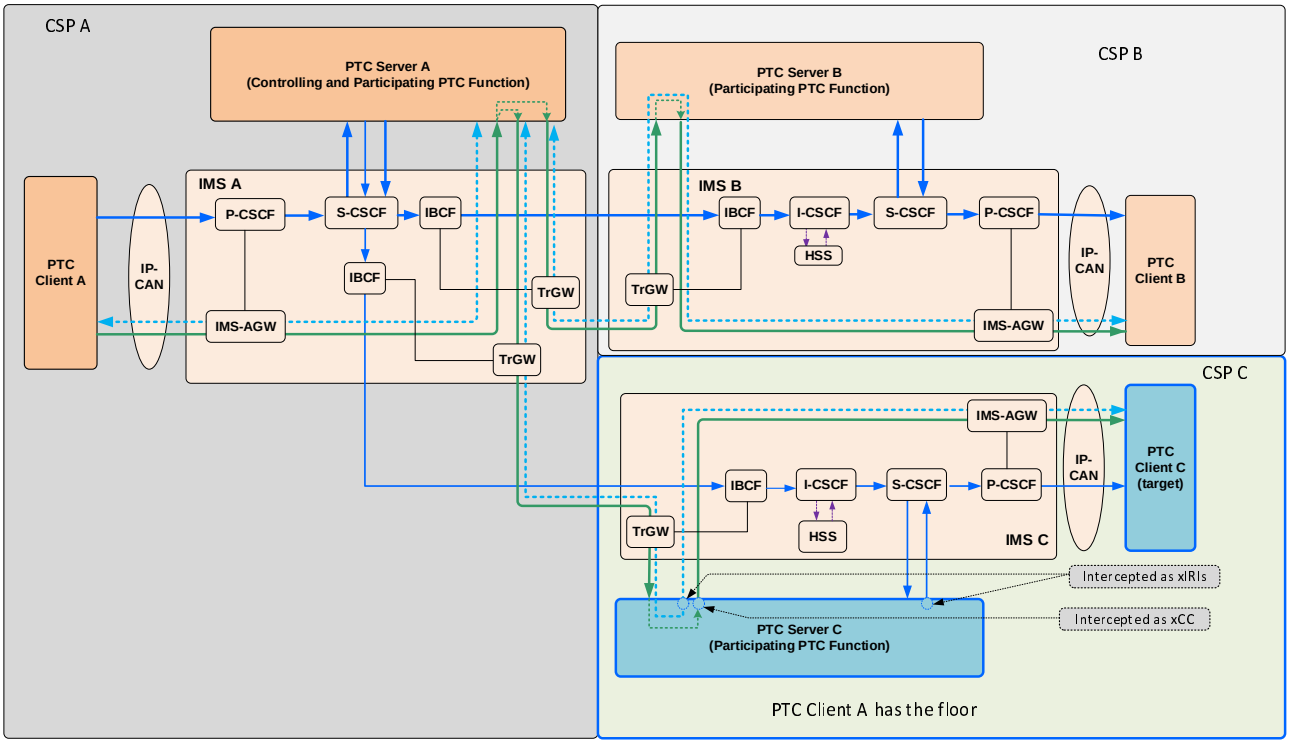


Figure 4.2.5.2.4-1: Ad-hoc group PTC session, PTC Client C is a target

When PTC Client C is a target, the PTC Server C that provides the participating PTC function for PTC Client C provides the LI functions. The scope of LI functions provided by PTC Server C is limited to the information available in PTC Server C. In this illustration, the controlling PTC function is provided by PTC Server A present in the CSP A domain. As shown in figure 4.2.5.2.4-1, the interception happens at the egress end of signalling and media flow. In this illustration, PTC Client A has the floor.

Figure 4.2.5.2.4-2 illustrates the topology with different PTC clients having the floor.

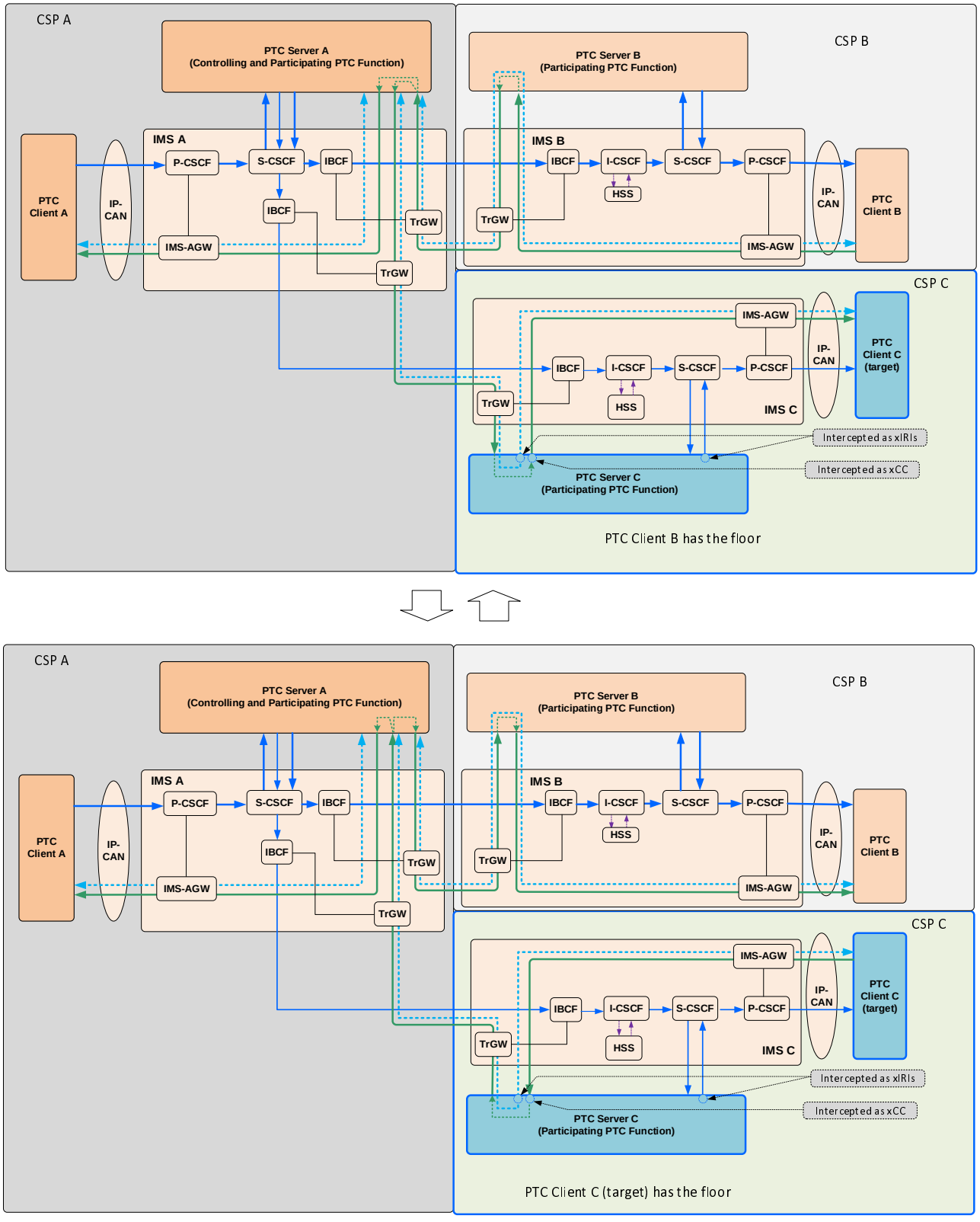


Figure 4.2.5.2.4-2: Ad-hoc group PTC session, PTC Client C is a target – floor control options

As shown in figure 4.2.5.2.4-2, either the PTC Client B or PTC Client C (target) can also have the floor at any given time in addition to PTC Client A having the floor (figure 4.2.5.2.4-1). The difference in the LI aspects is mainly the direction of media that is being intercepted. In all cases, the interception happens at the egress end of signalling and media flow with respect to the PTC Server C.

4.2.5.3 Similarities between PTC client B and PTC client C from an LI perspective

The illustrations shown in clauses 4.2.5.2.3 and 4.2.5.2.4 show a similarity between the cases when PTC Client B or PTC Client C is a target, respectively. When such similarities occur, the illustrations are not repeated in the remaining clauses of the present document.

In some instances, a single illustration may include all PTC clients as the targets.

4.2.5.4 Pre-arranged group PTC session

4.2.5.4.1 General

In the illustration, three PTC clients are shown. PTC Client A initiates the PTC session. The PTC Client B and PTC Client C are the other two PTC clients in the pre-arranged group. The illustrations also show the possibility of the Group Management Server in the same service provider domain of PTC session initiator.

Figure 4.2.5.4.1-1 illustrates the topology where the Group Management Server is in a different service provider domain.

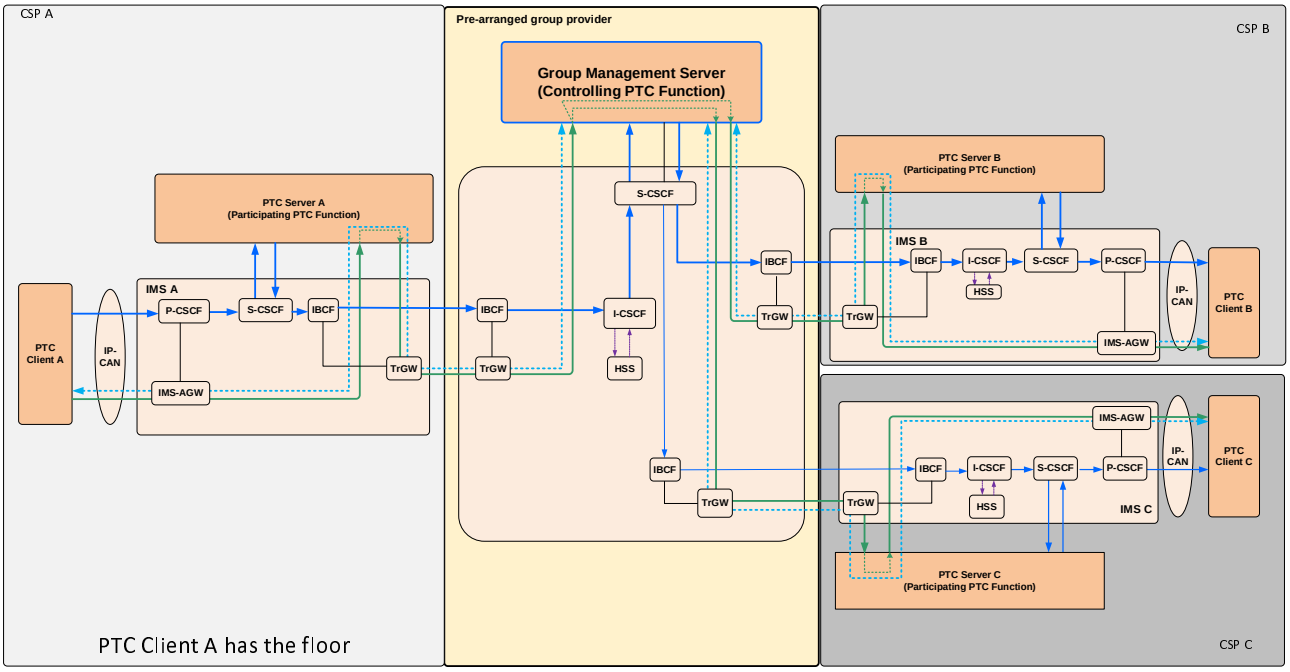


Figure 4.2.5.4.1-1: A pre-arranged group PTC session – a topology view – case 1

In the illustration, PTC Client A is in CSP A domain, PTC Client B is in CSP B domain and PTC Client C is in CSP C domain. The Group Management Server is in separate provider domain (shown as pre-arranged group provider).

Either two or all three of the PTC clients may also be in the same CSP domain. When the two PTC clients are in the same CSP domain, IBCFs and TrGWs are not present between the IMS domain serving those PTC clients (see figure 4.2.4.1-1).

Figure 4.2.5.4.1-2 illustrates the topology where the Group Management Server is in the same service provider domain as that of PTC session initiator.

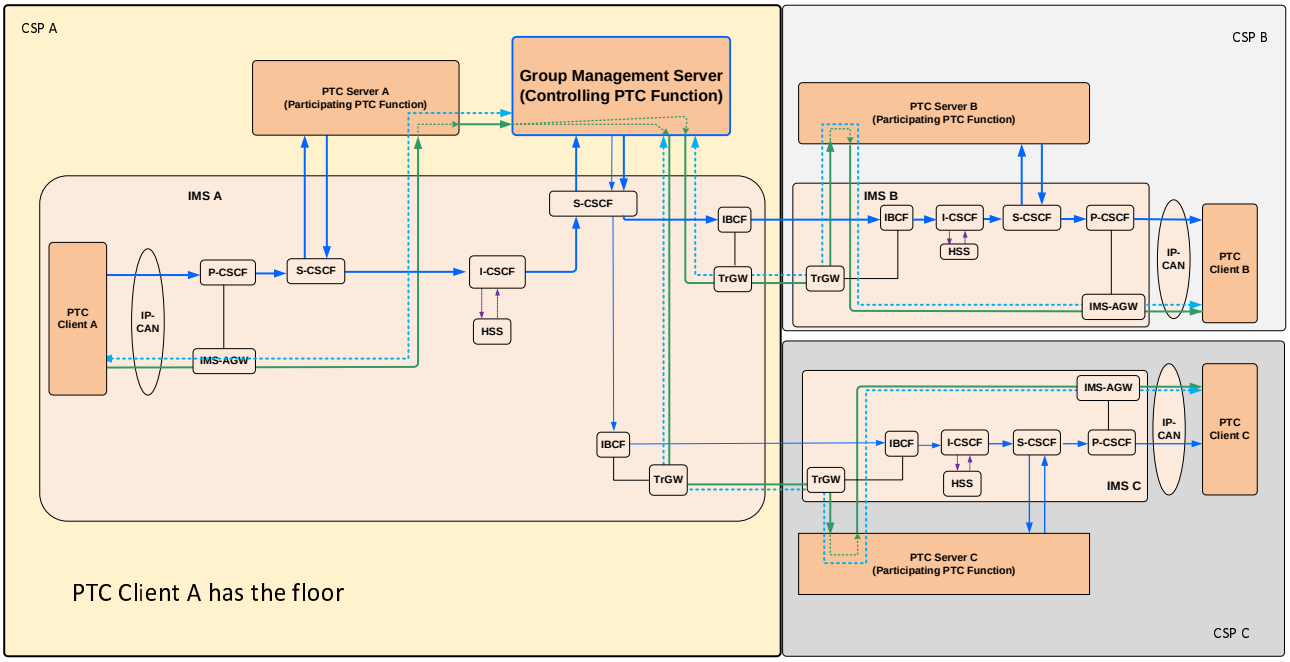


Figure 4.2.5.4.1-2: A pre-arranged group PTC session – a topology view – case 2

In the illustration, PTC Client A is in CSP A domain, PTC Client B is in CSP B domain and PTC Client C is in CSP C domain. The Group Management Server is in CSP A domain that serves the PTC session initiating client.

Either both or one of the destination PTC clients (B and C) may also be in the same CSP domain as that of PTC Client A. It is also possible that the two destination PTC clients (B and C) are in one CSP domain but different from the CSP A domain. When the two PTC clients are in the same CSP domain, IBCFs and TrGWs are not present between the IMS domain serving those PTC clients (see figure 4.2.4.1-1).

The topology shown in figures 4.2.5.4.1-1 and 4.2.5.4.1-2 are expanded with the LI aspects shown in the rest of the illustrations of this clause.

4.2.5.4.2 LI: Group management server is in different provider domain

Figure 4.2.5.4.2-1 illustrates a case of PTC Client A who initiated the pre-arranged group PTC session is a target. In the illustration, the LI is applicable only for PTC. The Group Management Server is in a different provider domain.

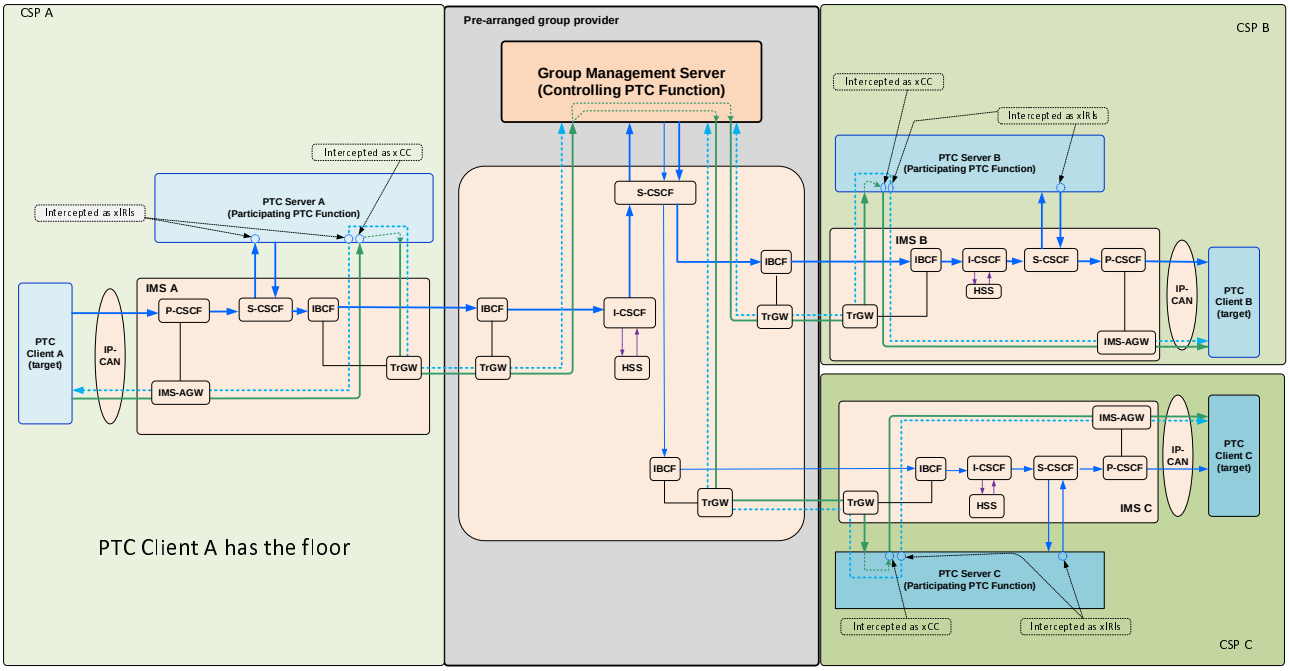


Figure 4.2.5.4.2-1: Pre-arranged group PTC session with LI – case 1

When PTC Client A is a target in CSP A domain, the PTC Server A that provides the participating PTC function for PTC Client A provides the LI functions. When PTC Client B is a target in CSP B domain, the PTC Server B that provides the participating PTC function for PTC Client B provides the LI functions. When PTC Client C is a target in CSP C domain, the PTC Server C that provides the participating PTC function for PTC Client C provides the LI functions.

For all three targets, the scope of LI functions provided by respective participating PTC Servers are limited to the information available in those servers.

Figure 4.2.5.4.2-2 illustrates the topologies with different PTC clients having the floor.

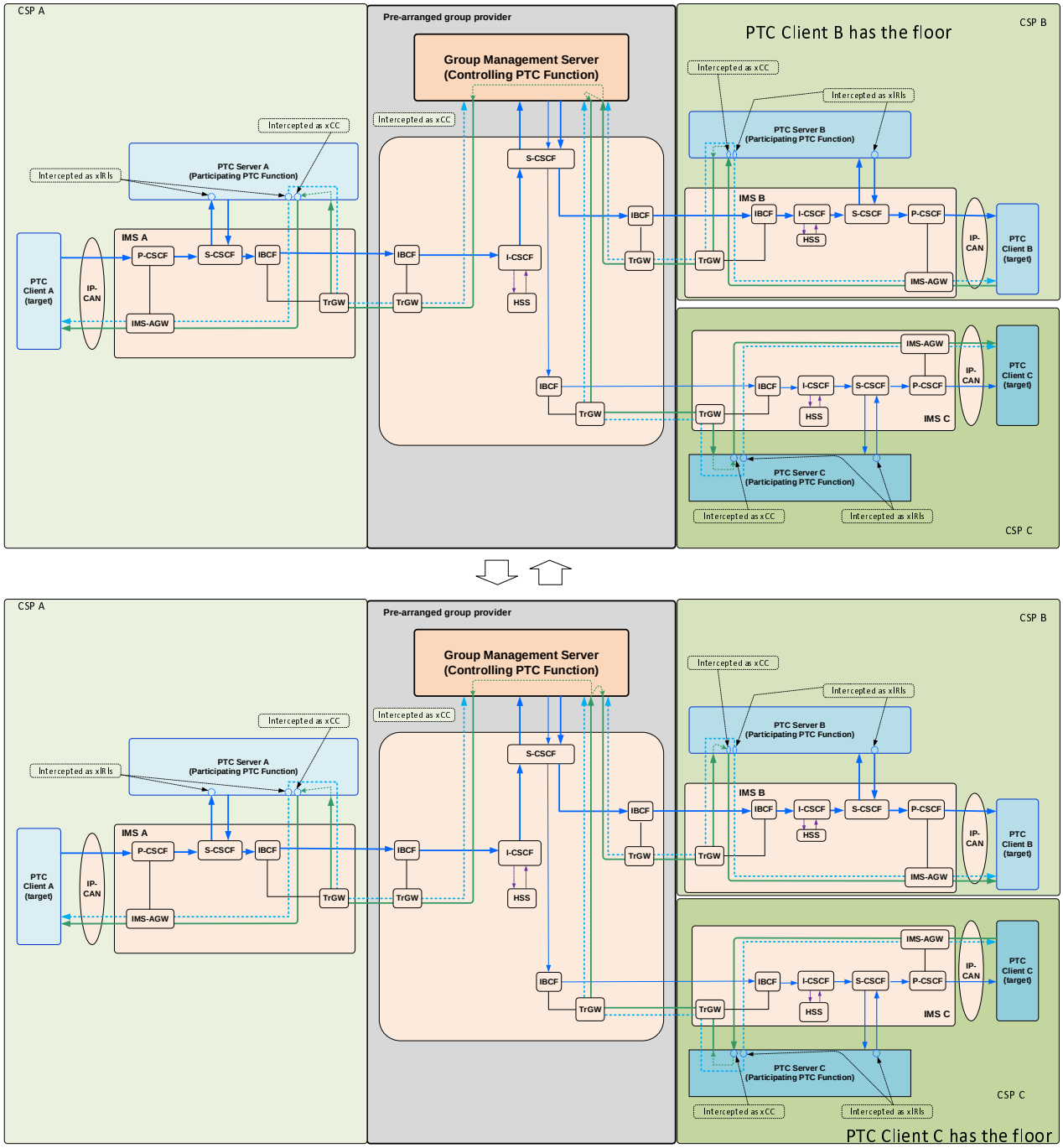


Figure 4.2.5.4.2-2: Pre-arranged group PTC session, case 1 – floor control options

As shown in figure 4.2.5.4.2-2, either PTC Client B or PTC Client C can also have the floor at any given time. The difference in the LI aspects is mainly the direction of media that is being intercepted.

4.2.5.4.3 Group management server is in same service provider domain of PTC session initiator

Figure 4.2.5.4.3-1 illustrates a case of PTC Client A who initiated the pre-arranged group PTC session is a target. In the illustration, the LI is applicable only for PTC. The Group Management Server is in the same service provider domain as that of PTC session initiator (PTC Client A).

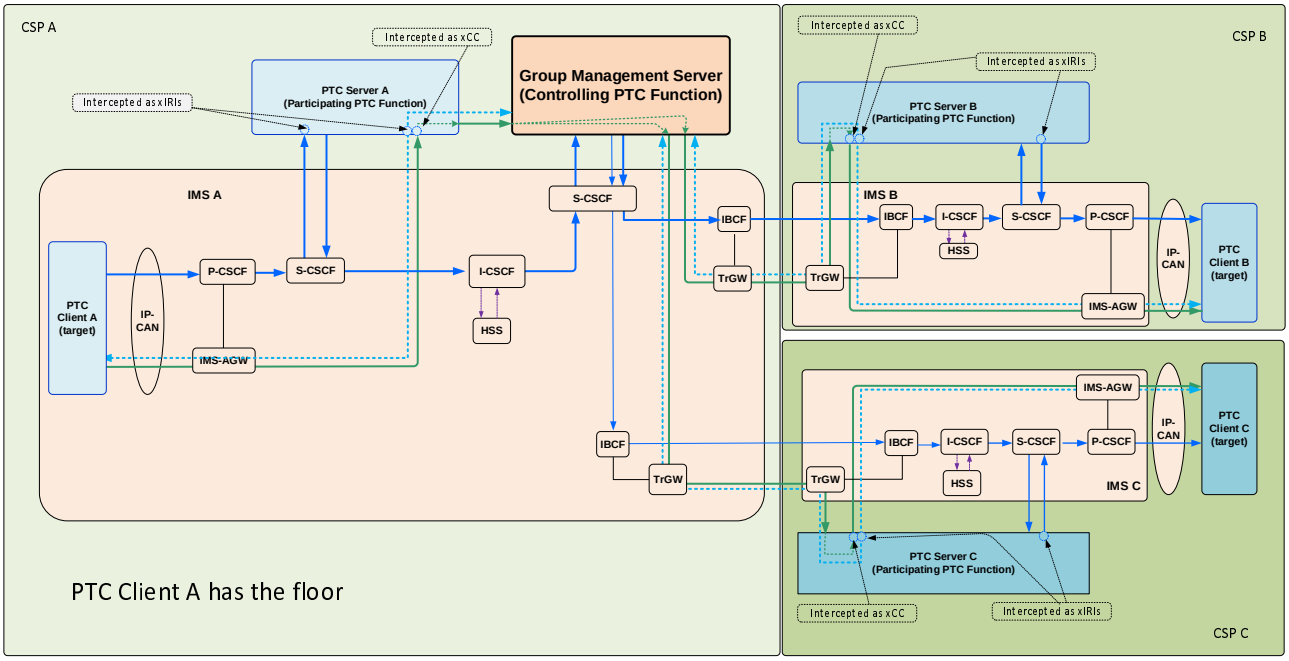


Figure 4.2.5.4.3-1: Pre-arranged group PTC session – case 2

When PTC Client A is a target in CSP A domain, the PTC Server A that provides the participating PTC function for PTC Client A provides the LI functions. When PTC Client B is a target in CSP B domain, the PTC Server B that provides the participating PTC function for PTC Client B provides the LI functions. When PTC Client C is a target in CSP C domain, the PTC Server C that provides the participating PTC function for PTC Client C provides the LI functions.

For all three targets, the scope of LI functions provided by respective participating PTC Servers are limited to the information available in those servers.

Figure 4.2.5.4.3-2 illustrates the topologies with different PTC clients having the floor.

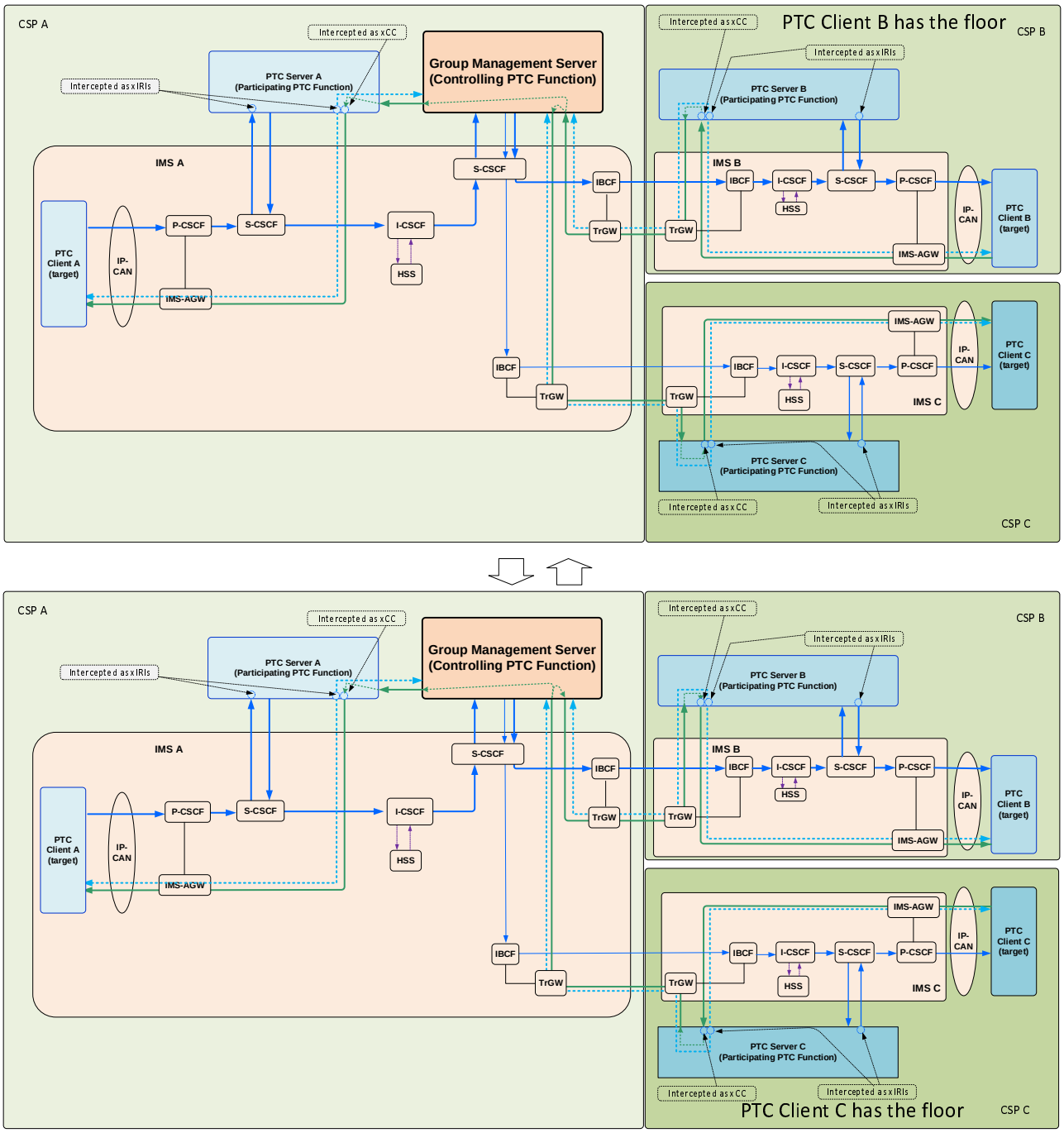


Figure 4.2.5.4.3-2: Pre-arranged group PTC session, case 2 – floor control options

As shown in figure 4.2.5.4.3-2, either PTC Client B or PTC Client C can also have the floor at any given time. The difference in the LI aspects is mainly the direction of media that is being intercepted.

As an implementation choice, the LI functions may also be provided by that Group Management Server when PTC Client A is a target since that server is providing the controlling PTC function. See figure 4.2.5.4.3-3 below.

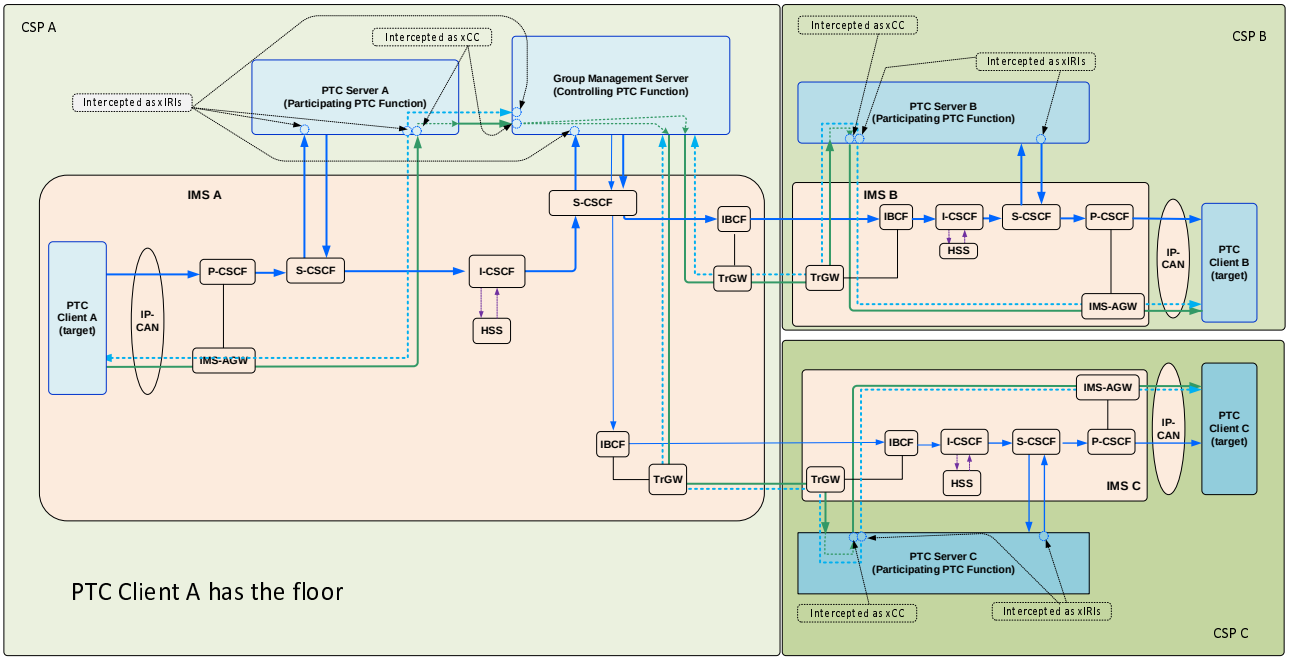


Figure 4.2.5.4.3-3: Pre-arranged group PTC session – case 2 (another implementation choice)

In the illustration shown in figure 4.3.5.4.3-3, as an implementation choice, the Group Management Server may provide the LI functions when the PTC Client A is a target. However, such an implementation choice, if made, results in a different system behaviour as compared to the system behaviour when the Group Management Server is in a different provider domain (figure 4.2.5.4.2-1). In addition, such an implementation choice, if made, may also result double interception (one at participating PTC function and controlling function).

NOTE: TS 33.127 [3] and TS 33.128 [4] do not support such an implementation choice.

Figure 4.2.5.4.3-4 illustrates the topologies with different PTC clients having the floor.

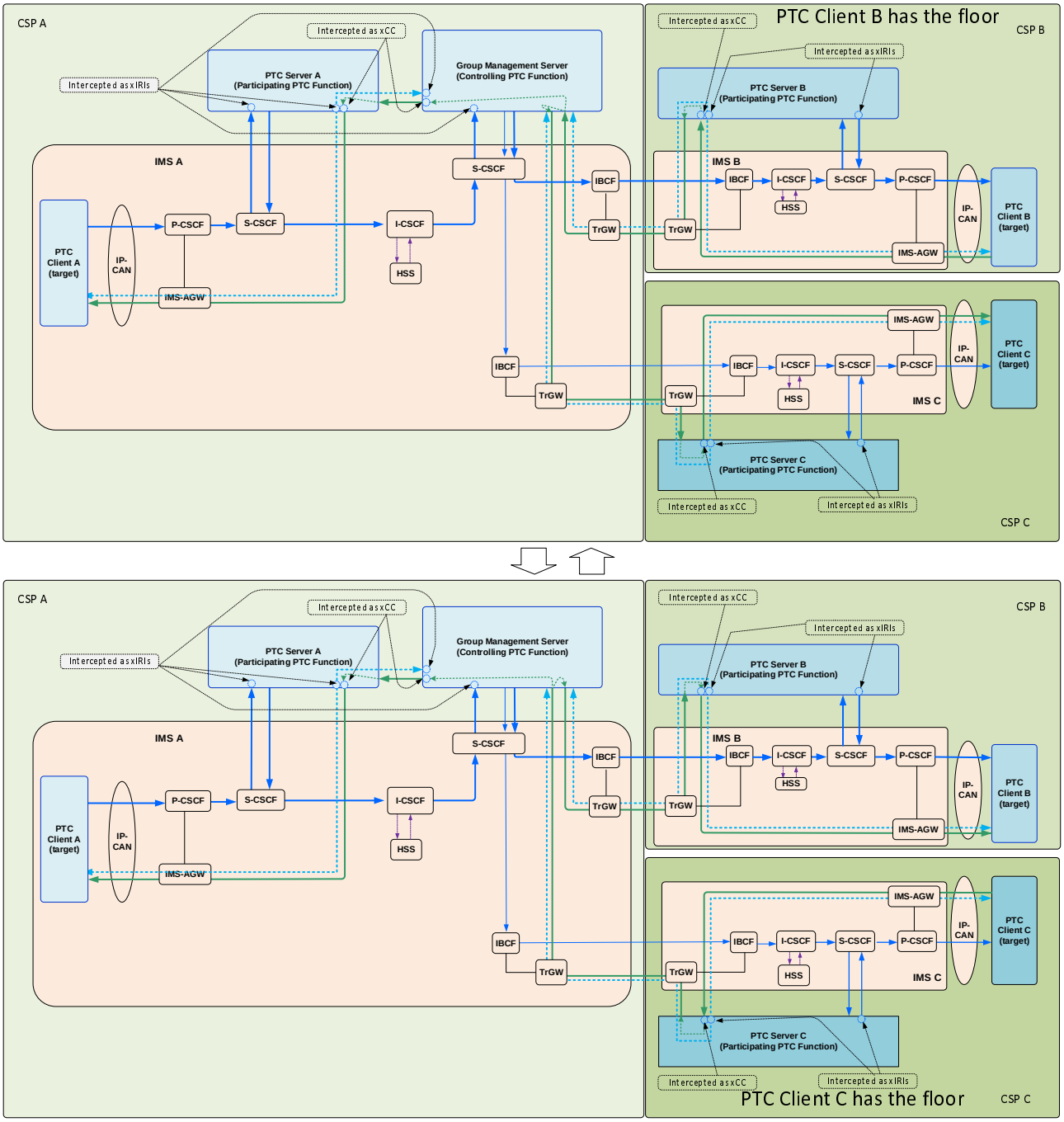


Figure 4.2.5.4.3-4: Pre-arranged group PTC session, case 2 – floor control options (another implementation choice)

As shown in figure 4.2.5.4.3-4, either PTC Client B or PTC Client C can also have the floor at any given time. The difference in the LI aspects is mainly the direction of media that is being intercepted.

4.2.5.5 Chat-group PTC session

4.2.5.5.1 General

In a chat-group PTC session, the PTC clients join the chat-group individually. In the illustrations shown in this clause, a separate PTC Server handles the chat-group PTC session (referred to as Chatgroup Server in the present document) and hence, provides the controlling PTC function.

The topology view would be similar to the case of a pre-arranged PTC session except that in a pre-arranged group PTC session, one of the PTC clients is initiating the PTC session. The direction of the SIP flow would be from the session-

initiating client in a pre-arranged group PTC session whereas the direction of SIP flow would be from each PTC client to the Chatgroup Server in a chat-group PTC session.

In the illustration, three PTC clients PTC Client A, PTC Client B and PTC Client C are in a chat-group PTC session.

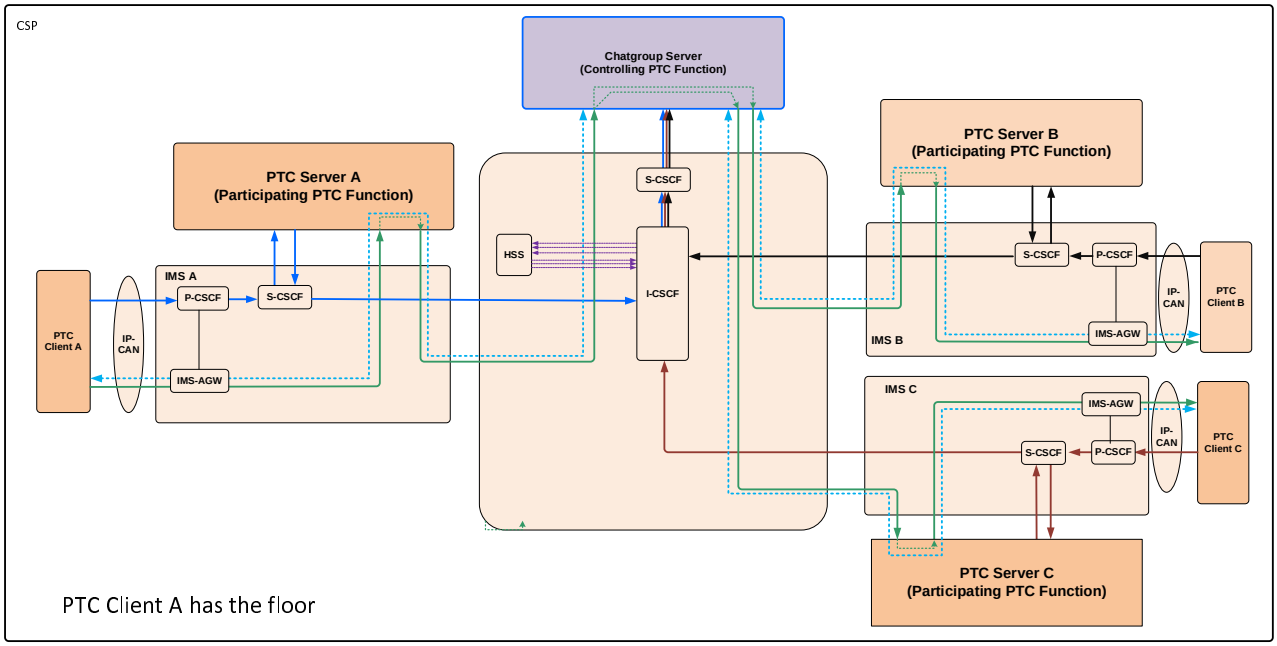


Figure 4.2.5.5.1-1: A chat-group PTC session – a topology view

In the illustration shown in figure 4.2.5.5.1-1, the PTC clients and the Chatgroup Server are in the same CSP domain. The PTC clients and the Chatgroup Server may be in different domains as shown in 4.2.5.5.1-2 below.

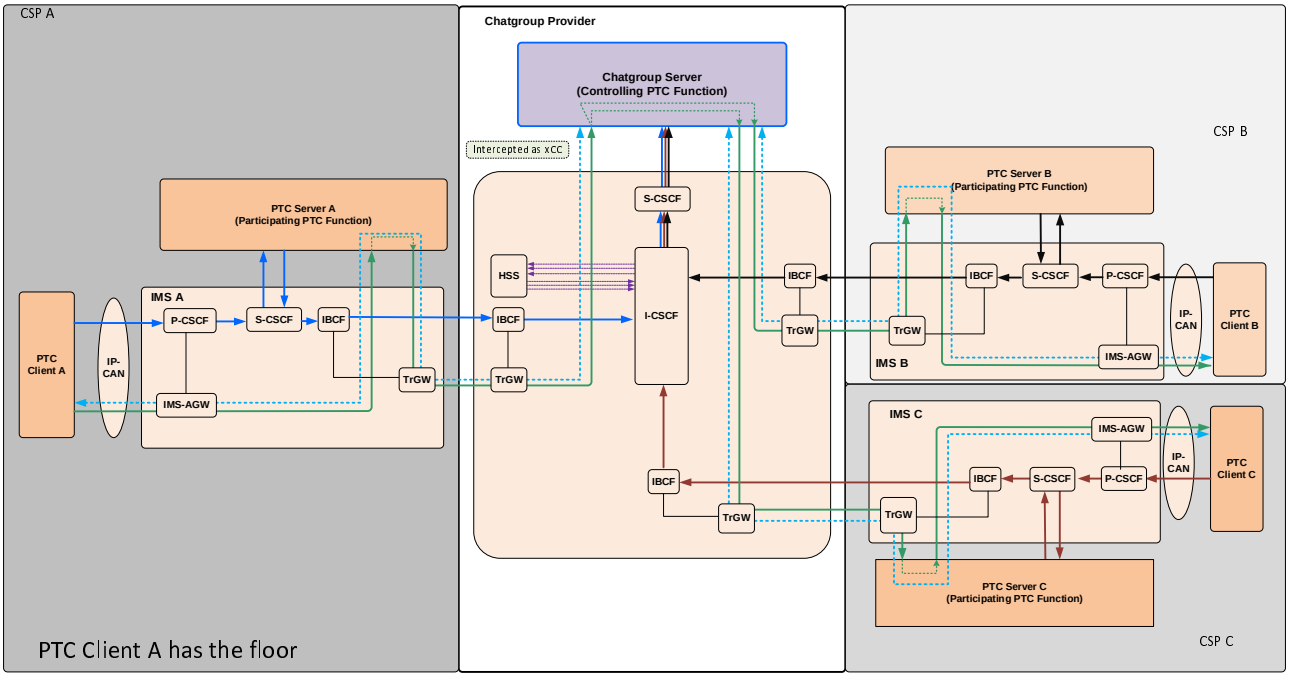


Figure 4.2.5.5.1-2: A chat-group PTC session – PTC clients and Chatgroup Server in different domains

It is also possible to have other mixed combinations:

- Either one or two of the PTC clients are in the same domain as that of Chatgroup Server.

- Either one or two or all the three PTC clients in one CSP domain with a different provider for the Chatgroup Server.
- Either one or two of the PTC clients in one CSP domain with two or one remaining PTC client in a different CSP domain with separate provider for the Chatgroup Server.

In the LI illustrations shown in the subsequent clauses, two cases expanded from figures 4.2.5.5.1-1 and 4.2.5.5.1-2 are considered.

4.2.5.5.2 One CSP domain scenarios

4.2.5.5.2.1 General

In the illustrations all PTC clients and the Chatgroup Server are in the same service provider domain.

4.2.5.5.2.2 PTC chat-group ID is the target

Figure 4.2.5.5.2.2-1 illustrates a case of PTC Chat-group ID is the target. In the illustration, the LI is applicable only for PTC.

The Chatgroup Server that provides the controlling PTC function for the chat-group PTC session provides the LI functions.

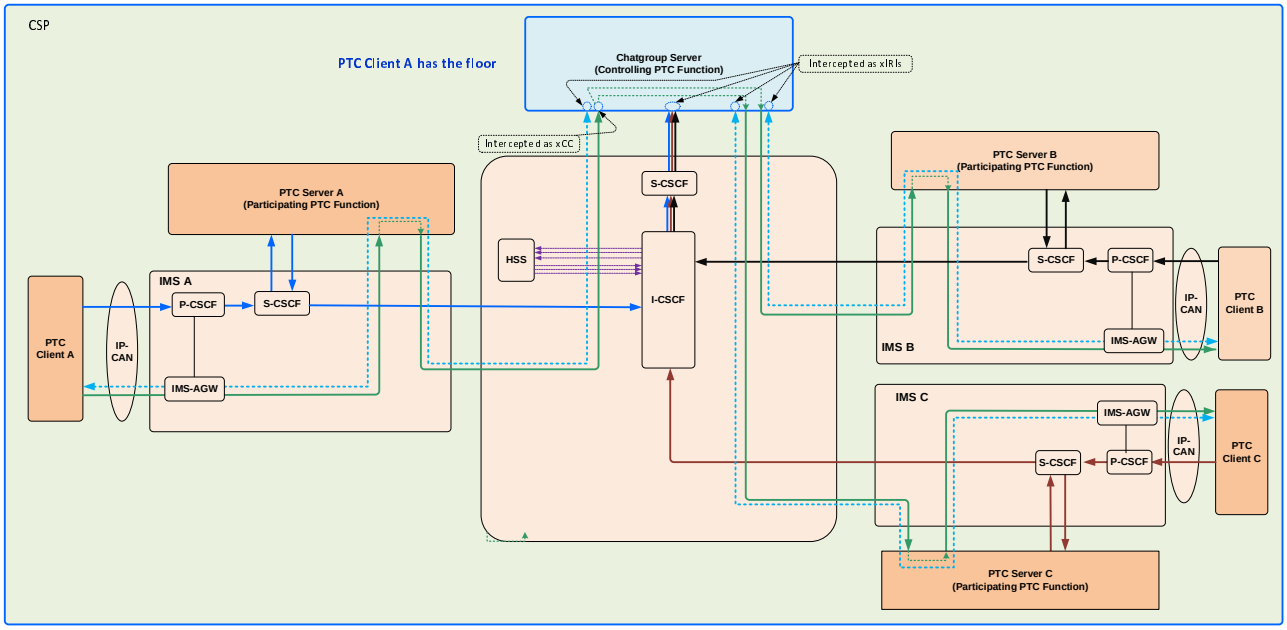


Figure 4.2.5.5.2.2-1: Chat-group PTC session with one CSP - PTC chat-group ID is a target

When PTC Chat-group ID is the target, the Chatgroup Server that handles the chat-group PTC session provides the controlling PTC function for the PTC session and therefore, provides the LI functions. As shown in figure 4.2.5.5.2.2-1, the interception happens at the ingress end of signalling and media flow. Signalling and media sent to, or received from, all the PTC clients in the chat-group PTC session are intercepted.

Figure 4.2.5.5.2.2-2 illustrates the topology with different PTC clients having the floor.

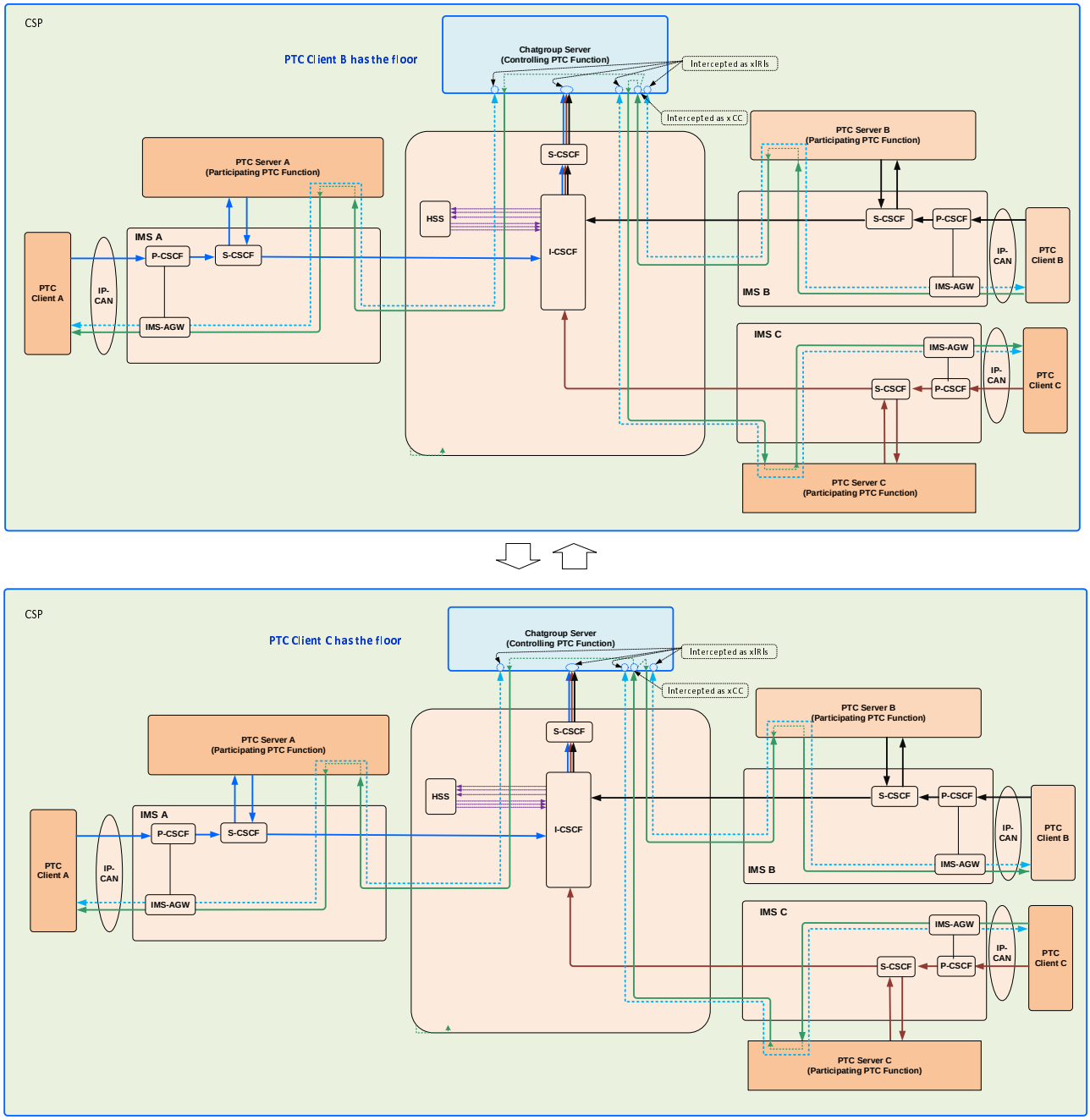


Figure 4.2.5.5.2.2-2: Chat-group PTC session with one CSP - PTC chat-group ID is a target – floor control options

As shown in figure 4.2.5.5.2.2-2, any one of the three PTC clients can have the floor at a given time. In all cases, the interception happens at the ingress end of signalling and media flow with respect to the Chatgroup Server. The difference in the LI aspects is mainly the direction of media that is being intercepted.

4.2.5.5.2.3 PTC clients are the targets

Figure 4.2.5.5.2.3-1 illustrates a case where all three PTC clients involved in a chat group PTC session are the targets. In the illustration, the LI is applicable only for PTC.

The PTC Server that provides the respective participating PTC function for PTC clients as targets provide the LI functions.

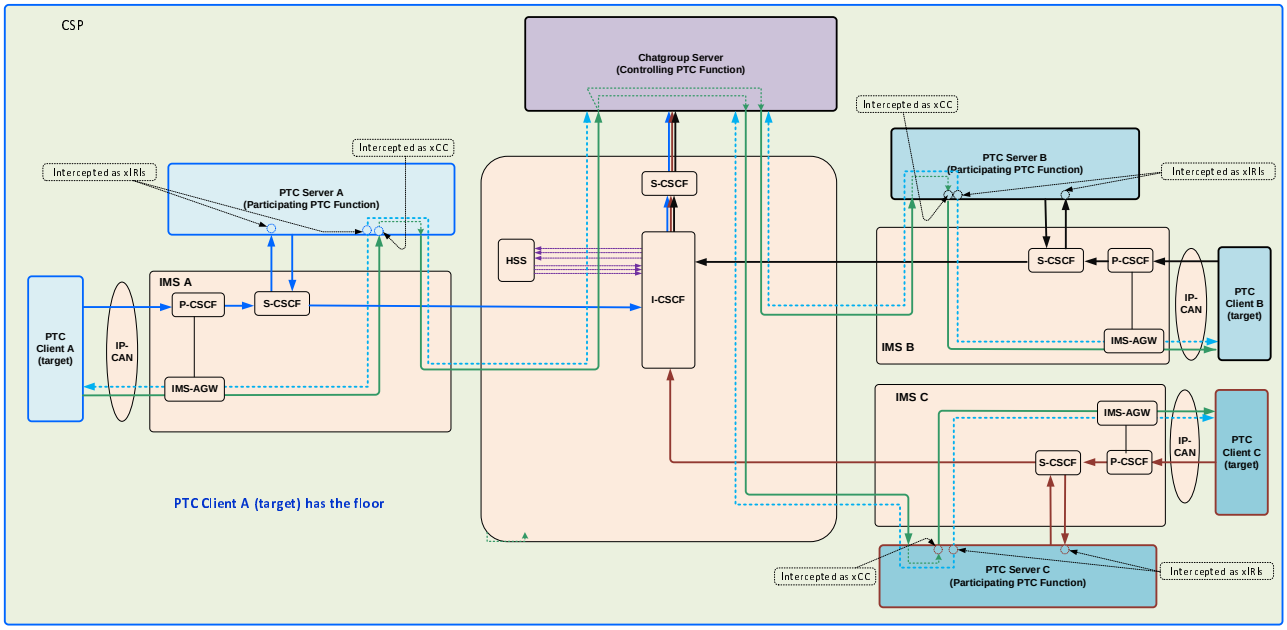


Figure 4.2.5.5.2.3-1: Chat-group PTC session within one CSP – all PTC clients are the targets

For PTC Client A as the target, the PTC Server A that provides the participating PTC function for PTC Client A provides the LI functions. Likewise, for PTC Client B as the target, the PTC Server B that provides the participating PTC function for PTC client B provides the LI functions. Similarly, for PTC Client C as the target, the PTC Server C that provides the participating PTC function for PTC client C provides the LI functions.

As shown in figure 4.2.5.5.2.3-1, the interception happens at the ingress side of signalling and media flow with respect to the PTC Server A and at the egress side of signalling and media flow with respect to the PTC Servers B and C.

NOTE: The option of LI at the Chatgroup Server providing the controlling PTC function for the PTC chat-group PTC session when one of the PTC Clients is a target is outside the scope the present document. TS 33.127 [3] and TS 33.128 [4] do not describe the case.

Figure 4.2.5.5.2.3-2 illustrates the topology with different PTC clients having the floor.

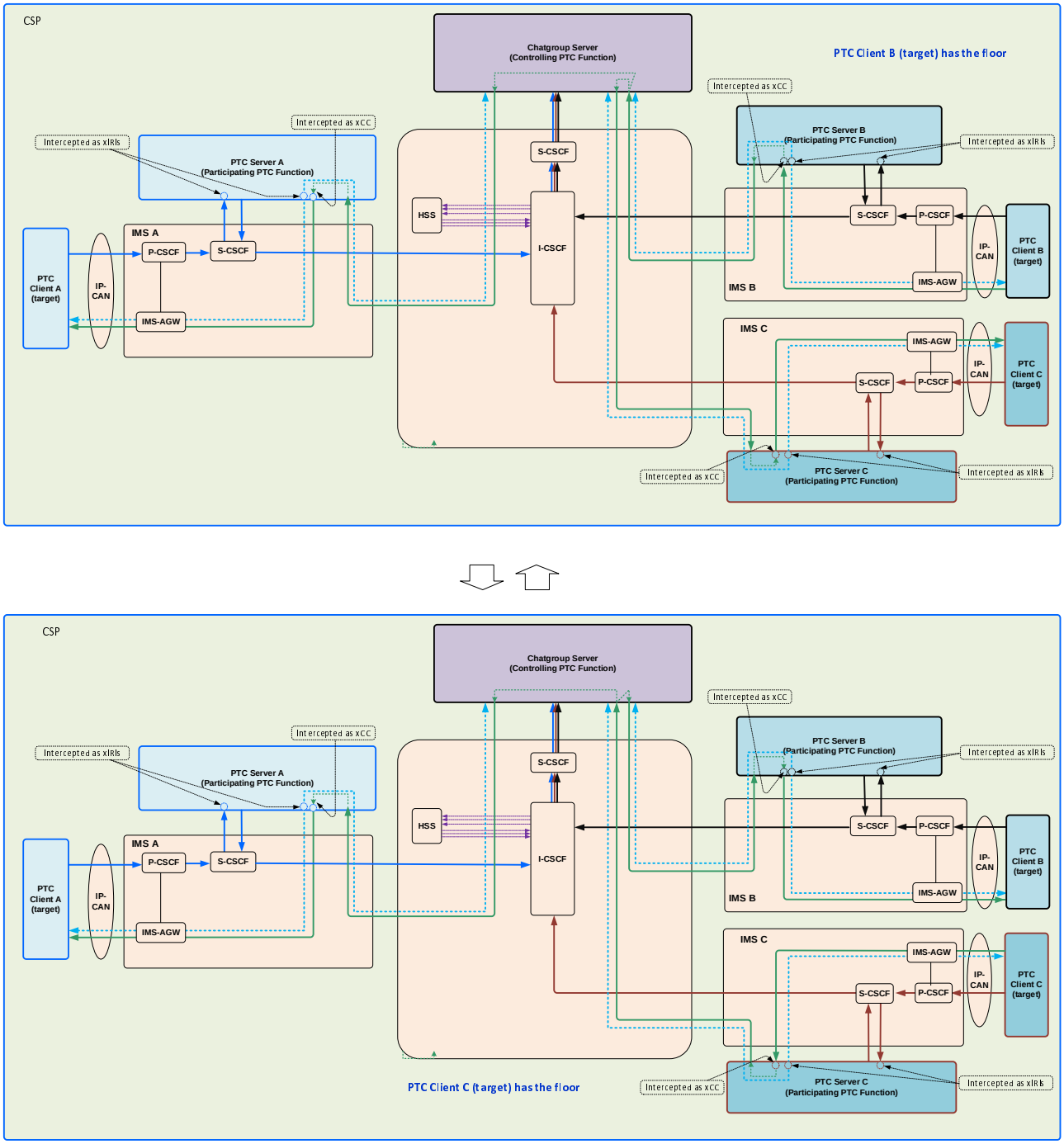


Figure 4.2.5.5.2.3-2: Chat-group PTC session within one CSP – all PTC clients are the targets – floor control options

As shown in figure 4.2.5.5.2.3-2, any one of the three PTC clients in the chat-group PTC session can have the floor at any given time. The difference in the LI aspects is mainly the direction of media that is being intercepted.

As an implementation choice, the LI functions may also be provided by the Chatgroup Server when one of the PTC clients is a target and the Chatgroup Server is in the same CSP domain as that of target. In the illustration, all PTC clients happen (even though it does not have to be that way) to be in the same CSP domain. See figure 4.2.5.5.3-3 below:

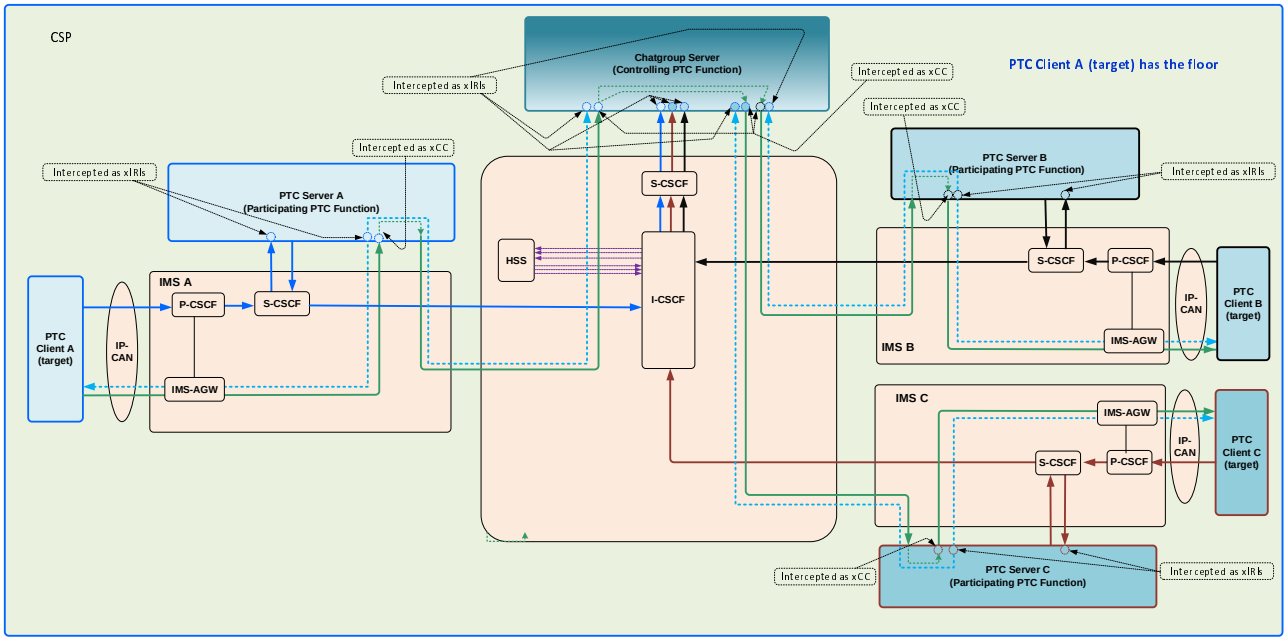


Figure 4.2.5.5.2.3-3: Chat-group PTC session (an implementation choice)

In the illustration shown in figure 4.3.5.5.2.3-3, as an implementation choice, the Chatgroup Server may provide the LI functions when the PTC clients are the targets. However, such an implementation choice, if made, results in a different system behaviour as compared to the system behaviour when the Chatgroup Server is in a different provider domain (figure 4.2.5.5.2.3-1). In addition, such an implementation choice, if made, may also result double interception (one at participating PTC function and controlling function).

NOTE: TS 33.127 [3] and TS 33.128 [4] do not support such an implementation choice.

Figure 4.2.5.5.2.3-4 illustrates the topologies with different PTC clients having the floor.

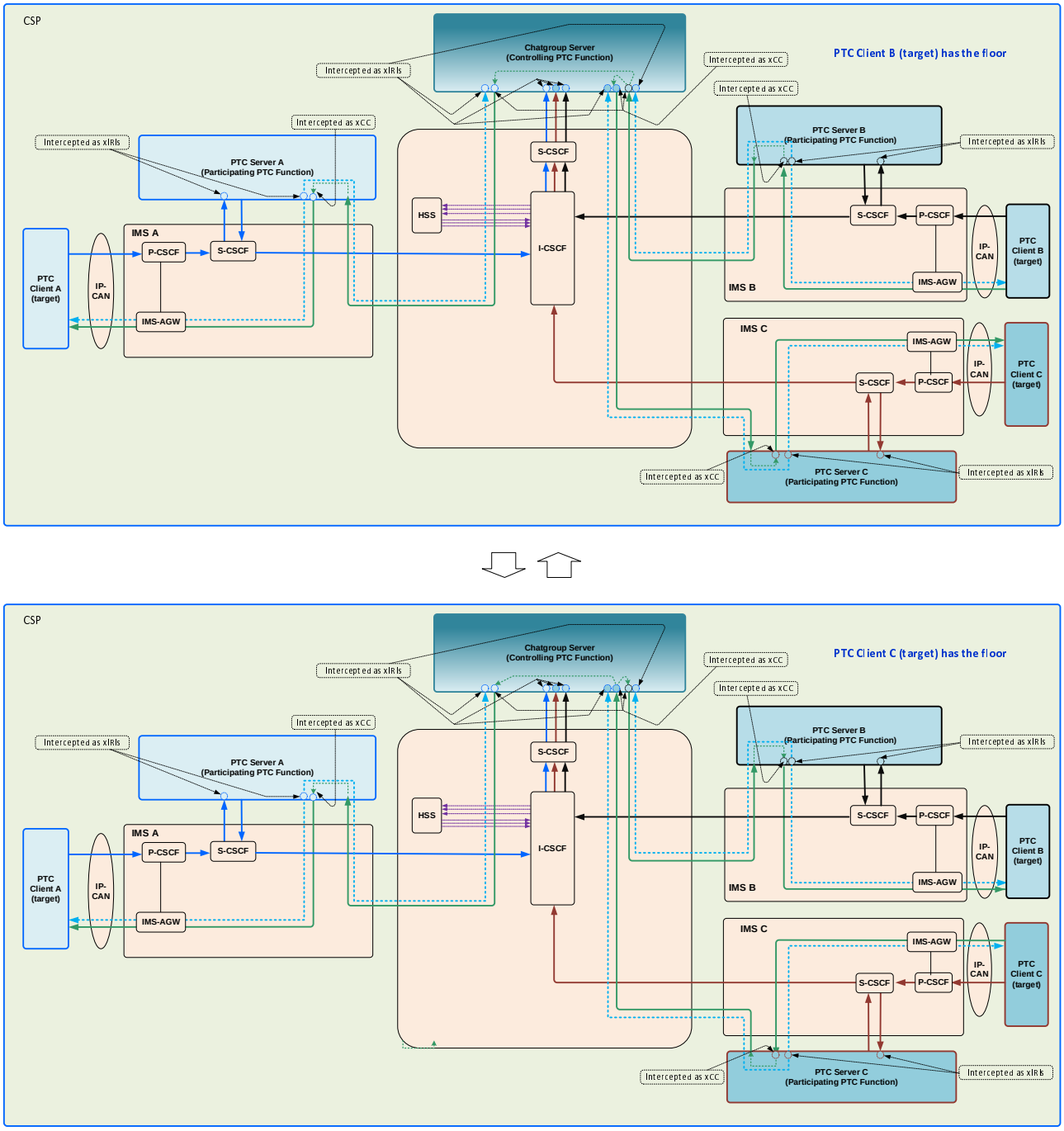


Figure 4.2.5.5.2.3-4: Chat-group PTC session – floor control options (an implementation choice)

As shown in figure 4.2.5.5.2.3-4, either PTC Client B or PTC Client C can also have the floor at any given time. The difference in the LI aspects is mainly the direction of media that is being intercepted.

4.2.5.5.3 Different CSP domain scenarios

4.2.5.5.3.1 General

In the illustrations all PTC clients and the Chatgroup Server are in the different provider domains.

4.2.5.5.3.2 PTC chat-group ID is the target

Figure 4.2.5.5.3.2-1 illustrates a case of PTC Chat-group ID is the target. In the illustration, the LI is applicable only for PTC.

The Chatgroup Server in the chatgroup provider domain provides the controlling PTC function for the PTC chat-group PTC session and therefore, provides the LI functions when the PTC Chat-group ID is the target.

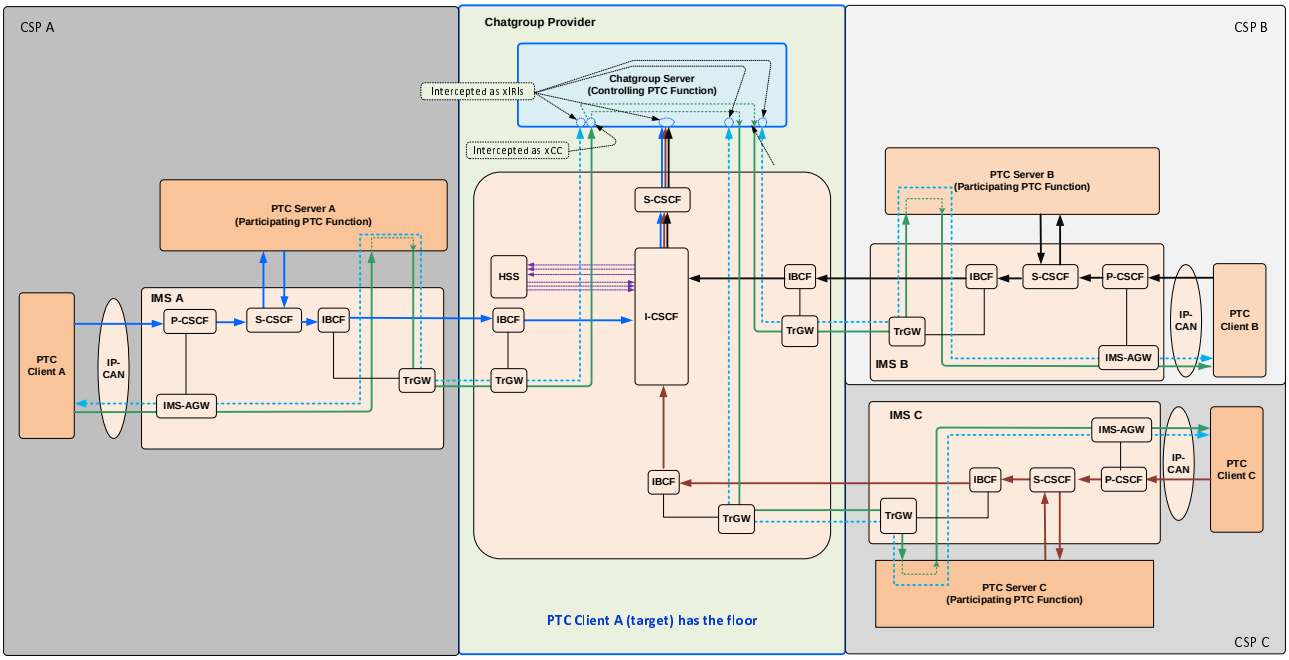


Figure 4.2.5.5.3.2-1: Chat-group PTC session with different CSPs - PTC chat-group ID is a target

When PTC Chat-group ID is the target the Chatgroup Server in the chatgroup provider domain that handles the chat-group PTC session provides the controlling PTC function for the PTC session and therefore, provides the LI functions. As shown in figure 4.2.5.5.3.2-1, the interception happens at the ingress end of signalling and media flow.

Figure 4.2.5.5.3.2-2 illustrates the topology with different PTC clients having the floor.

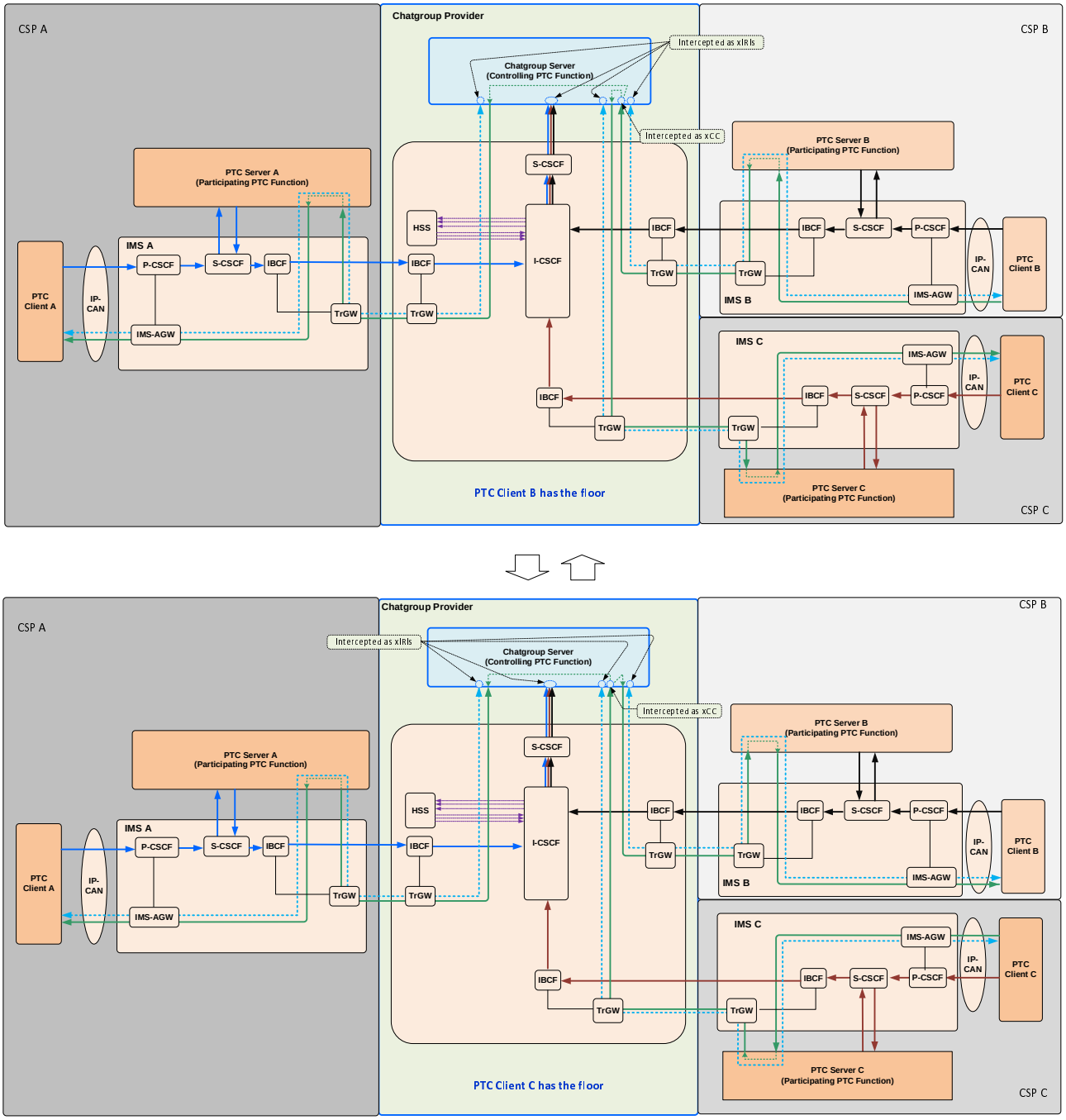


Figure 4.2.5.5.3.2-2: Chat-group PTC session with different CSPs - PTC chat-group ID is a target – floor control options

As shown in figure 4.2.5.5.3.2-2, any one of the three PTC clients have the floor at a given time. In all cases, the interception happens at the ingress end of signalling and media flow with respect to the Chatgroup Server. The difference in the LI aspects is mainly the direction of media that is being intercepted.

4.2.5.5.3.3 PTC clients are the targets

Figure 4.2.5.5.3.3-1 illustrates a case where all three PTC clients involved in a PTC chat group are the targets. In the illustration, the LI is applicable only for PTC.

The PTC Server that provides the respective participating PTC function for PTC clients as targets provide the LI functions.

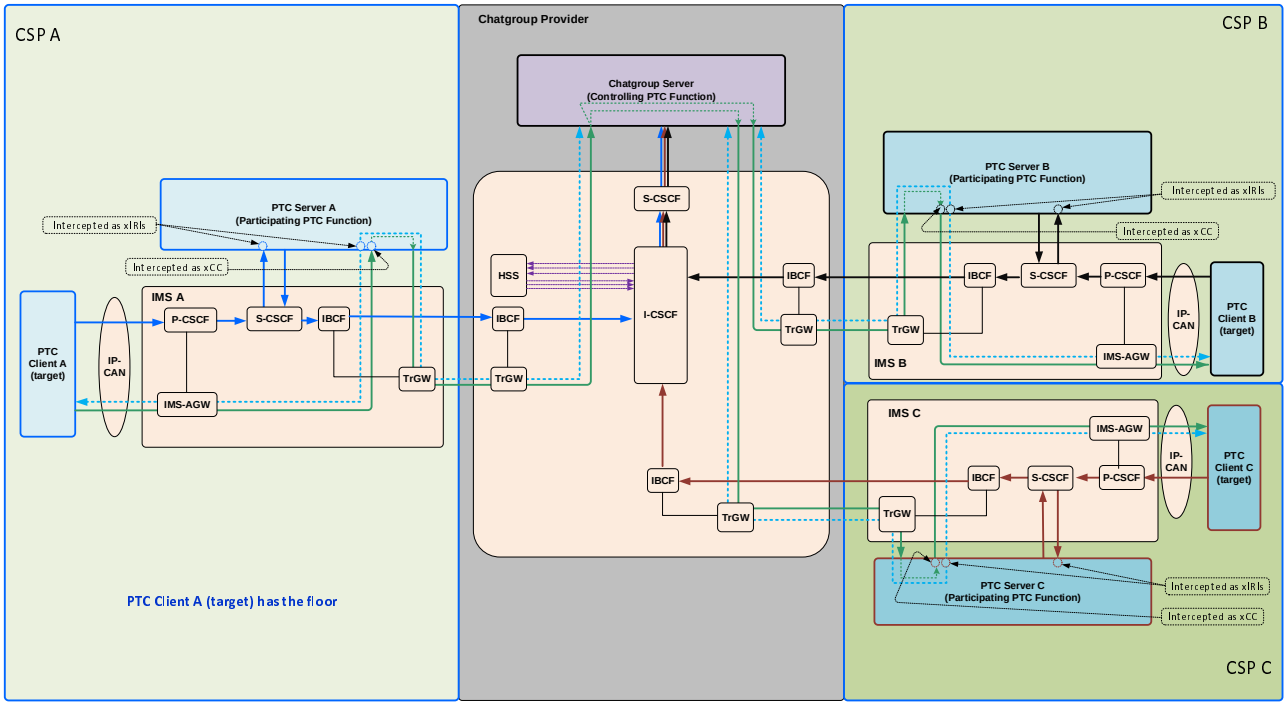


Figure 4.2.5.5.3.3-1: Chat-group PTC session with different CSPs – all PTC clients are the targets

PTC Client A is a target in CSP A domain, PTC Client B is a target in CSP B domain and PTC Client C is a target in CSP C domain. The PTC Server A that provides the participating PTC function for PTC Client A provides the LI functions for target PTC Client A in CSP A domain. Likewise, the PTC Server B that provides the participating PTC function for PTC Client B in CSP B domain provides the LI functions for target PTC Client B in CSP B domain. Similarly, the PTC Server C that provides the participating PTC function for PTC Client C in CSP C domain provides the LI functions for target PTC Client C in CSP C domain.

The scope of xIRIs generated at the POIs present in the PTC Servers that provide the participating PTC functions may be limited to the information available at that PTC servers. For example, the PTC Server that provides the participating PTC function may not have the information about the floor-control details.

As shown in figure 4.2.5.5.3.3-1, the interception happens at the ingress side of signalling and media flow with respect to the PTC Server A and at the egress side of signalling and media flow with respect to the PTC Servers B and C.

Figure 4.2.5.5.3.3-2 illustrates the topology with different PTC clients having the floor.

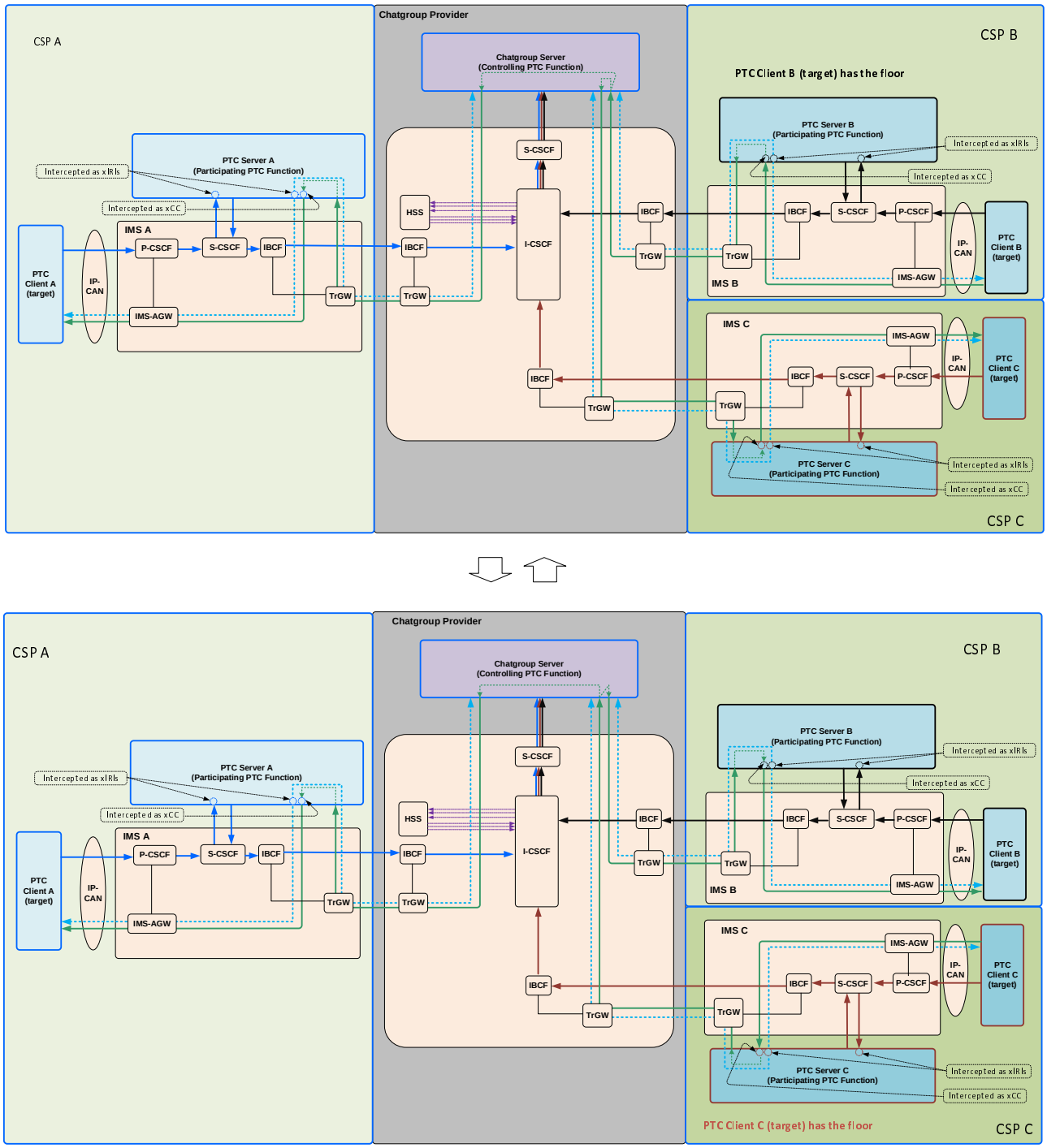


Figure 4.2.5.5.3.3-2: Chat-group PTC session with different CSPs – all PTC clients are the targets – floor control options

As shown in figure 4.2.5.5.3.3-2, any one of the three PTC clients in the chat-group PTC session can have the floor at any given time. The difference in the LI aspects is mainly the direction of media that is being intercepted.

4.2.6 PTC one-to-many-to-one session

4.2.6.1 General

The one-to-many-one PTC session is an extension of a pre-arranged group PTC session. In a PTC One-to-Many-to-One session as well, multiple PTC clients are involved as group in a PTC session. The speech from PTC client that initiates the pre-arranged group PTC session is heard by all whereas the speech from other PTC clients is heard by only by the PTC client who initiated the PTC session.

In the illustration, three PTC clients are shown. PTC Client A initiates the pre-arranged group one-to-many-one PTC session. The PTC Client B and PTC Client C are the other two PTC clients in the group.

From a topology perspective, this is similar to the illustrations shown in clause 4.2.5.4 and therefore, to illustrate the difference, only the topology view where the Group Management Server is at a different provider is shown.

4.2.6.2 PTC Client A is a target

Figure 4.2.6.2-1 illustrates a case of PTC Client A who initiated the One-to-Many-to-One PTC session to PTC Client B and PTC Client C is a target. In the illustration, the LI is applicable only for PTC.

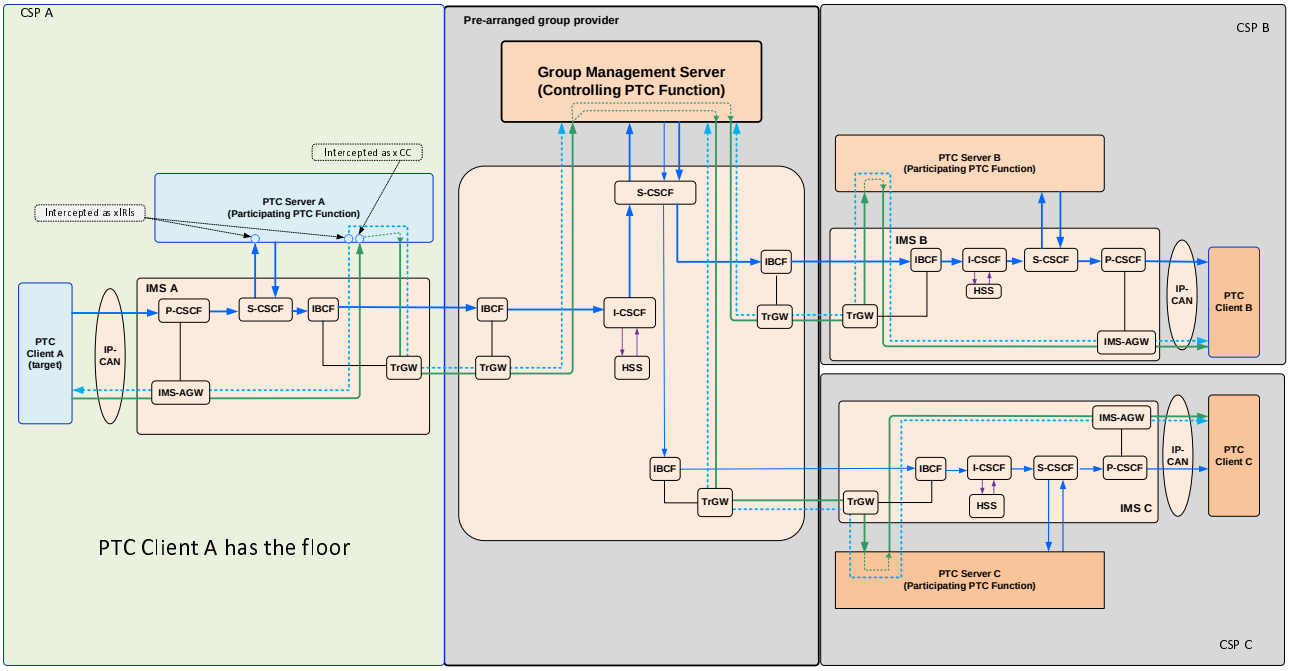


Figure 4.2.6.2-1: One-to-Many-One PTC session, PTC Client A is a target

When PTC Client A is a target, the PTC Server A that provides the participating PTC function for PTC Client A provides the LI functions. As shown in figure 4.2.6.2-1, the interception happens at the ingress side of signalling and media flow in the PTC Server A.

Figure 4.2.6.2-2 illustrates the topology with different PTC clients having the floor.

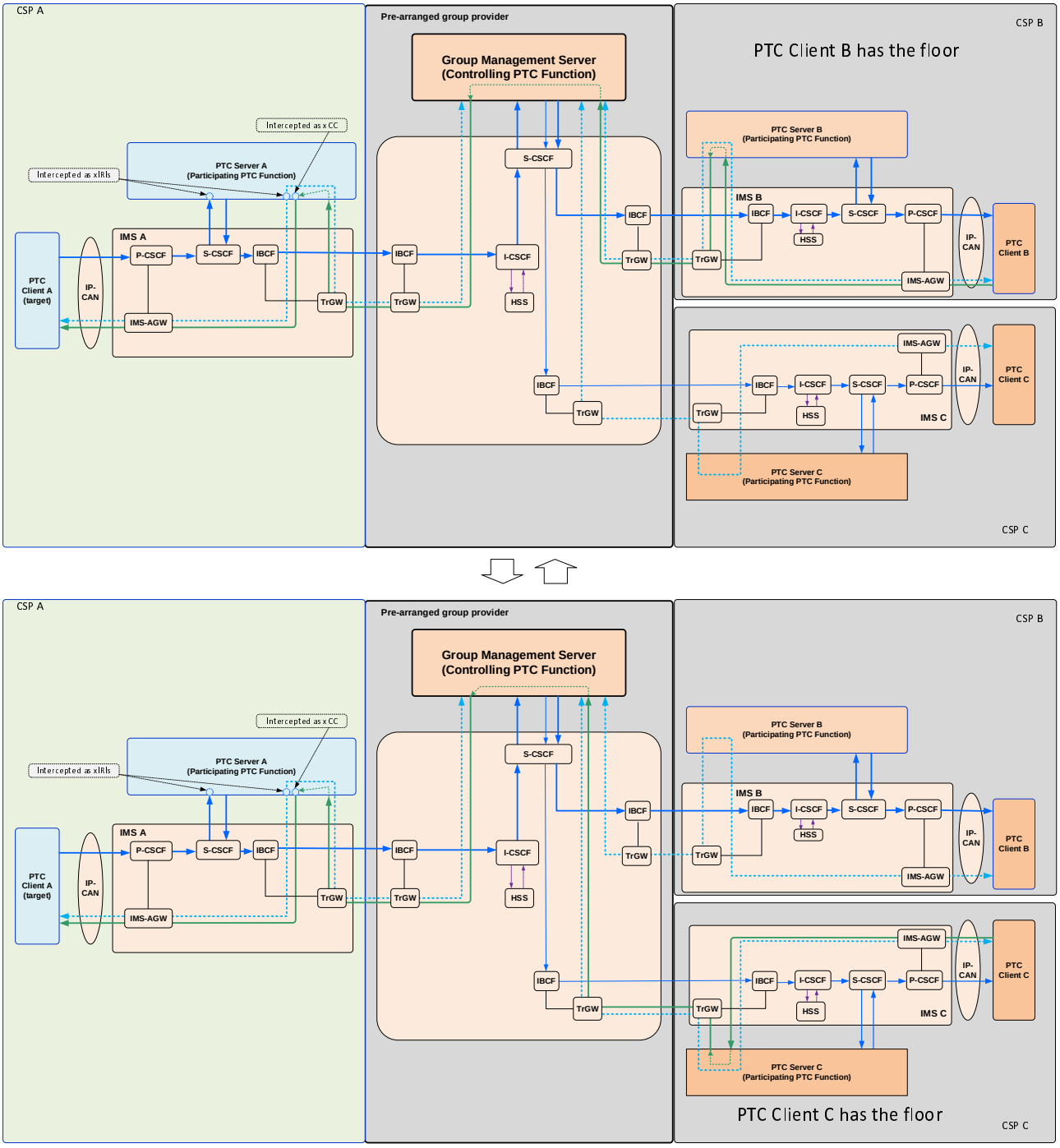


Figure 4.2.6.2-2: One-to-Many-One PTC session, PTC Client A is a target – other floor control options

As shown in figure 4.2.6.2-2, either the PTC Client B or PTC Client C can also have the floor at any given time. However, when PTC Client B or PTC Client C have the floor, the PTC Client A (target) only hears their speech.

4.2.6.3 PTC Client B is a target

Figure 4.2.6.3-1 illustrates a case of PTC Client B one of the destination PTC client of One-to-Many-One PTC session is the target. In the illustration, the LI is applicable only for PTC.

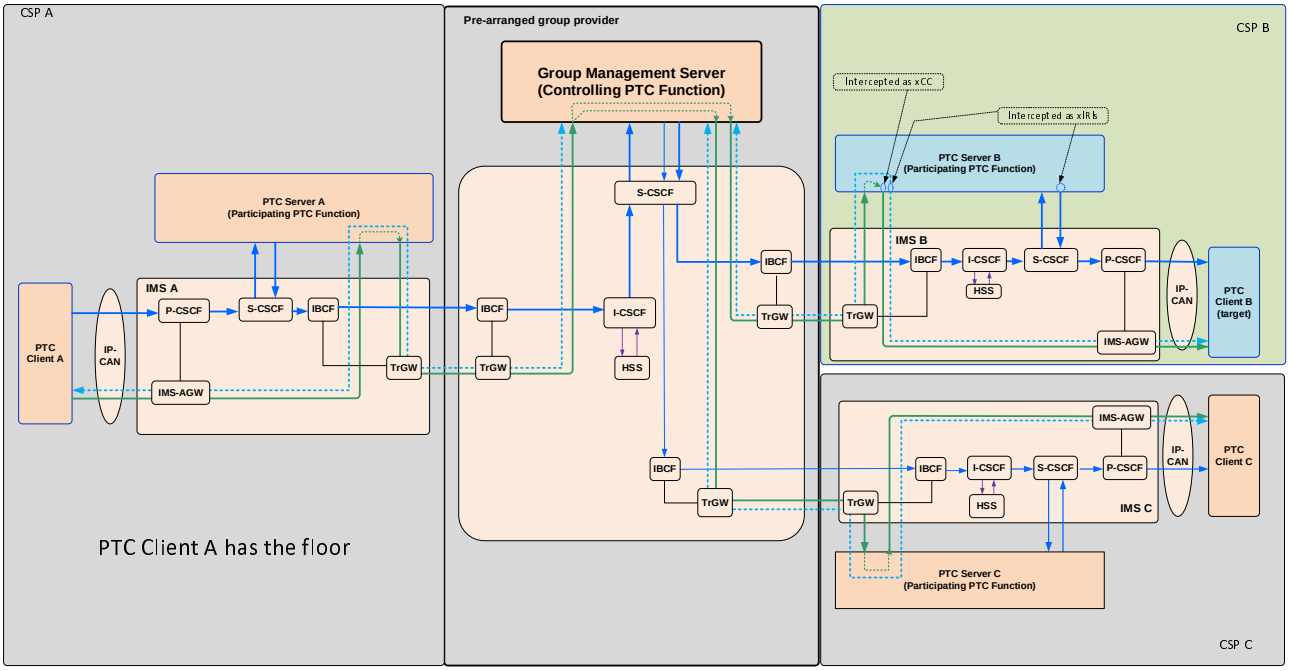


Figure 4.2.6.3-1: One-to-Many-One PTC session, PTC Client B is a target

When PTC Client B is a target, the PTC Server B that provides the participating PTC function for PTC Client B provides the LI functions. The scope of LI functions provided by PTC Server B is limited to the information available in PTC Server B. As shown in figure 4.2.6.3-1, the interception happens at the egress side of signalling and media flow in the PTC Server A.

Figure 4.2.6.3-2 illustrates the topology with different PTC clients having the floor.

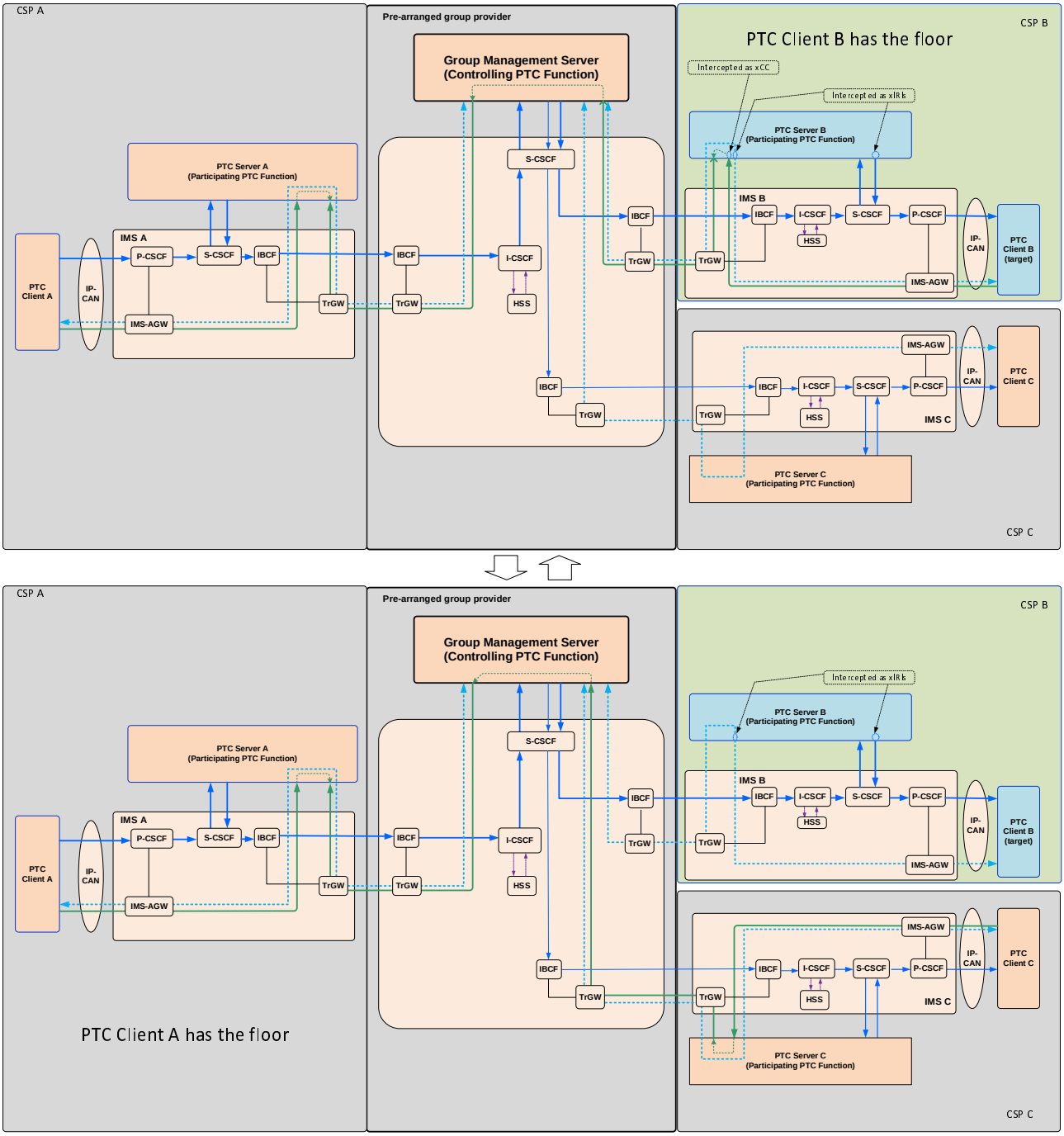


Figure 4.2.6.3-2: One-to-Many-One PTC session, PTC Client B is a target – floor control options

As shown in figure 4.2.6.3-2, either the PTC Client B (target) or PTC Client C can also have the floor at any given time. However, when PTC Client B or PTC Client C have the floor, the PTC Client A. only hears their speech Therefore, when PTC Client C has the floor, the media from PTC Client C to PTC Client A cannot be intercepted in the PTC Server B.

4.2.6.4 PTC Client C is a target

Figure 4.2.6.4-1 illustrates a case of PTC Client C one of the destination PTC Client of One-to-Many-to-One PTC session is the target. In the illustration, the LI is applicable only for PTC.

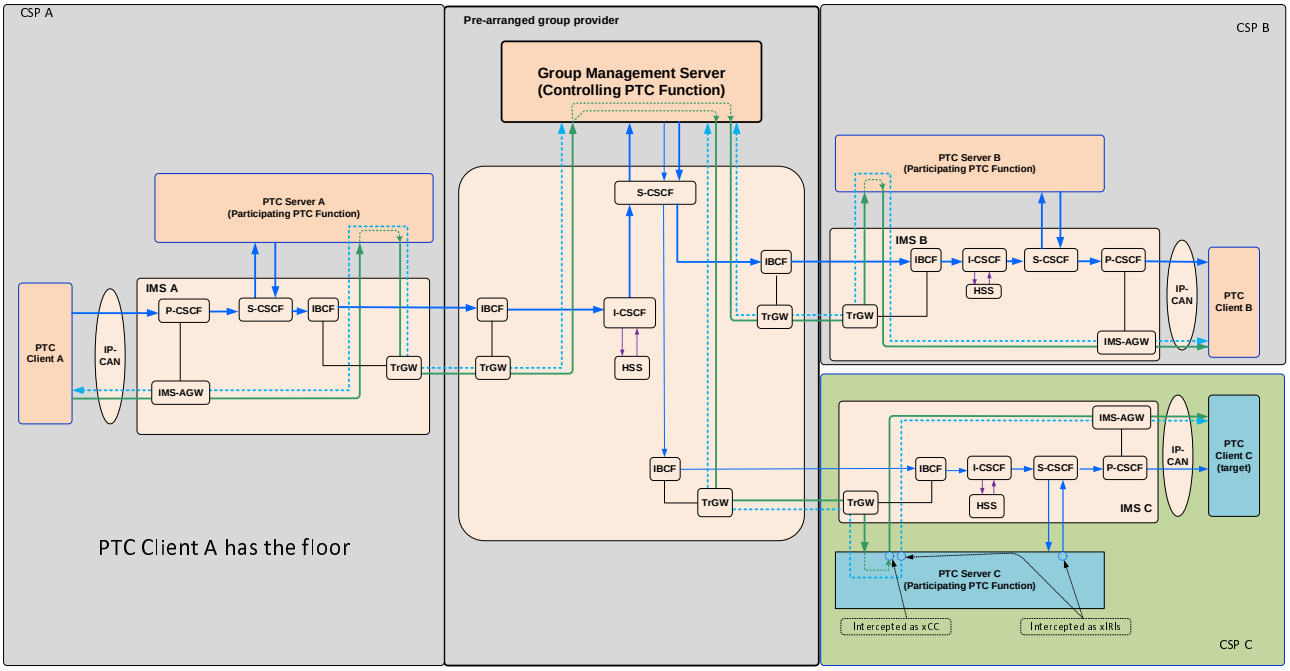


Figure 4.2.6.4-1: One-to-Many-One PTC session, PTC Client C is a target

When PTC Client C is a target, the PTC Server C that provides the participating PTC function for PTC Client C provides the LI functions. The scope of LI functions provided by PTC Server C is limited to the information available in PTC Server C. As shown in figure 4.2.6.4-1, the interception happens at the ingress side of signalling and media flow in the PTC Server A.

Figure 4.2.6.4-2 illustrates the topology with different PTC clients having the floor.

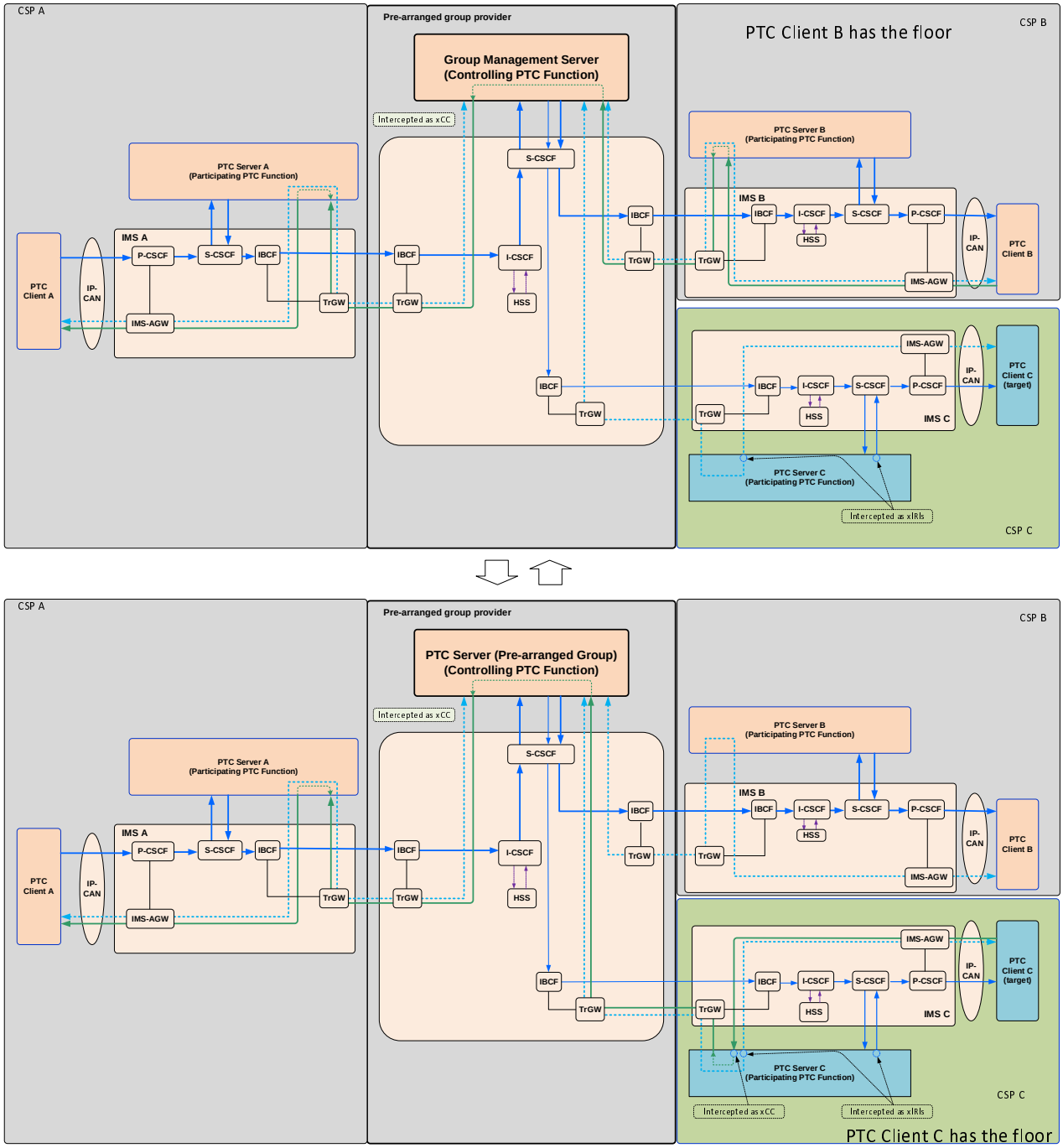


Figure 4.2.6.4-2: One-to-Many-One PTC session, PTC Client C is a target.

As shown in figure 4.2.6.4-2, either the PTC Client B or PTC Client C (target) can also have the floor at any given time. However, when PTC Client B or PTC Client C have the floor, the PTC Client A. only hears their speech Therefore, when PTC Client B has the floor, the media from PTC Client B to PTC Client A cannot be intercepted in the PTC Server C.

5 Example call flows of LI for PTC

5.1 General

The following sub-clauses give a series of flow-diagrams to illustrate the LI for PTC. In doing so, only the PTC specific NFs are included in the flow. In other words, the IMS NFs through which the PTC signalling and media flow through are not shown to have a better view of the flows. For example, when PTC Client A and PTC Client B are in the different CSP domains, as per the topology views shown in clause 4.2, the following NFs are present in the PTC signalling and media flow:

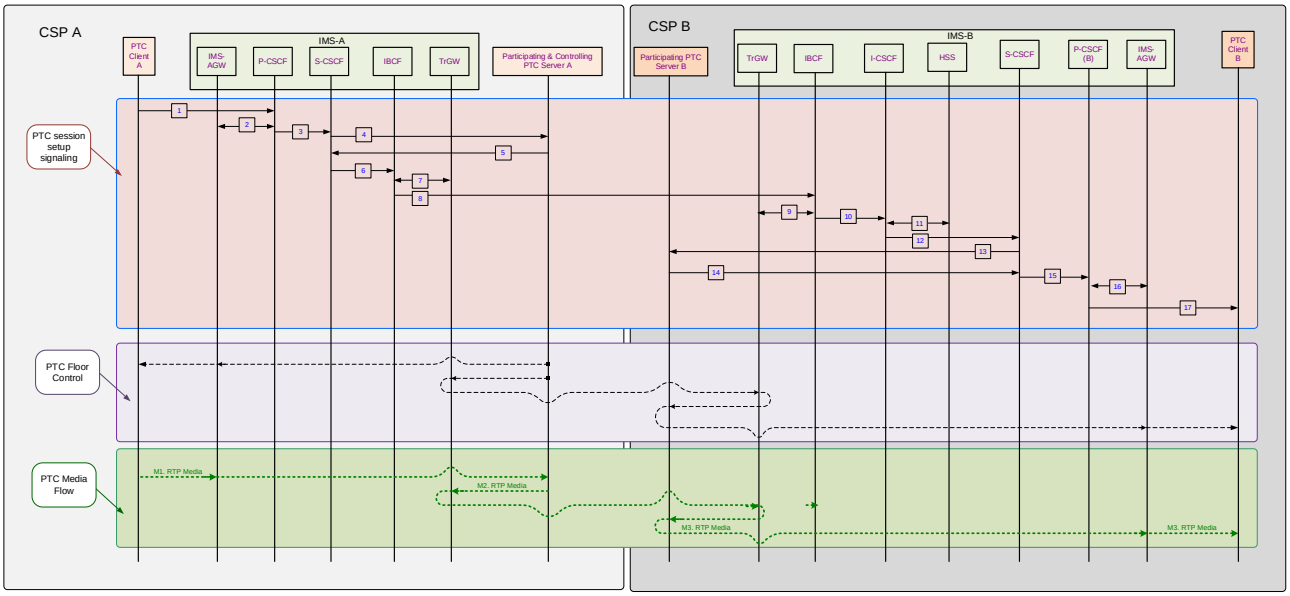


Figure 5.1-1: NFs involved in a PTC signalling and media flow – two CSP domains

When PTC Client A and B are in the same CSP domain, as per the topology views shown in clause 4.2, the following NFs are present in the PTC signalling and media flow.

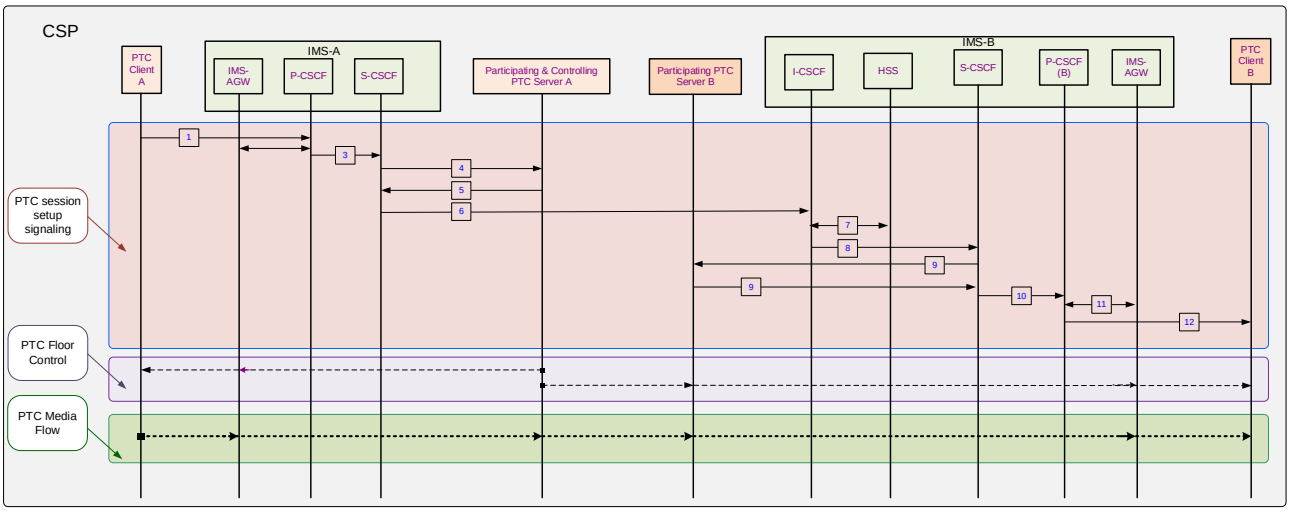


Figure 5.1-2: NFs involved in a PTC signalling and media flow – two CSP domains

The PTC session setup signalling includes SIP messages, H.248 interactions and Diameter interactions. The interworking signalling (SIP), media (RTP) and floor-control (RTCP) flow through the IBCF and TrGW. Since no PTC specific functions are performed in those NFs, from a PTC flow perspective, the presence of those NFs is insignificant. The illustrations shown in the subsequent clauses, therefore, remove those NFs from the flow-diagrams and instead have only the following:

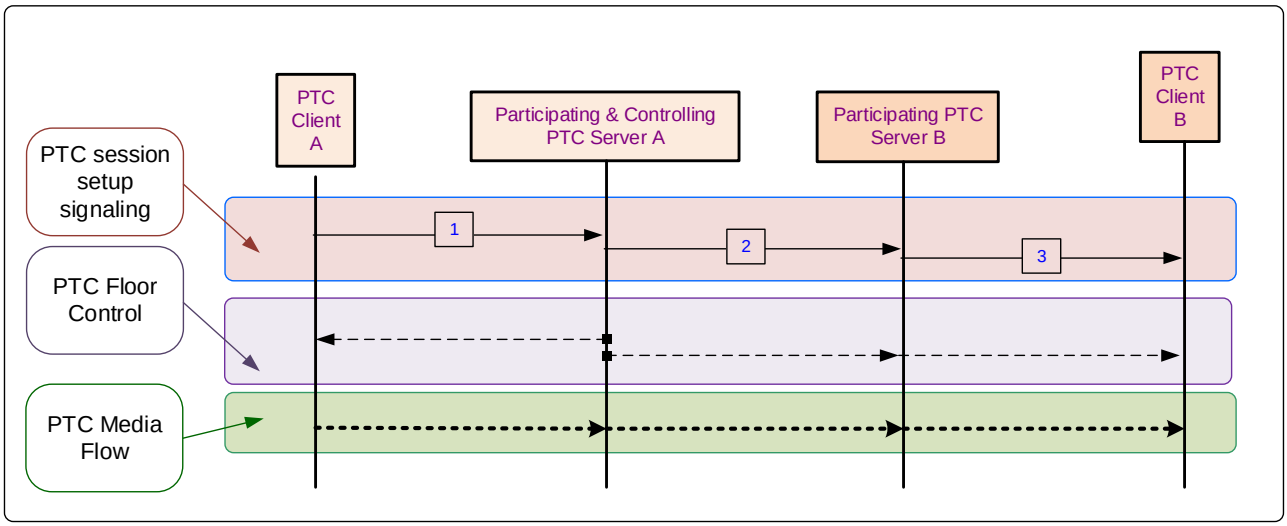


Figure 5.1-3: NFs included in the flow diagrams

The above indicated approach applies to both single CSP case as well as multiple CSP cases unless target non-local ID related illustrations are needed.

The PTC session may involve two PTC clients or multiple PTC clients. The illustrations show the presence of only two PTC clients (except for one-to-many-to-one cases) due to similarities of flows to the other PTC clients. Floor Control related flows are similar for all cases.

The illustrations are specific to LI and therefore, also include the MDF2, MDF3 and LEMF. To show the user-interactions, User as a separate box is also shown. The flow-diagrams illustrated in the present document use the flow-diagrams in ATIS-0700016 [22] and OMA architecture document [24] as a reference.

5.2 Pre-established session setup

Figure 5.2-1 illustrates flow diagram for a pre-established session setup for a target.

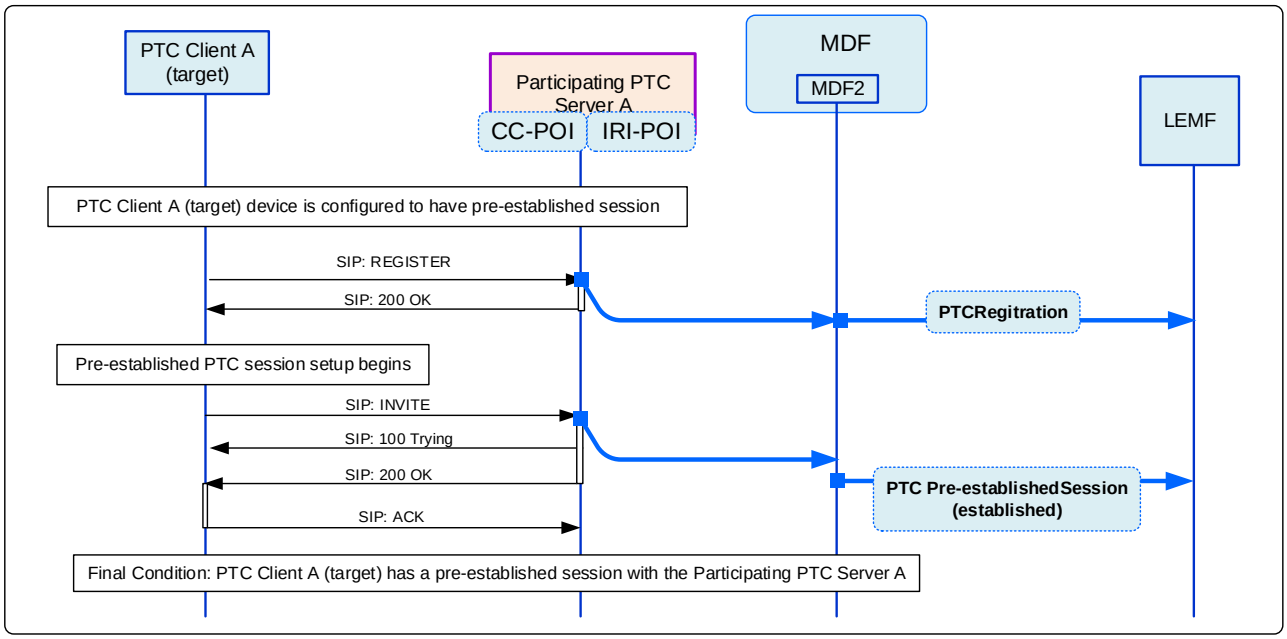


Figure 5.2-1: Pre-established session setup for PTC client A (target)

The flow shown in figure 5.2-1 is per the requirements specified in TS 33.128 [4]. As an alternative implementation, the flow shown in figure 5.2-2 ensures that xIRIs are sent at the successful completion of registration and session setup.

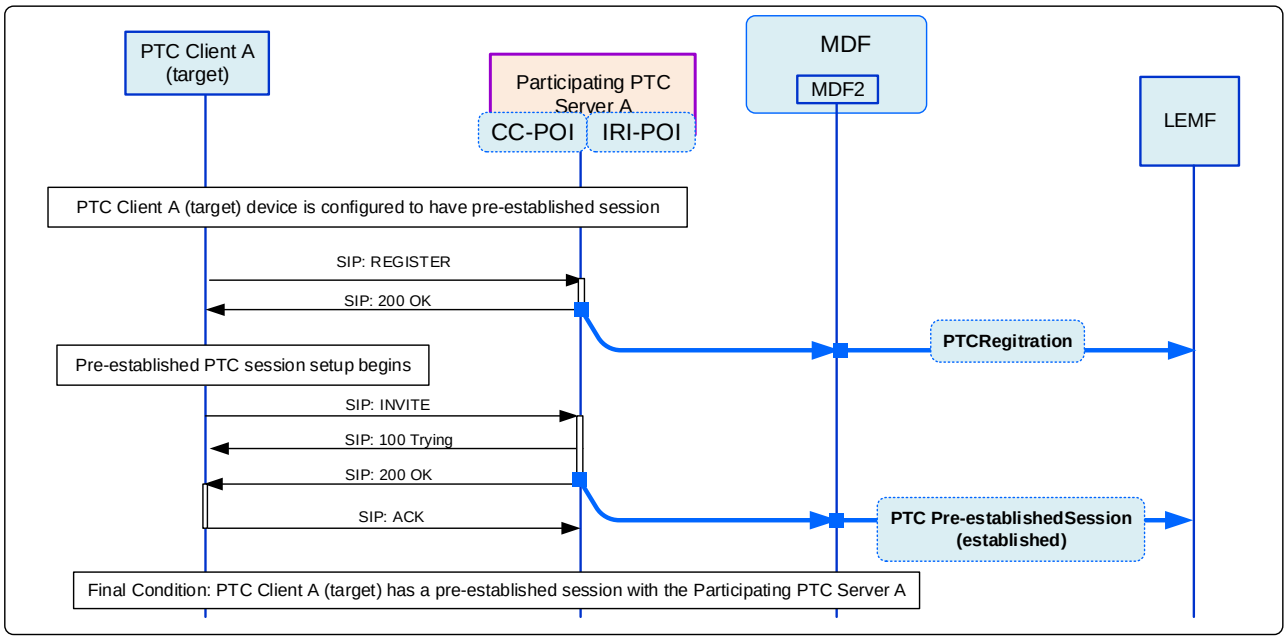


Figure 5.2-2: Pre-established session setup for PTC client A (target) - alternative

Timing-wise, the flows shown in figures 5.2-1 and 5.2-2 appear to be the same. However, with implementation approach shown in figure 5.2-2, the xIRIs are not sent if the PTC session setup is not successful.

5.3 On-demand PTC 1:1 and ad-hoc group sessions

5.3.1 General

The flows included in this clause show on-demand PTC sessions from PTC Client A to PTC Client B. Separate flows are shown for PTC Client A as a target and PTC Client B as a target. For a PTC session setup perspective, the flow applies to on-demand 1:1 PTC session and on demand ad-hoc PTC session. In the latter case, there can be multiple destinations, however, signalling to all destinations follow the same flow. In both cases, at the PTC session initiating end, the PTC Server that provides the participating PTC function to the PTC Client A also provides the controlling PTC function.

The floor control variations which would also apply to PTC sessions started from or to PTC clients with the pre-established PTC sessions. In all the illustrations, the TBCP message names are shown from a functional perspective.

5.3.2 PTC session initiated from a target

5.3.2.1 PTC session start

In the flow-diagram shown in figure 5.3.2.1-1, PTC Client A (target) initiates an on-demand PTC session to PTC Client B.

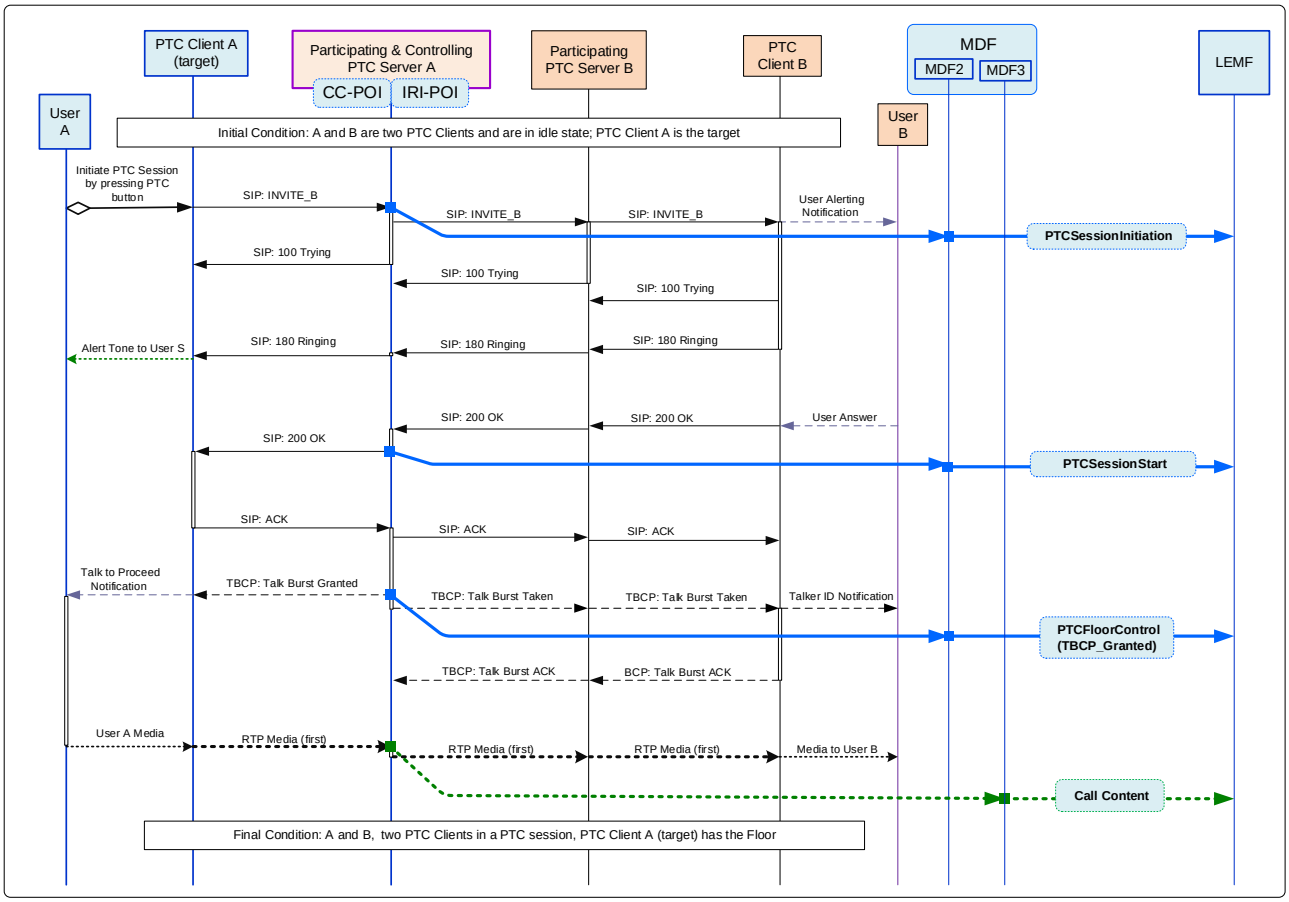


Figure 5.3.2.1-1: On-demand PTC session initiation from PTC client A (target)

5.3.2.2 PTC session end

In the flow-diagram shown in figure 5.3.2.2-1, PTC Client A (target) ends the PTC session previously setup to PTC Client B.

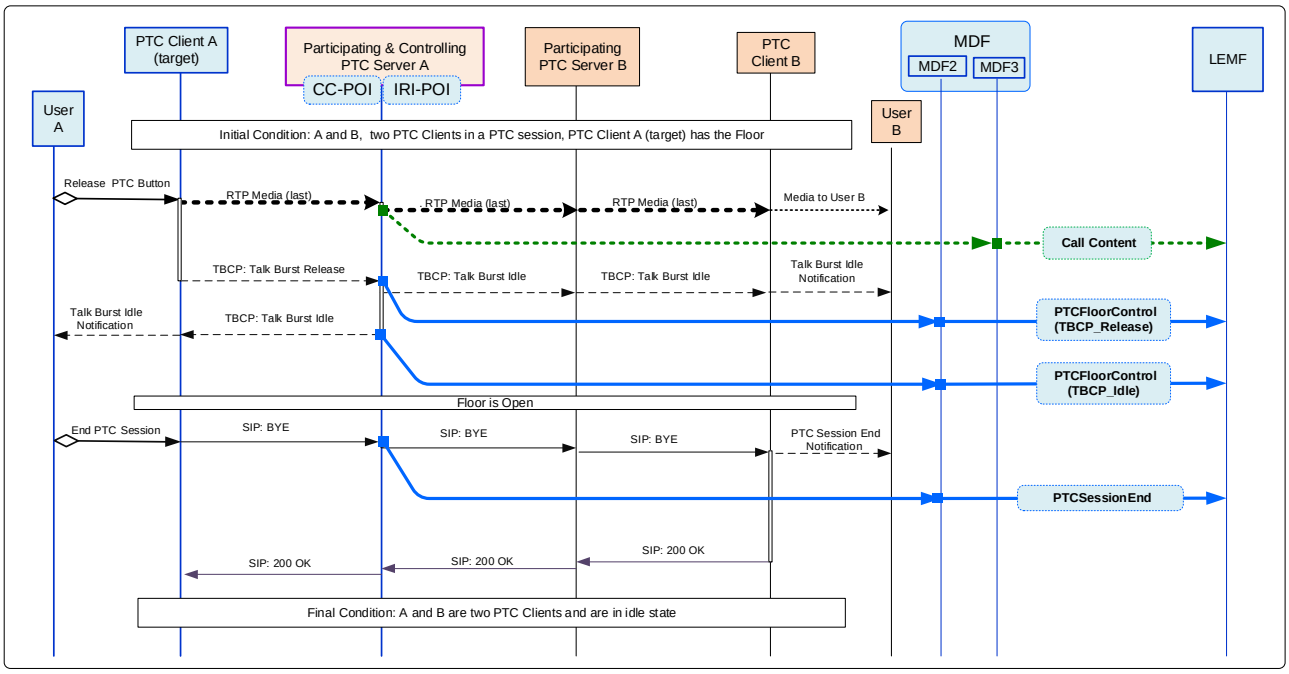


Figure 5.3.2.2-1: On-demand PTC session from PTC client A (target) is ended

The PTC session ending flow shown in figure 5.3.2.2-1 is an example. There can be other ways how a PTC session can be ended.

5.3.2.3 Floor control switching

The flow-diagram shown in figure 5.3.2.3-1 illustrates the floor-control switching.

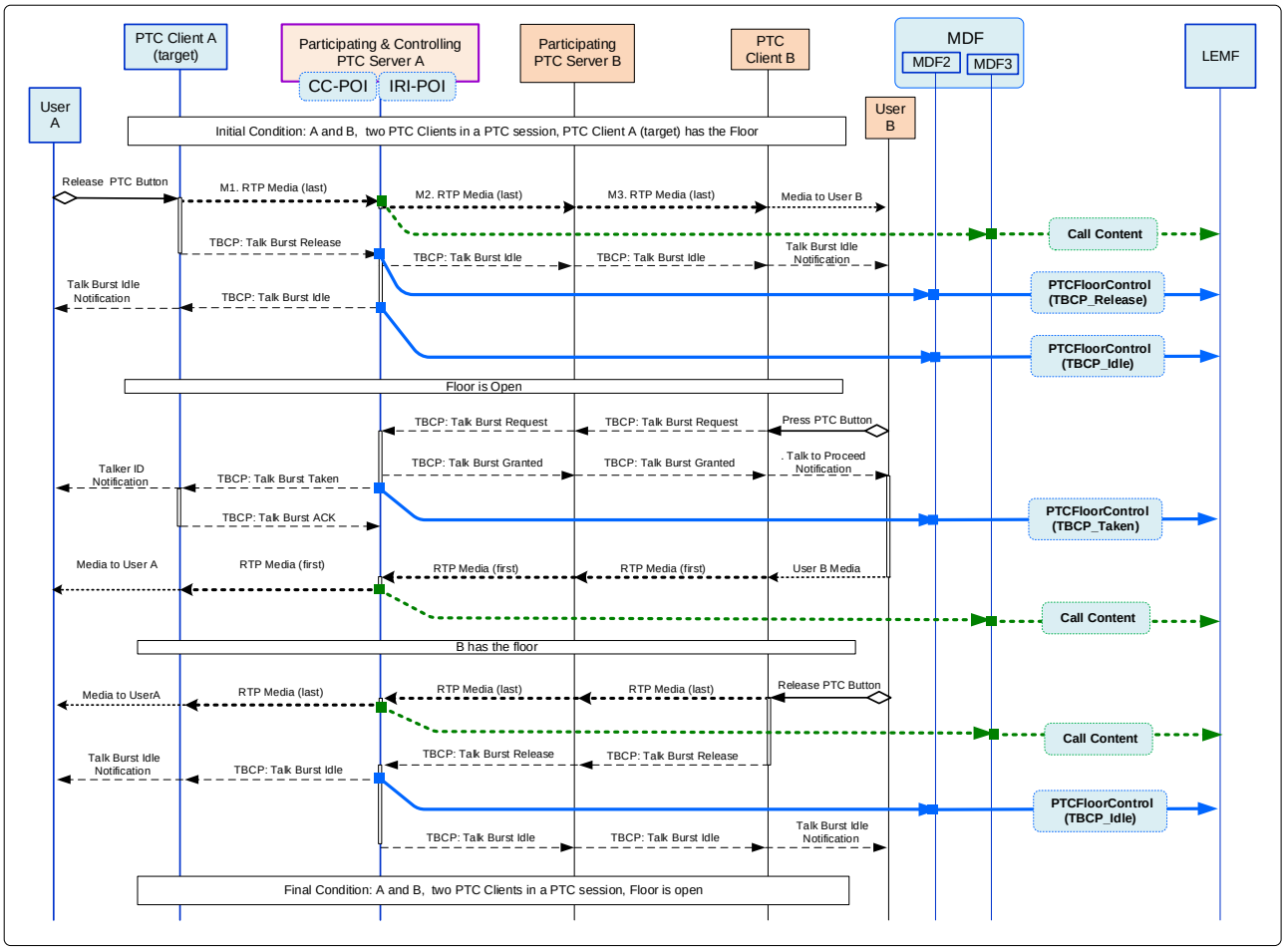


Figure 5.3.2.3-1: On-demand PTC session from PTC client A (target) – floor control switching

At the end, as shown, the floor is open and can be acquired by either of the two PTC clients.

5.3.2.4 Floor control request queued

The flow-diagram shown in figure 5.3.2.4-1 illustrates an example of floor-control request is being queued.

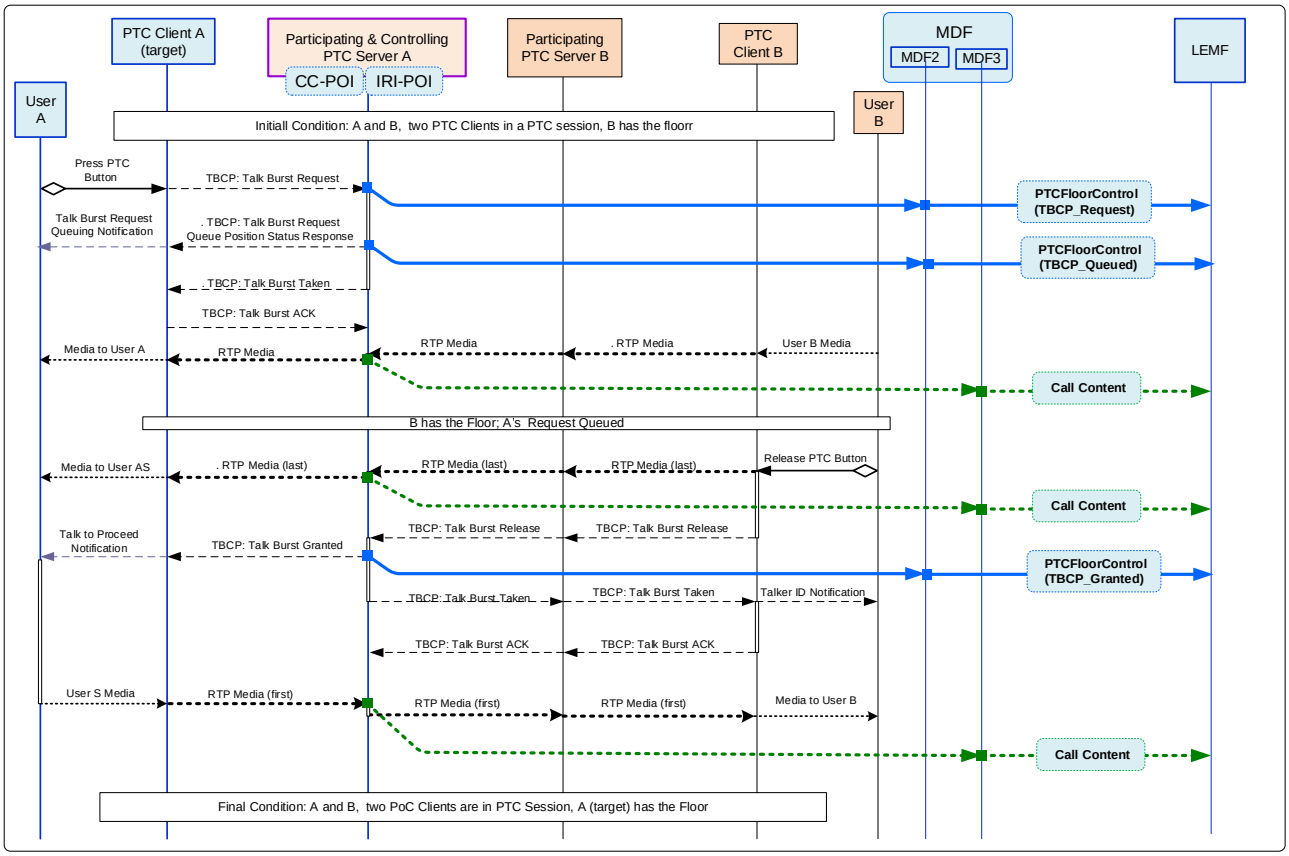


Figure 5.3.2.4-1: On-demand PTC session from PTC client A (target) – floor control request is queued

The illustration shows that a floor-control request received when the floor is not open can be queued. Such a queuing can happen for floor control request received from any of the PTC clients present on the PTC session.

5.3.2.5 Floor control request denied

The flow-diagram shown in figure 5.3.2.5-1 illustrates an example of floor-control request is being denied.

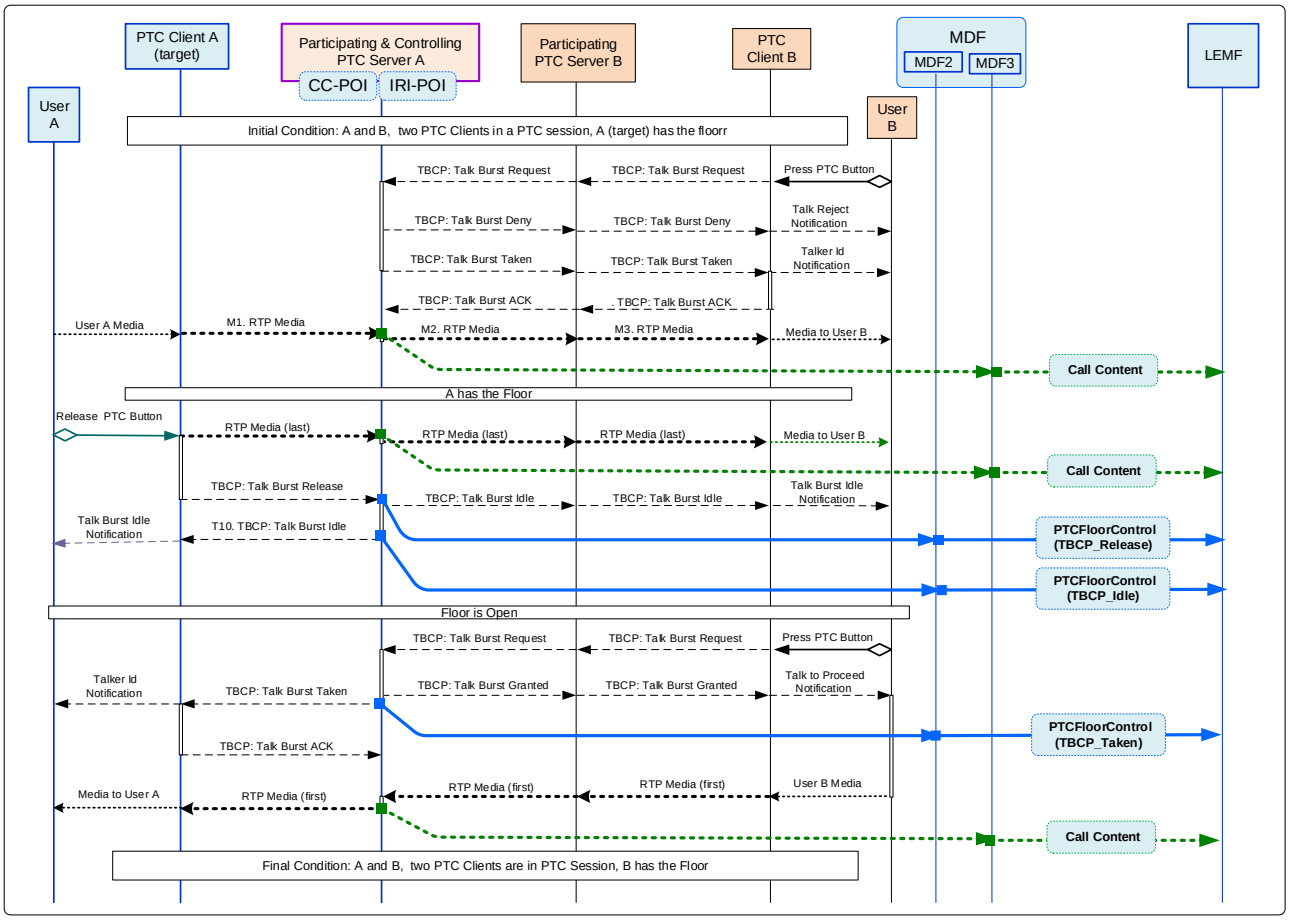


Figure 5.3.2.5-1: On-demand PTC session from PTC client A (target) – floor control request is denied

The illustration shows that a floor-control request received when the floor is not open can be denied. Such a denial can happen for floor control request received from any of the PTC clients present on the PTC session. When the denial is made to a request received from a PTC client who is not a target, as shown, no xIRI is generated.

5.3.2.6 Floor control request revoked

The flow-diagram shown in figure 5.3.2.6-1 illustrates an example of floor-control request is being revoked.

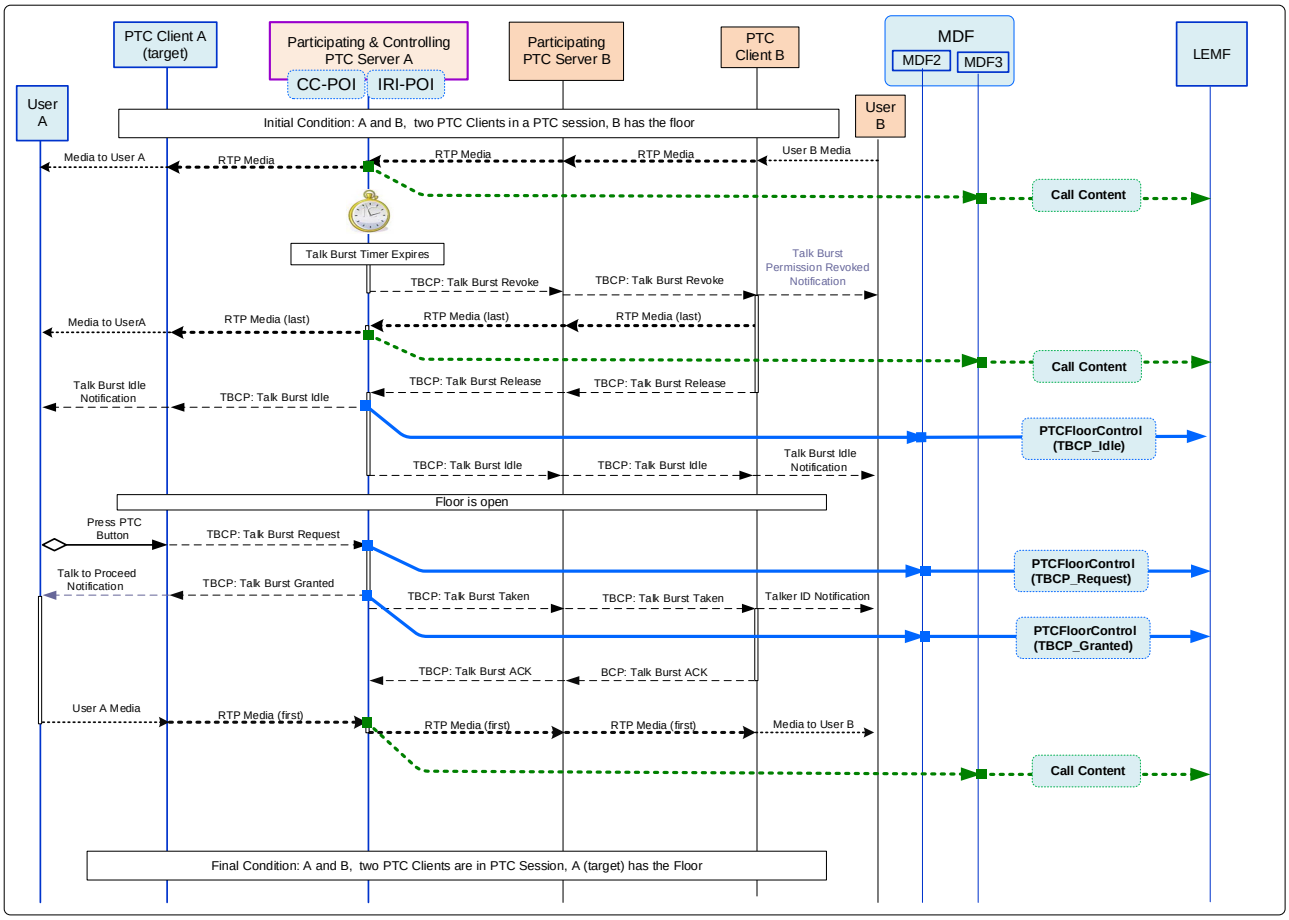


Figure 5.3.2.6-1: On-demand PTC session from PTC client A (target) – floor control request is revoked

The illustration shows that a floor-control that a PTC client has can be revoked. Such a revocation can happen when any of the PTC clients have the floor. When the floor control is revoked for a PTC client who is not a target, as shown, no xIRI is generated.

5.3.3 PTC sessions terminated to a target

5.3.3.1 PTC session start

In the flow-diagram shown in figure 5.3.3.1-1, PTC Client A initiates an on-demand PTC session to PTC Client B (target).

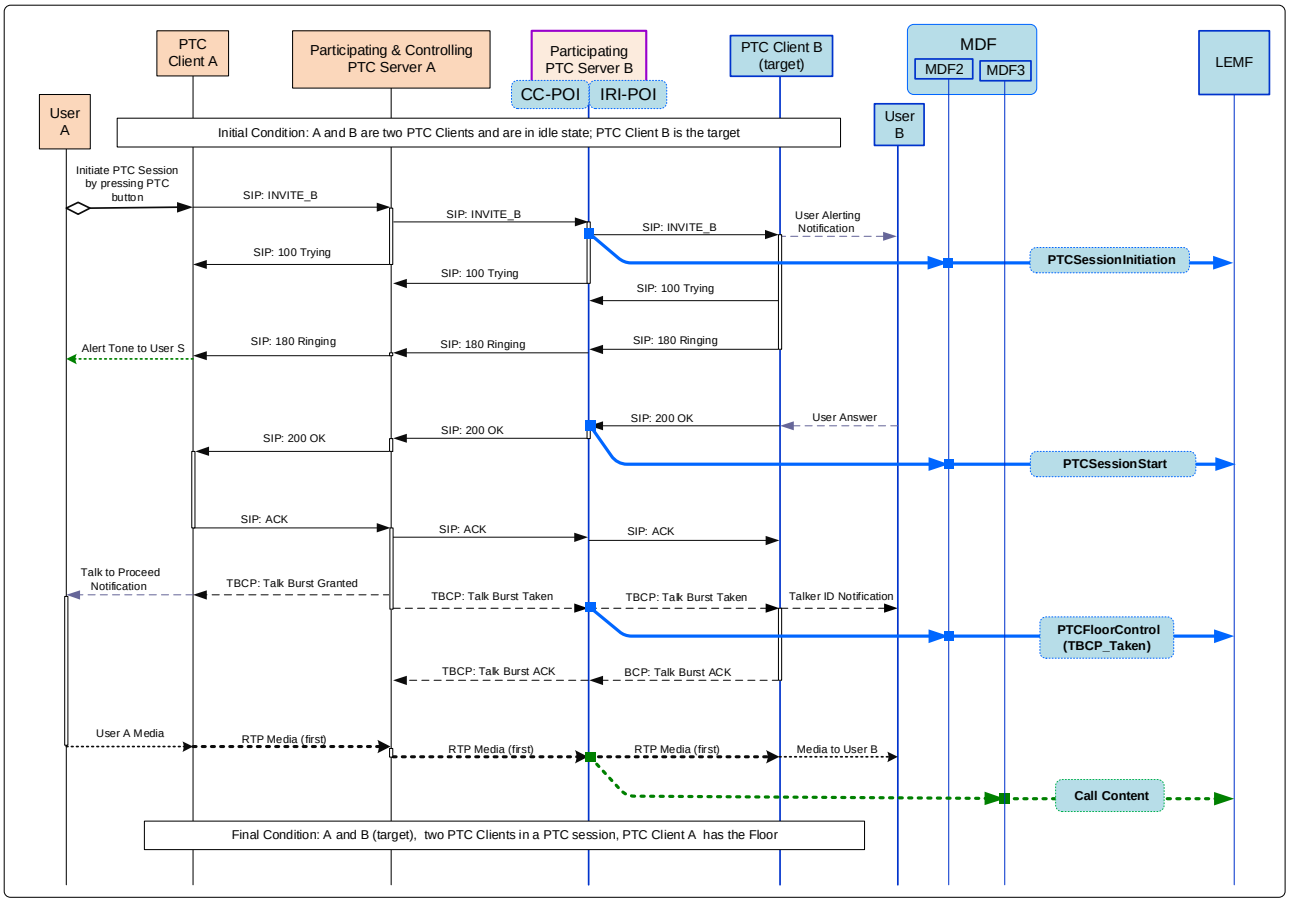


Figure 5.3.3.1-1: On-demand PTC session to PTC client B (target)

As shown, the LI functions are provided by the IRI-POI present in the PTC Server B that provides the participating PTC function.

5.3.3.2 PTC session end

In the flow-diagram shown in figure 5.3.3.2-1, PTC Client A ends the PTC session previously setup to PTC Client B (target).

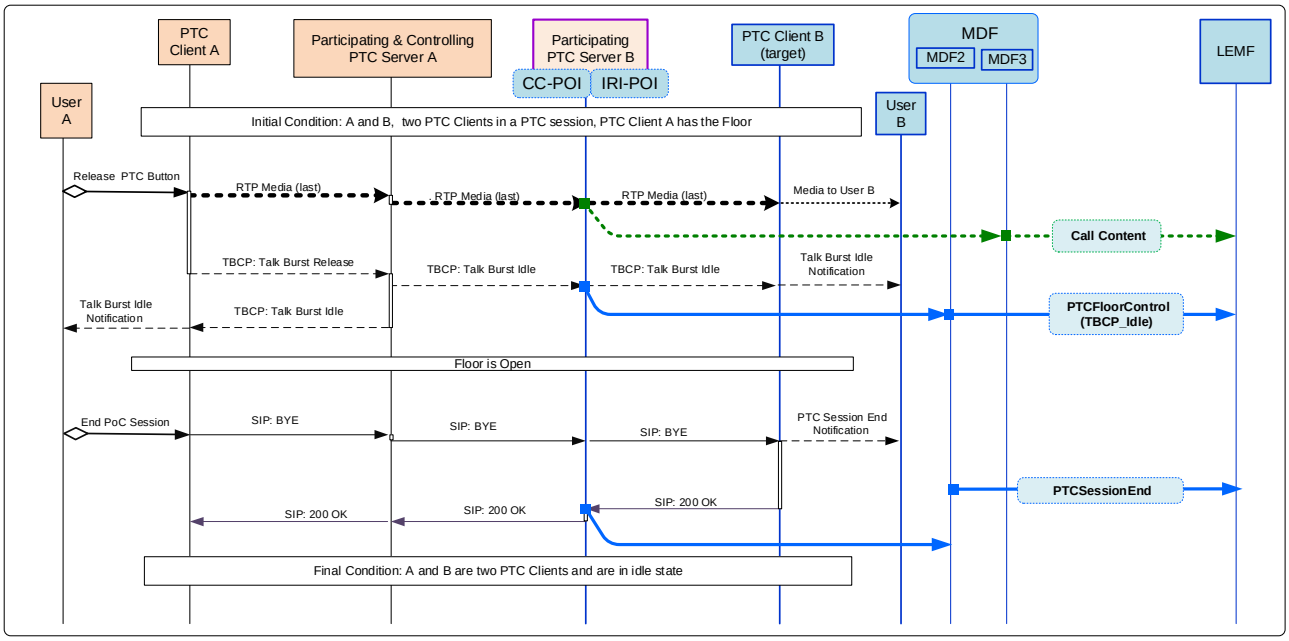


Figure 5.3.3.2-1: On-demand PTC session to PTC client B (target) is ended

The PTC session ending flow shown in figure 5.3.3.2-1 is an example. There can be other ways how a PTC session can be ended. As shown, the LI functions are provided by the IRI-POI present in the PTC Server B that provides the participating PTC function.

5.3.3.3 Floor control switching

The flow-diagram shown in figure 5.3.3.3-1 illustrates the floor-control switching.

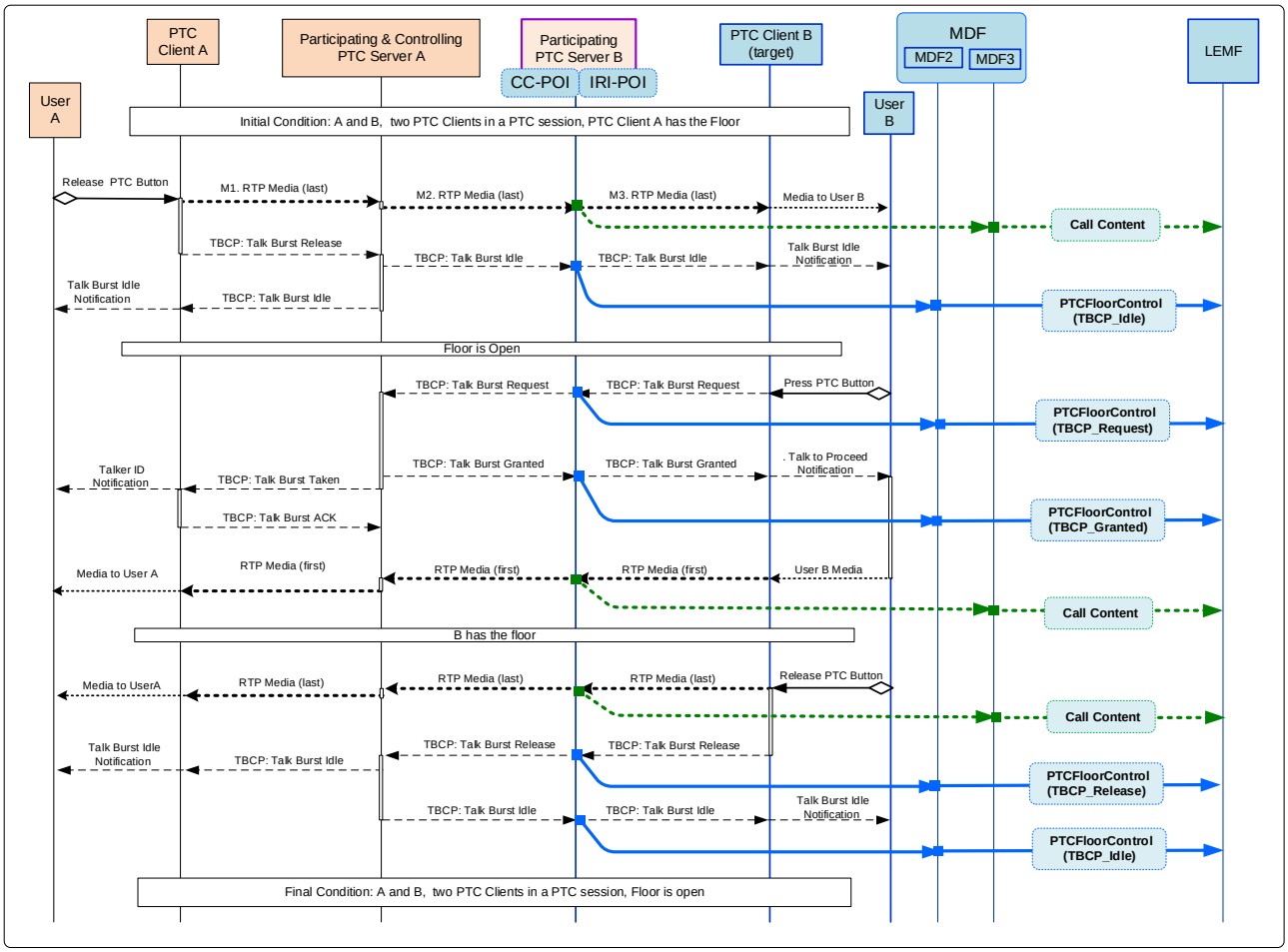


Figure 5.3.3.3-1: On-demand PTC session to PTC client B (target) – floor control switching

At the end, as shown, the floor is open and can be acquired by either of the two PTC clients. As shown, the floor control related xIRIs are generated from the IRI-POI present in the PTC Server B that provides the participating PTC function.

5.3.3.4 Floor control request queued

The flow-diagram shown in figure 5.3.3.4-1 illustrates an example of floor-control request is being queued.

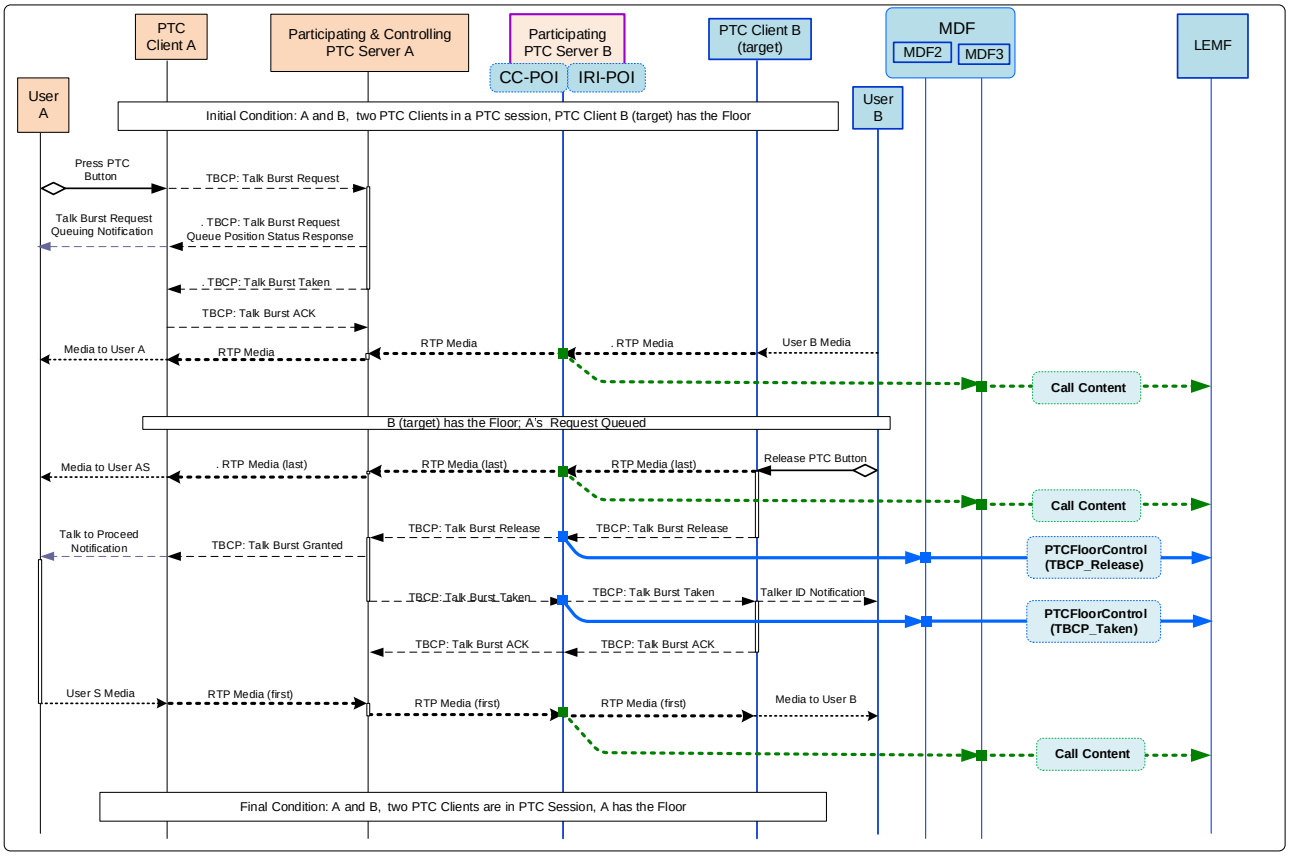


Figure 5.3.3.4-1: On-demand PTC session to PTC client B (target) – floor control request is queued

The illustration shows that a floor-control request received when the floor is not open can be queued. Such a queuing can happen for floor control request received from any of the PTC clients present on the PTC session. In the illustration, a floor control request from PTC Client A (not a target) is queued and hence, no specific xIRI is generated when the queuing happens.

5.3.3.5 Floor control request denied

The flow-diagram shown in figure 5.3.3.5-1 illustrates an example of floor-control request is being denied.

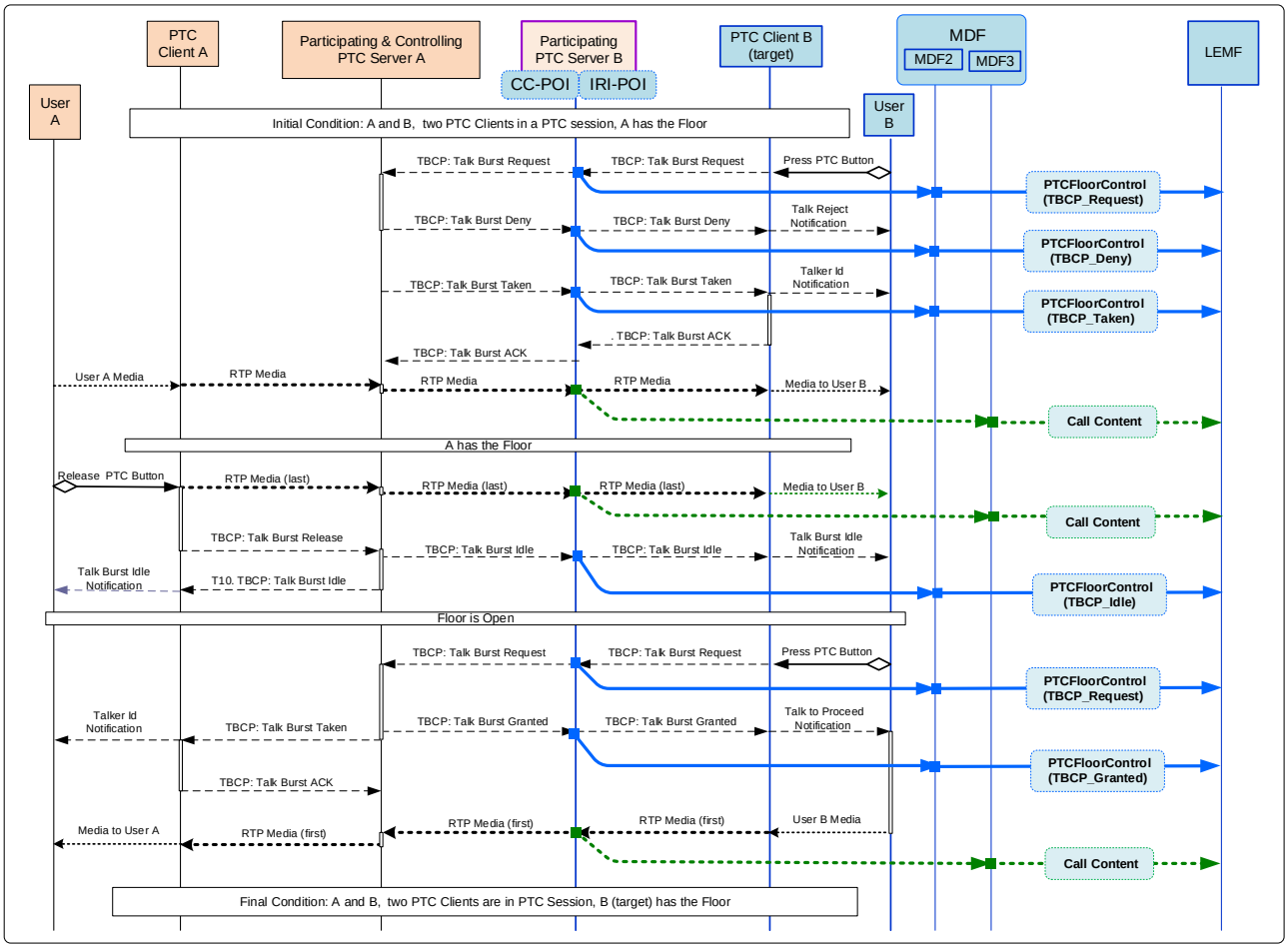


Figure 5.3.3.5-1: On-demand PTC session to PTC client B (target) – floor control request is denied

The illustration shows that a floor-control request received when the floor is not open can be denied. Such a denial can happen for floor control request received from any of the PTC clients present on the PTC session. When the denial is made to a request received from a target, as shown, an xIRI is generated.

As shown, the xIRI is generated from the IRI-POI present in the PTC Server B that provides a participating PTC function. The denial of the floor control is made at the PTC Server A that provides the controlling PTC function.

5.3.3.6 Floor control request revoked

The flow-diagram shown in figure 5.3.3.6-1 illustrates an example of floor-control request is being revoked.

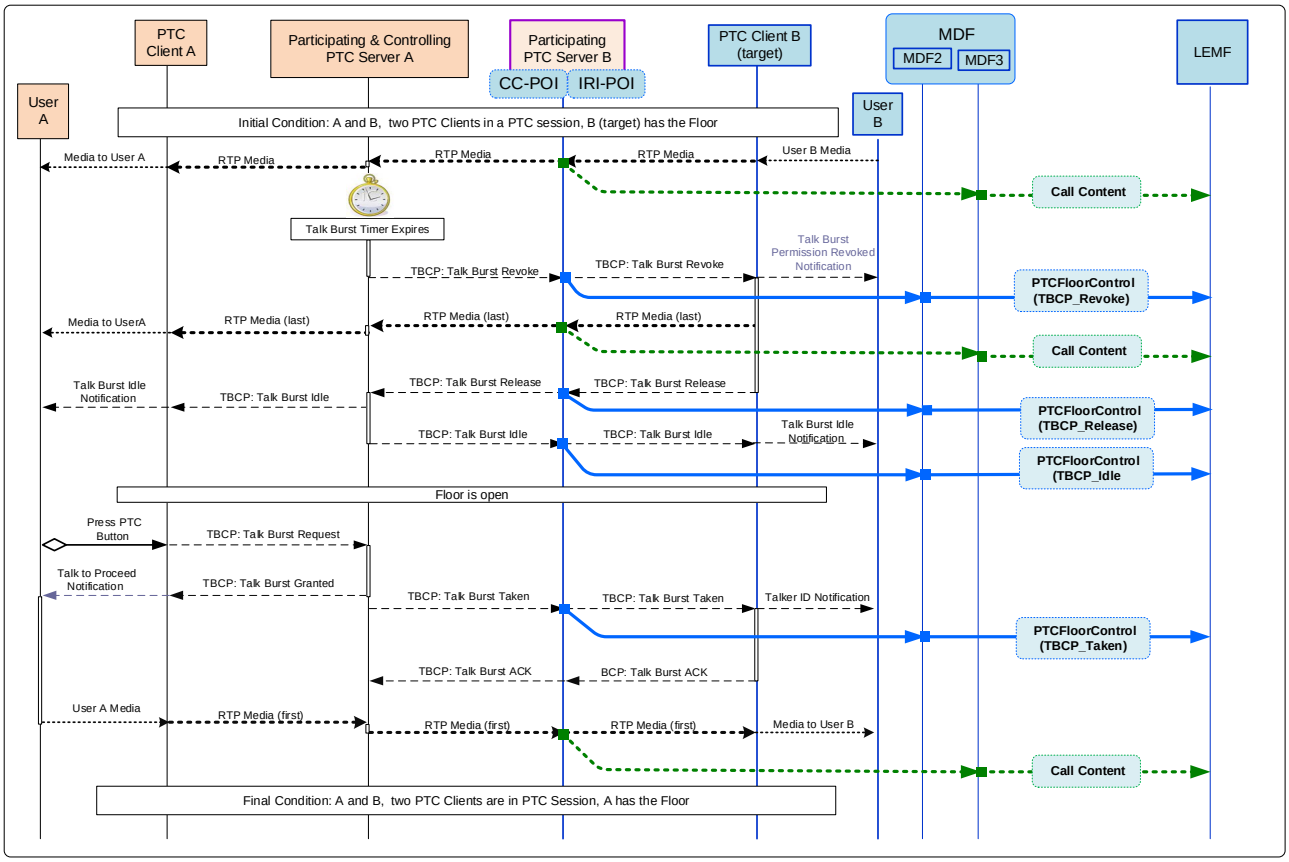


Figure 5.3.3.6-1: On-demand PTC session to PTC client B (target) – floor control request is revoked

The illustration shows that a floor-control that a PTC client has can be revoked. Such a revocation can happen when any of the PTC clients have the floor. When the floor control is revoked for a PTC client who is a target, as shown, xIRI is generated.

As shown, the xIRI is generated from the IRI-POI present in the PTC Server B that provides a participating PTC function. The revocation of the floor control is made at the PTC Server A that provides the controlling PTC function.

5.4 PTC sessions (1:1 or ad-hoc group) from pre-established sessions

5.4.1 General

The flows included in this clause show PTC sessions start with one or more PTC clients having the pre-established session. Separate flows are shown for PTC Client A as a target and PTC Client B as a target. For a PTC sessions perspective, the flows apply to 1:1 PTC session and ad-hoc PTC session. In the latter case, there can be multiple destinations, however, signalling to all destinations follow the same flow. In both cases, at the PTC session originating end, the PTC Server that provides the participating PTC function to the PTC Client A also provides the controlling PTC function.

Once the PTC session starts, the floor control variations are same as the on-demand PTC sessions and hence, not included here. In all the illustrations, the TBCP message names are shown from a functional perspective.

5.4.2 PTC session start from a target

5.4.2.1 Scenario 1: Pre-established to Pre-established

5.4.2.1.1 PTC session start

In the flow-diagram shown in figure 5.4.2.1.1-1, PTC Client A (target) with a pre-established session starts a PTC session to PTC Client B with also a pre-established session.

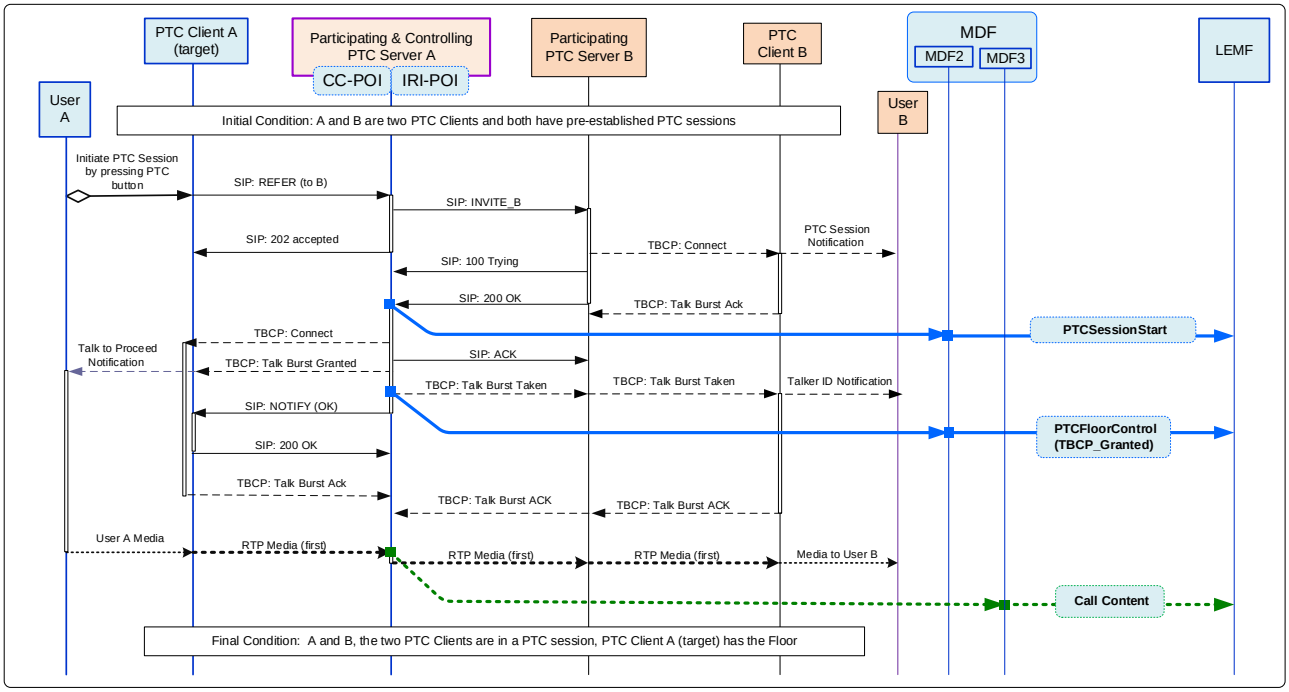


Figure 5.4.2.1.1-1: PTC session from PTC client A (target)

As shown, the LI functions are provided by the IRI-POI present in the PTC Server A that provides the participating PTC function and controlling PTC function.

5.4.2.1.2 PTC session end

In the flow-diagram shown in figure 5.4.2.1.2-1, PTC Client A (target) ends the PTC session previously setup to PTC Client B.

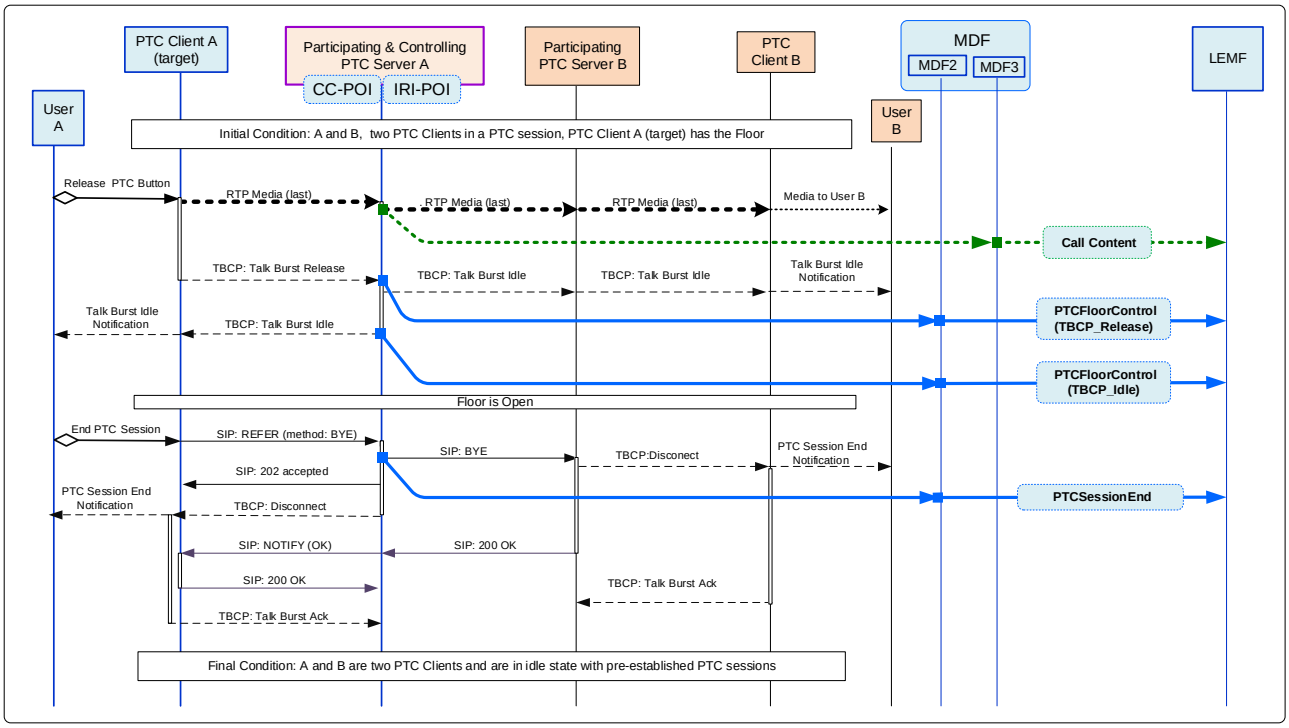


Figure 5.4.2.1.2-1: PTC session from PTC client A (target) is ended

The PTC session ending flow shown in figure 5.4.2.1.2-1 is an example. There can be other ways how a PTC session can be ended. As shown, the LI functions are provided by the IRI-POI present in the PTC Server A that provides the participating PTC function and controlling PTC function.

5.4.2.2 Scenario 2: Pre-established to on-demand

5.4.2.2.1 PTC session start

In the flow-diagram shown in figure 5.4.2.2.1-1, PTC Client A (target) with a pre-established session starts a PTC session to PTC Client B without a pre-established session.

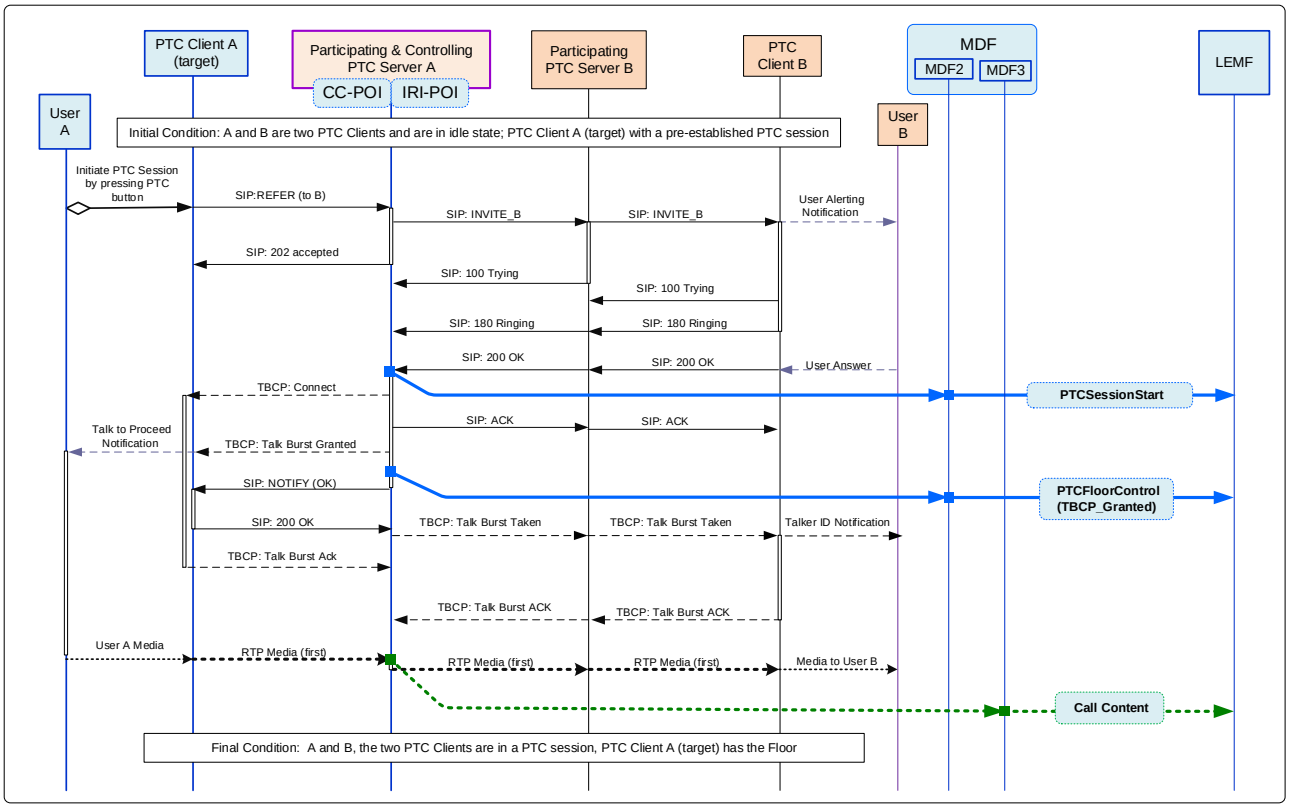


Figure 5.4.2.2.1-1: PTC session from PTC client A (target)

As shown, the LI functions are provided by the IRI-POI present in the PTC Server A that provides the participating PTC function and controlling PTC function.

5.4.2.2.2 PTC session end

In the flow-diagram shown in figure 5.4.2.2.2-1, PTC Client A (target) ends the PTC session previously setup to PTC Client B.

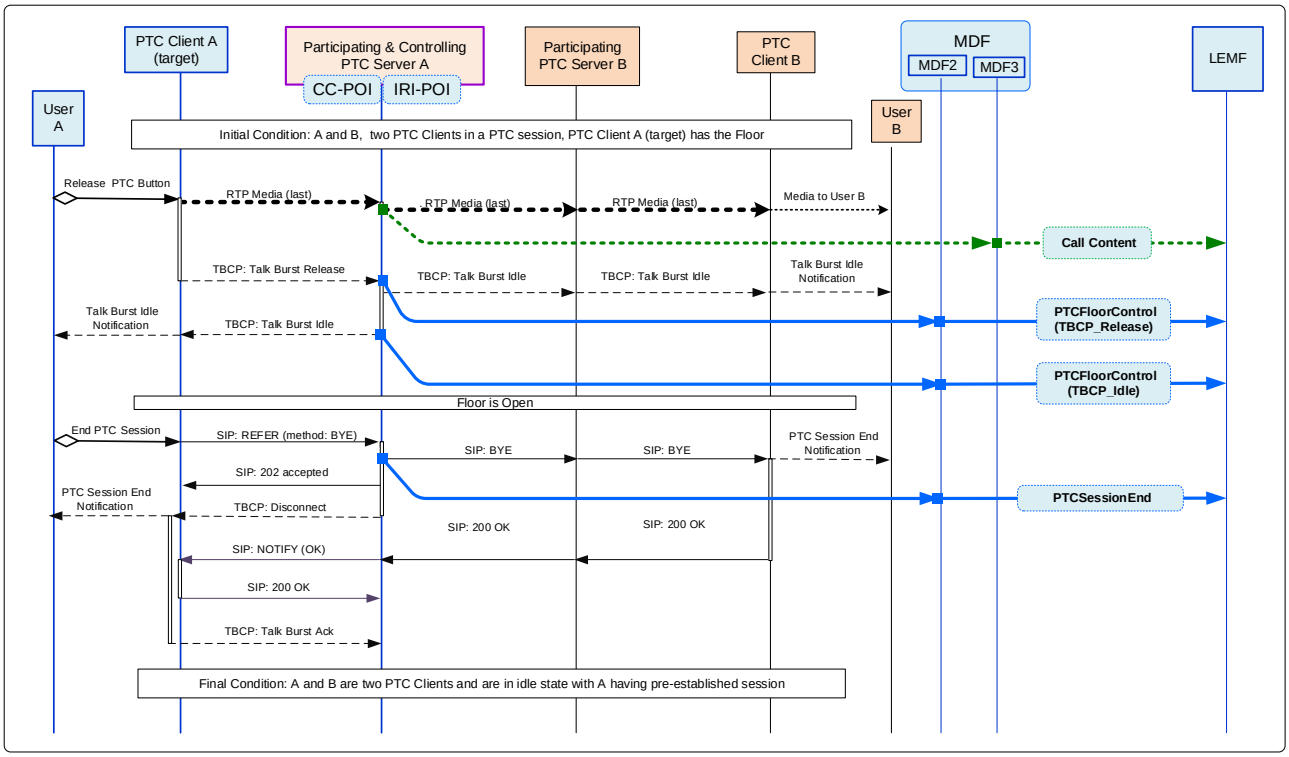


Figure 5.4.2.2.2-1: PTC session from PTC client A (target) is ended

The PTC session ending flow shown in figure 5.4.2.2.2-1 is an example. There can be other ways how a PTC session can be ended. As shown, the LI functions are provided by the IRI-POI present in the PTC Server A that provides the participating PTC function and controlling PTC function.

5.4.2.3 Scenario 3: On-demand to pre-established

5.4.2.3.1 PTC session start

In the flow-diagram shown in figure 5.4.2.3.1-1, PTC Client A (target) without a pre-established session starts a PTC session to PTC Client B with a pre-established session.

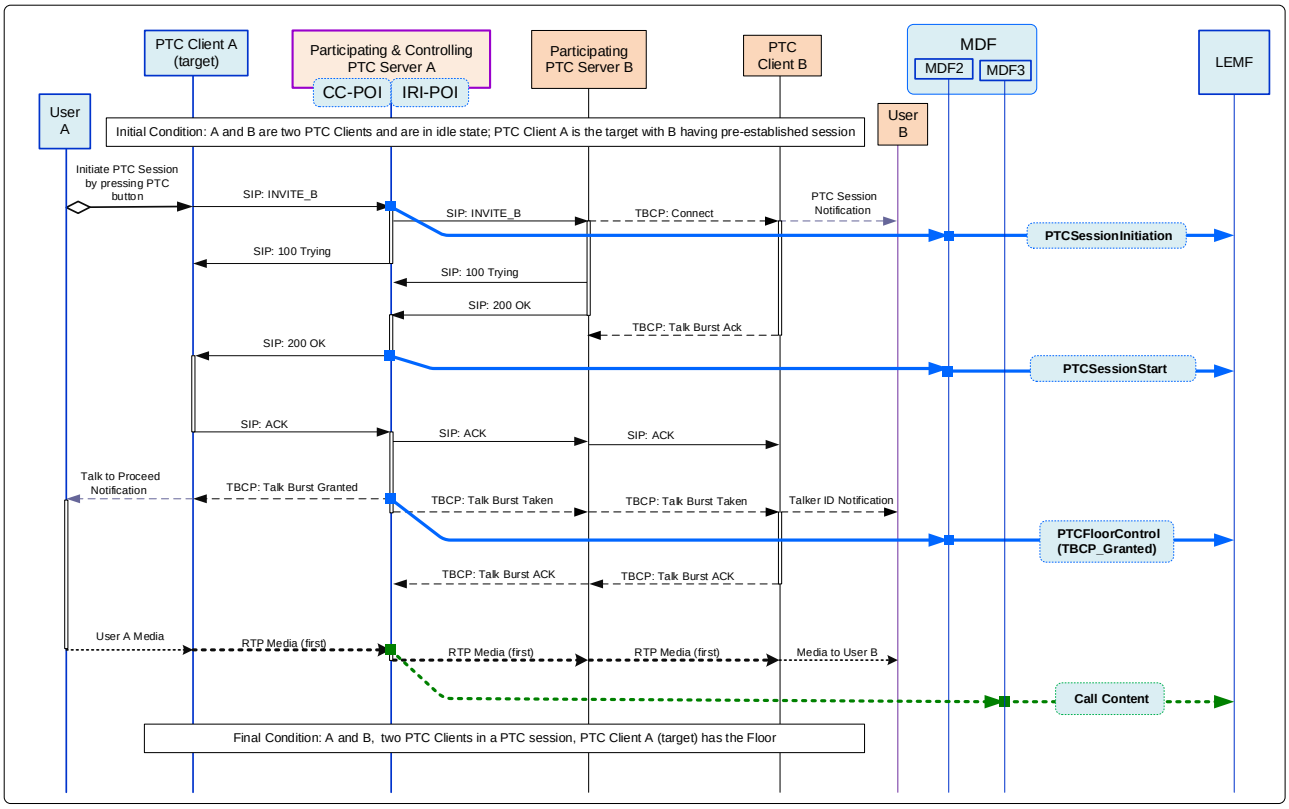


Figure 5.4.2.3.1-1: PTC session from PTC client A (target)

As shown, the LI functions are provided by the IRI-POI present in the PTC Server A that provides the participating PTC function and controlling PTC function.

5.4.2.3.2 PTC session end

In the flow-diagram shown in figure 5.4.2.3.2-1, PTC Client A (target) ends the PTC session previously setup to PTC Client B.

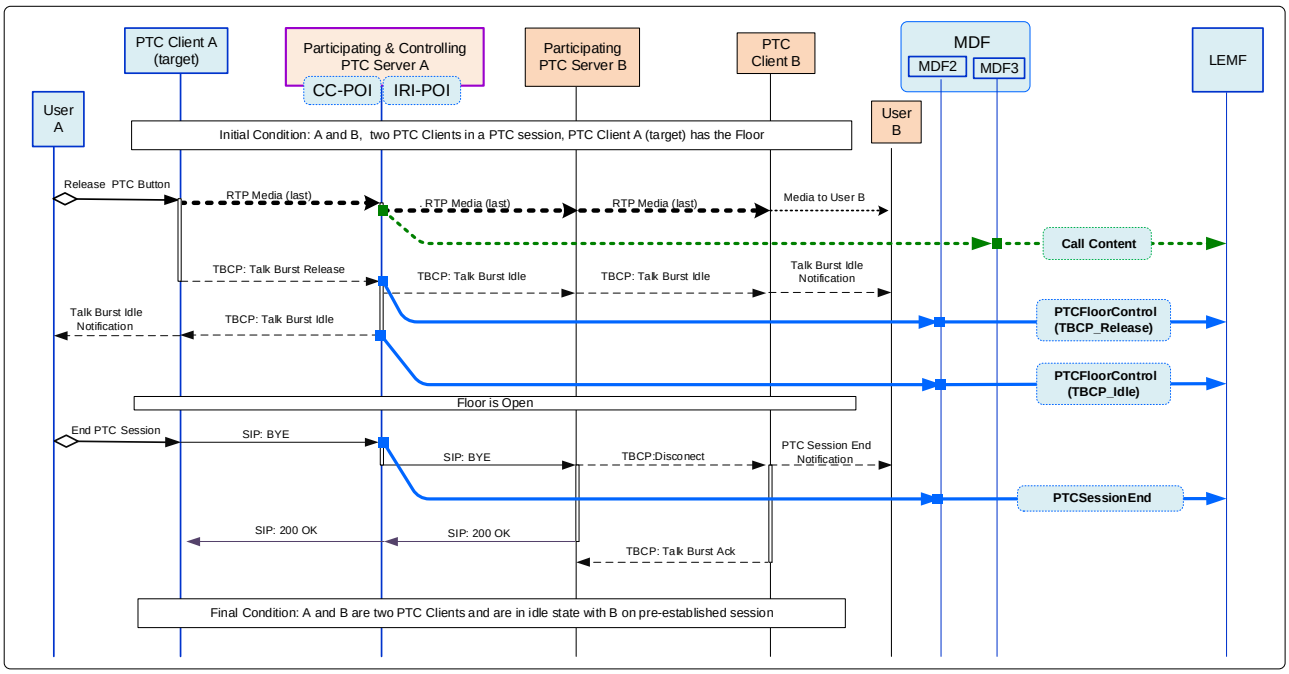


Figure 5.4.2.3.2-1: PTC session from PTC client A (target) is ended

The PTC session ending flow shown in figure 5.4.2.3.2-1 is an example. There can be other ways how a PTC session can be ended. As shown, the LI functions are provided by the IRI-POI present in the PTC Server A that provides the participating PTC function and controlling PTC function.

5.4.3 PTC session terminated to a target

5.4.3.1 Scenario 1: Pre-established to Pre-established

5.4.3.1.1 PTC session start

In the flow-diagram shown in figure 5.4.3.1.1-1, PTC Client A with a pre-established session starts a PTC session to PTC Client B (target) with also a pre-established session.

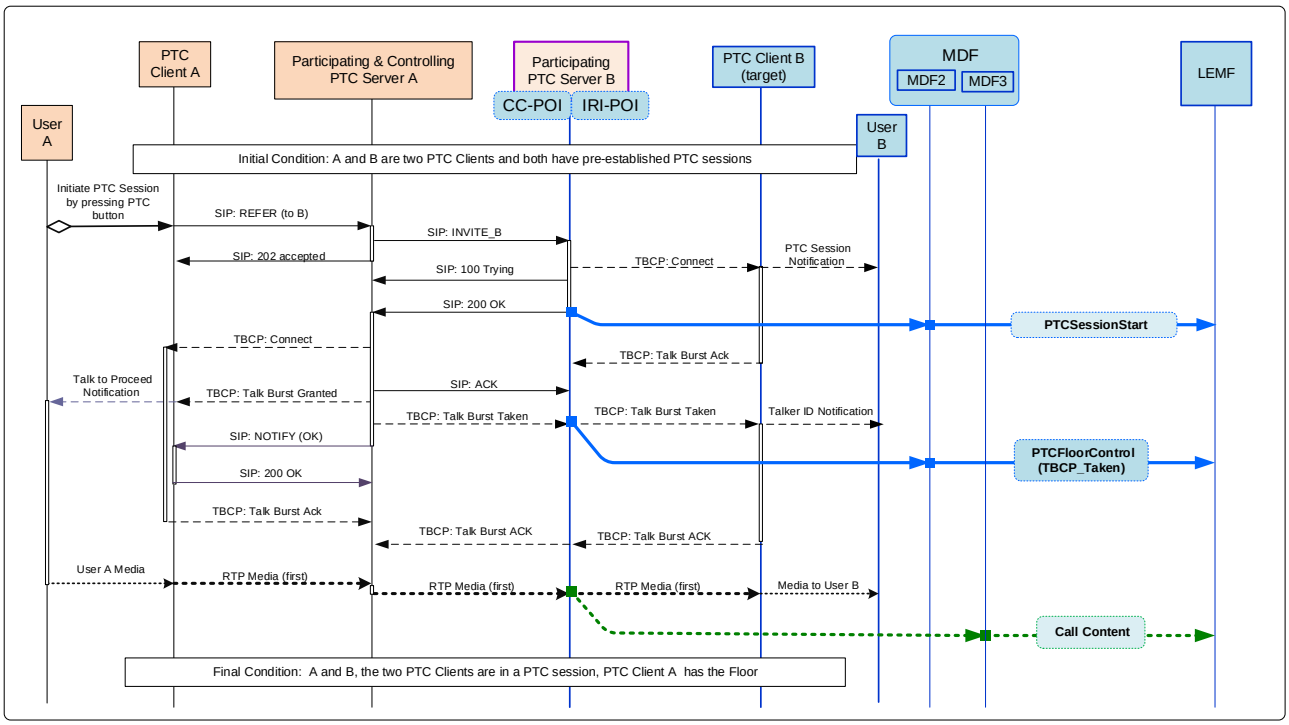


Figure 5.4.3.1.1-1: PTC session to PTC client B (target)

As shown, the LI functions are provided by the IRI-POI present in the PTC Server B that provides the participating PTC function.

5.4.3.1.2 PTC session end

In the flow-diagram shown in figure 5.4.3.1.2-1, PTC Client A ends the PTC session previously setup to PTC Client B (target).

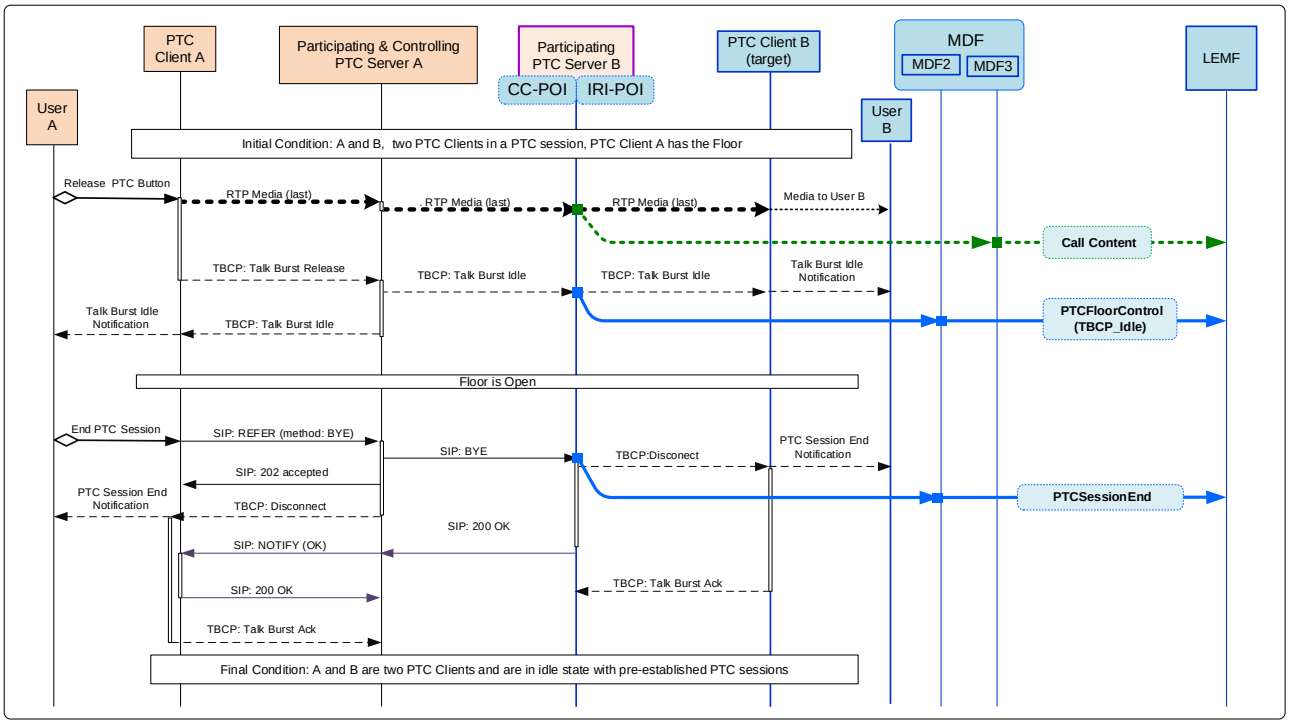


Figure 5.4.3.1.2-1: PTC session to PTC client B (target) is ended

The PTC session ending flow shown in figure 5.4.3.1.2-1 is an example. There can be other ways how a PTC session can be ended. As shown, the LI functions are provided by the IRI-POI present in the PTC Server B that provides the participating PTC function.

5.4.3.2 Scenario 2: Pre-established to on-demand

5.4.3.2.1 PTC session start

In the flow-diagram shown in figure 5.4.3.2.1-1, PTC Client A with a pre-established session starts a PTC session to PTC Client B (target) without a pre-established session.

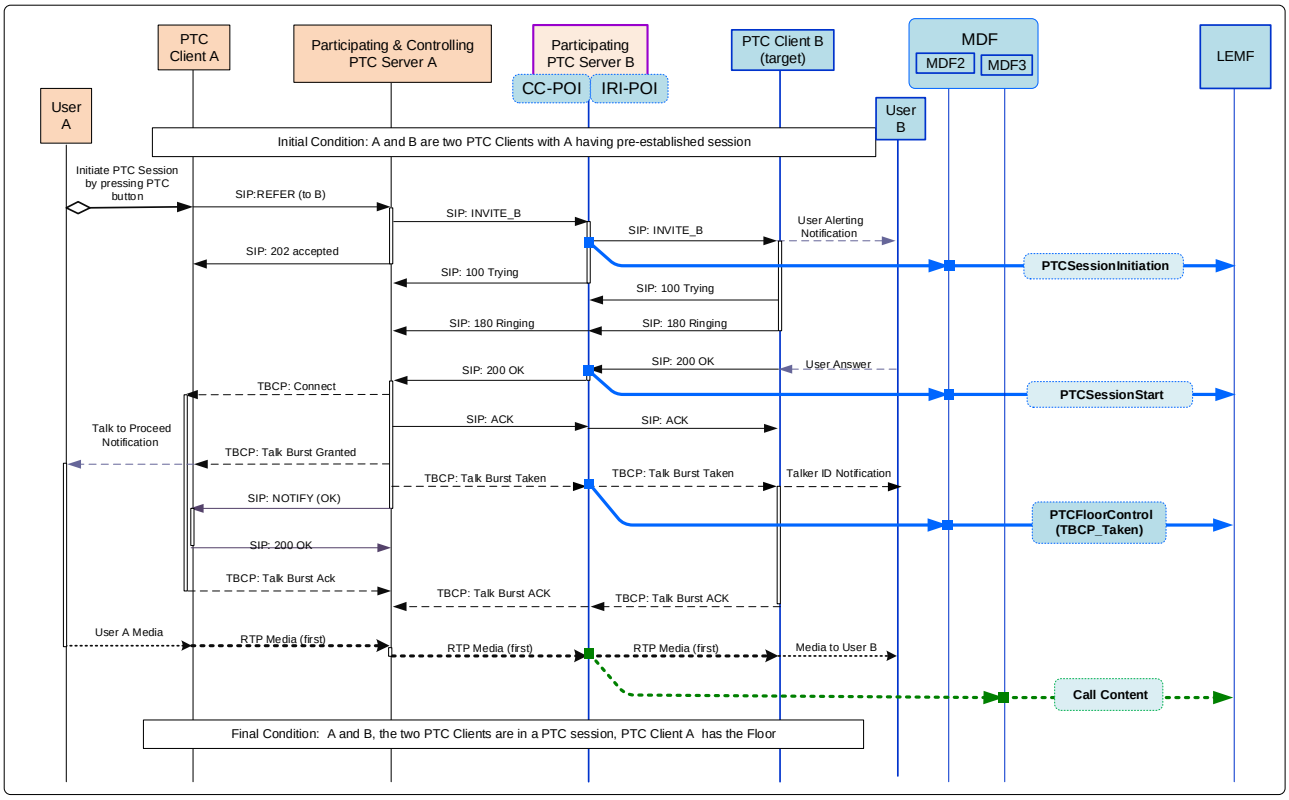


Figure 5.4.3.2.1-1: PTC session to PTC client B (target)

As shown, the LI functions are provided by the IRI-POI present in the PTC Server B that provides the participating PTC function.

5.4.3.2.2 PTC session end

In the flow-diagram shown in figure 5.4.3.2.2-1, PTC Client A ends the PTC session previously setup to PTC Client B (target).

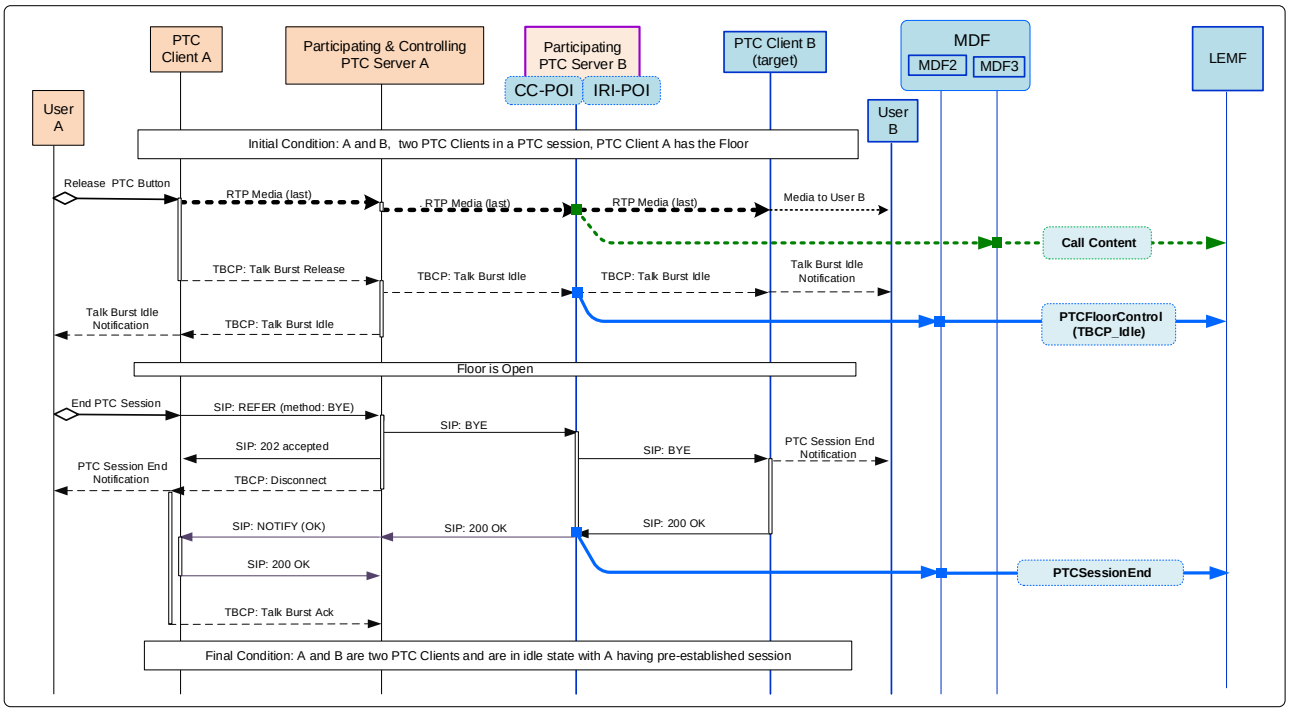


Figure 5.4.3.2.2-1: PTC session to PTC client B (target) is ended

The PTC session ending flow shown in figure 5.4.3.2.2-1 is an example. There can be other ways how a PTC session can be ended. As shown, the LI functions are provided by the IRI-POI present in the PTC Server B that provides the participating PTC function.

5.4.3.3 Scenario 3: On-demand to pre-established

5.4.3.3.1 PTC session start

In the flow-diagram shown in figure 5.4.3.3.1-1, PTC Client A without a pre-established session starts a PTC session to PTC Client B (target) with a pre-established session.

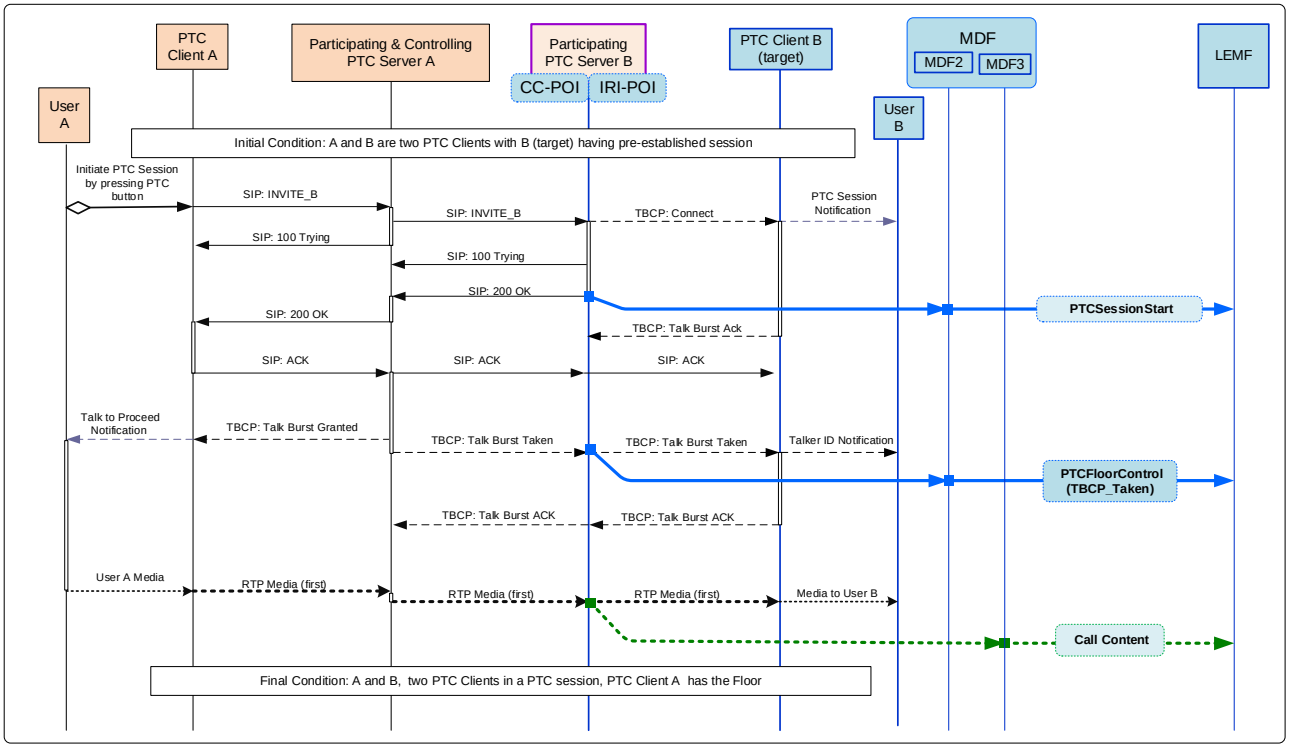


Figure 5.4.3.3.1-1: PTC session to PTC client B (target)

As shown, the LI functions are provided by the IRI-POI present in the PTC Server B that provides the participating PTC function.

5.4.3.3.2 PTC session end

In the flow-diagram shown in figure 5.4.3.3.2-1, PTC Client A (target) ends the PTC session previously setup to PTC Client B.

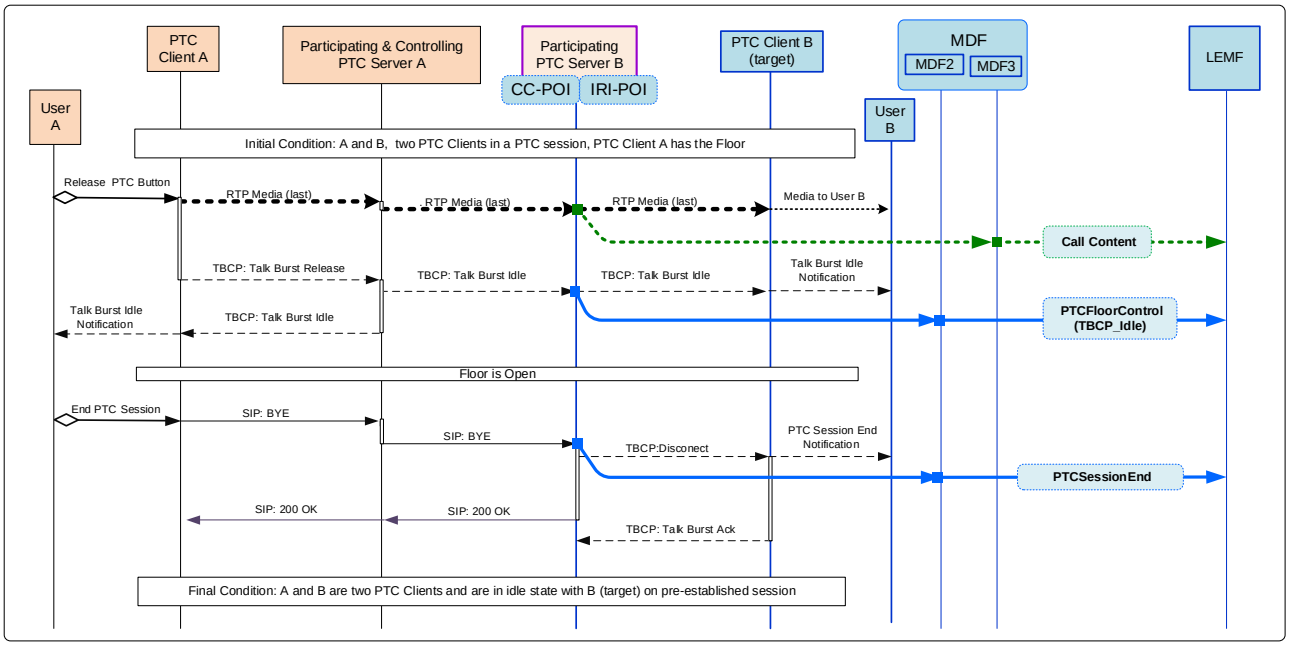


Figure 5.4.3.3.2-1: PTC session to PTC client B (target) is ended

The PTC session ending flow shown in figure 5.4.3.3.2-1 is an example. There can be other ways how a PTC session can be ended. As shown, the LI functions are provided by the IRI-POI present in the PTC Server B that provides the participating PTC function.

5.5 On-demand PTC pre-arranged group sessions

5.5.1 General

The flows included in this clause show on-demand pre-arranged PTC sessions from PTC Client A. For simplicity of the drawing only one destination PTC Client B is shown. Furthermore, the signalling to all destinations follow the same flow. No separate flows are shown when a destination PTC client is a target since from a destination client's perspective, the flow would be same as 1:1 or an ad-hoc group PTC session.

At the originating end, the PTC Server A provides the participating PTC function for PTC Client A and Group Management Server that has the list of all destination PTC Clients in the group provides the controlling PTC function. The Group Management Server may be hosted by a different provider.

The LI functions for target (PTC Client A) is provided by the PTC Server A. In all the illustrations, the TBCP message names are shown from a functional perspective.

5.5.2 PTC session start

In the flow-diagram shown in figure 5.5.2-1, PTC Client A (target) initiates an on-demand pre-arranged PTC session. Only one destination client (PTC Client B) is shown.

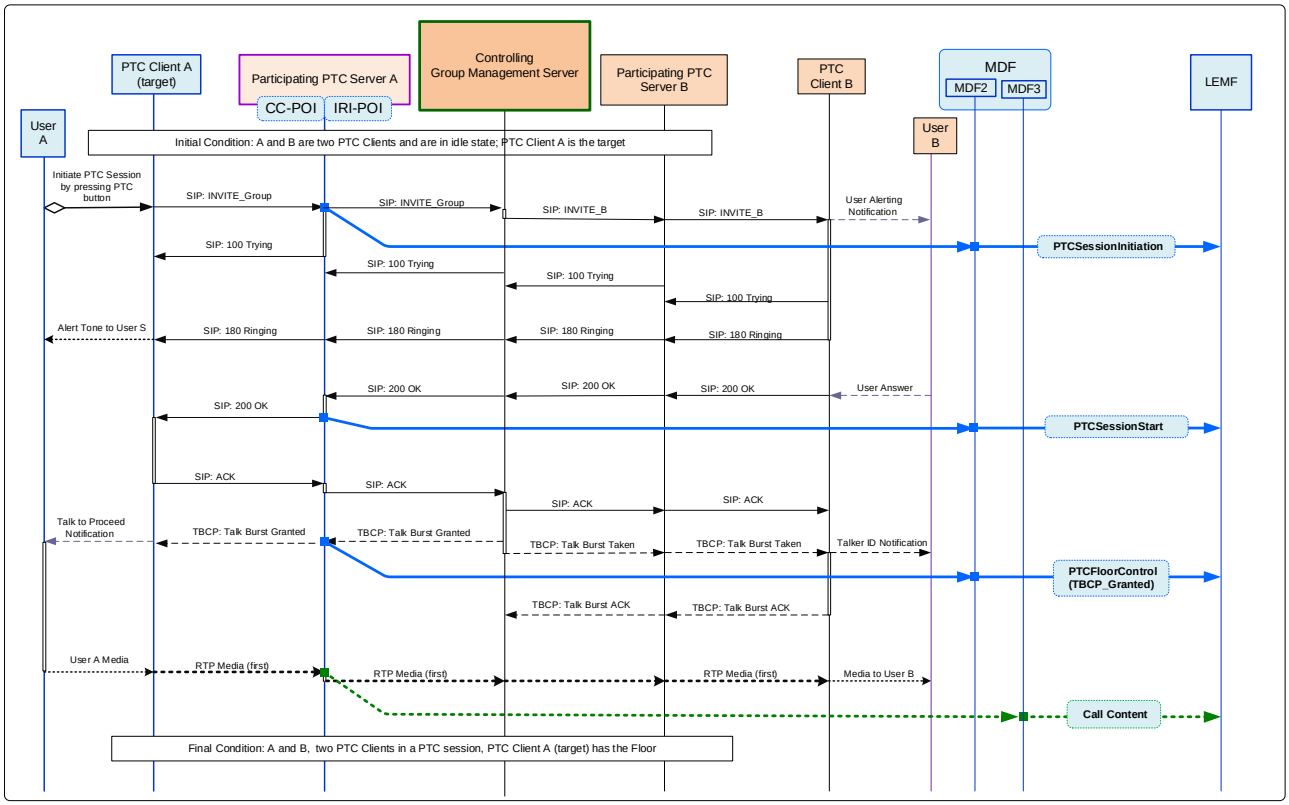


Figure 5.5.2-1: On-demand pre-arranged PTC session initiation from PTC client A (target)

5.5.3 PTC session end

In the flow-diagram shown in figure 5.5.3-1, PTC Client A (target) ends the previously started pre-arranged PTC session.

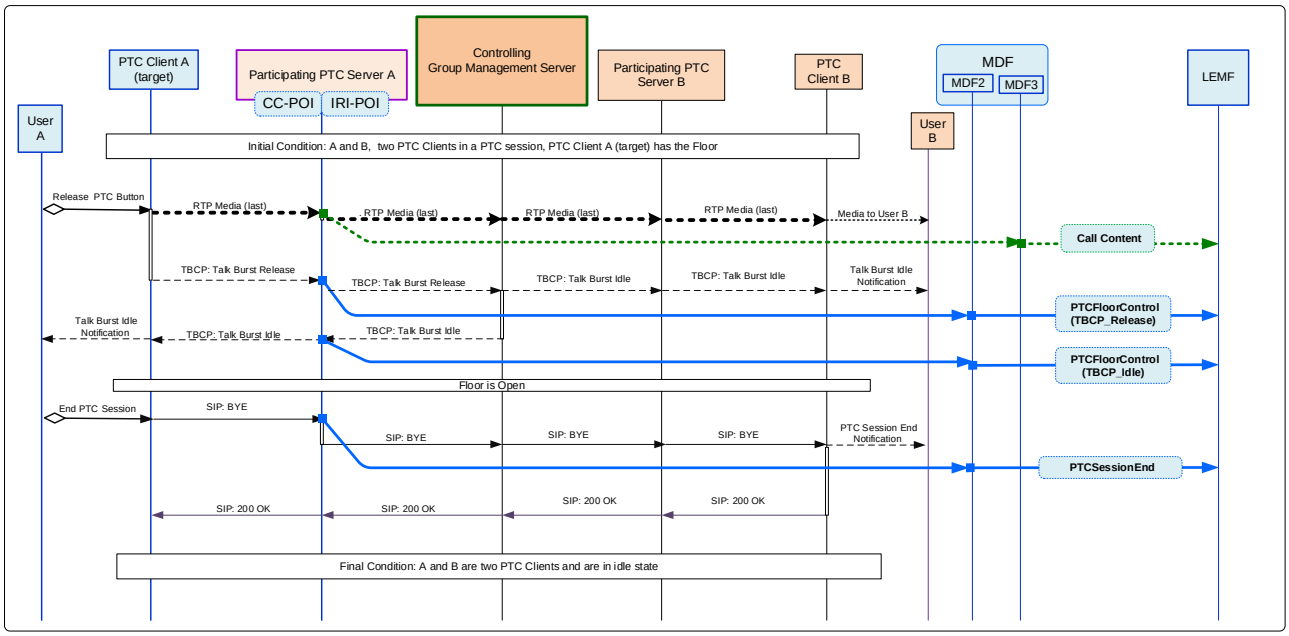


Figure 5.5.3-1: On-demand pre-arranged PTC session from PTC client A (target) is ended

A pre-arranged PTC session ends when the session initiator (PTC Client A in the illustration) ends the session. When any of the destination client may just self-end the session.

5.5.4 Floor control switching

All the floor control flows illustrated in clause 5.3 apply to a pre-arranged PTC session with one difference. The PTC Server A that provides the participating PTC function provides the LI functions for the floor control related events. One example is shown below to illustrate the concept.

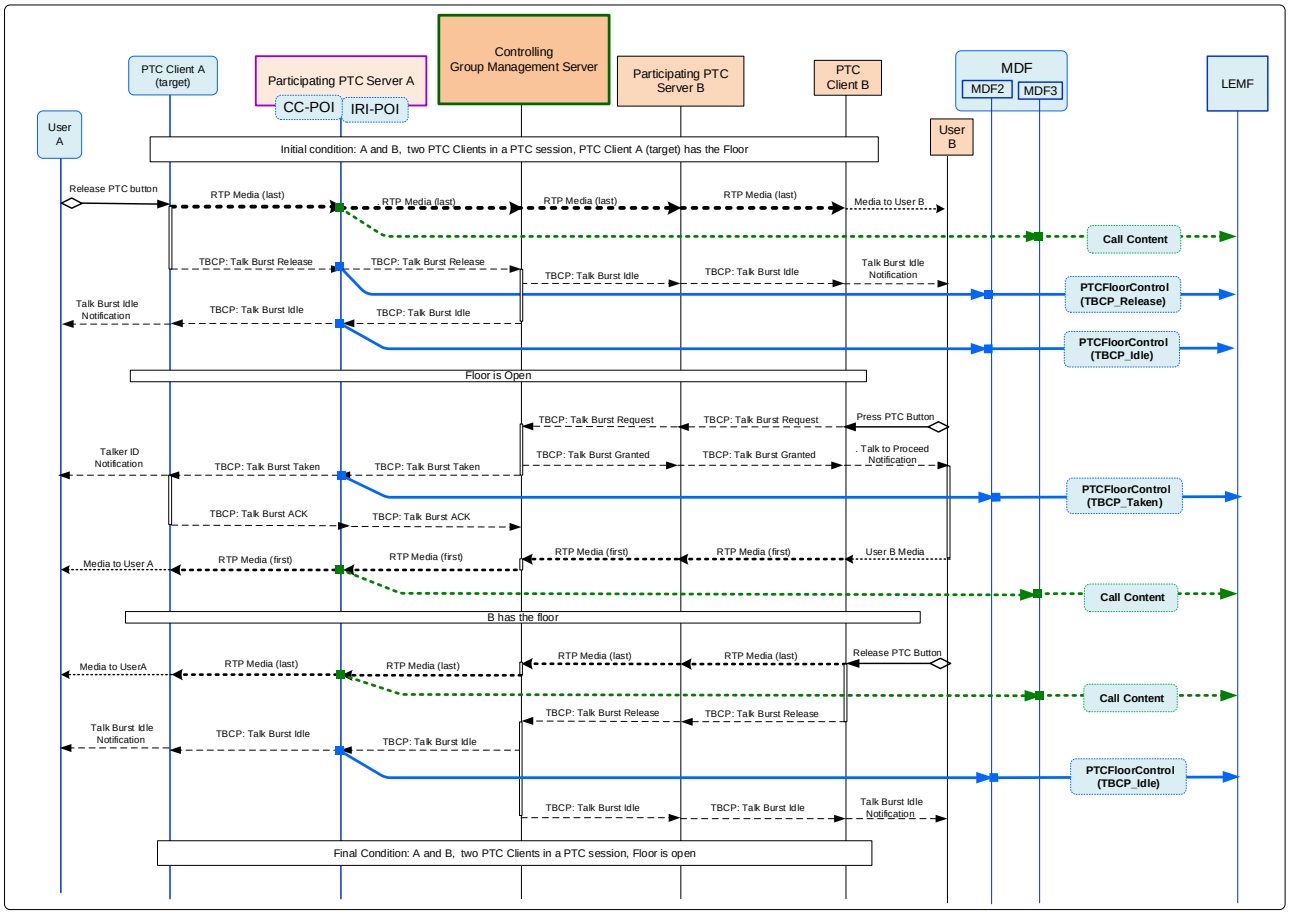


Figure 5.5.4-1: On-demand pre-arranged PTC session from PTC client A (target) – floor control switching

At the end, as shown, the floor is open and can be acquired by either of the two PTC clients. As shown in figure 5.5.4-1, the floor control decisions are made by the Group Management Server that provides the controlling PTC function.

All the TBCP messages pass through the PTC Server A that provides the controlling PTC function. From those TBCP messages, the LI functions in the PTC Server A that provides the participating PTC functions generate the floor control related xIRIs.

5.6 Pre-arranged PTC session with pre-established sessions

5.6.1 General

The flows included in this clause show pre-arranged PTC session from PTC Client A with one or more PTC clients have pre-established sessions. For simplicity of the drawing only one destination PTC Client B is shown. Furthermore, the signalling to all destinations follow the same flow. No separate flows are shown when a destination PTC client is a target since from a destination client's perspective, the flow would be same as 1:1 or an ad-hoc group PTC session.

At the originating end, the PTC Server A provides the participating PTC function for PTC Client A and Group Management Server that has the list of all destination PTC Clients in the group provides the controlling PTC function. The Group Management Server may be hosted by a different provider.

Once the PTC session starts, the floor control variations are same as the on-demand PTC sessions and hence, not included here. In all the illustrations, the TBCP message names are shown from a functional perspective.

5.6.2 PTC session start

In the flow-diagram shown in figure 5.6.2-1, PTC Client A (target) initiates a pre-arranged PTC session with pre-established session. Only one destination client (PTC Client B) is shown. In the illustration, the PTC Client B is also with pre-established session.

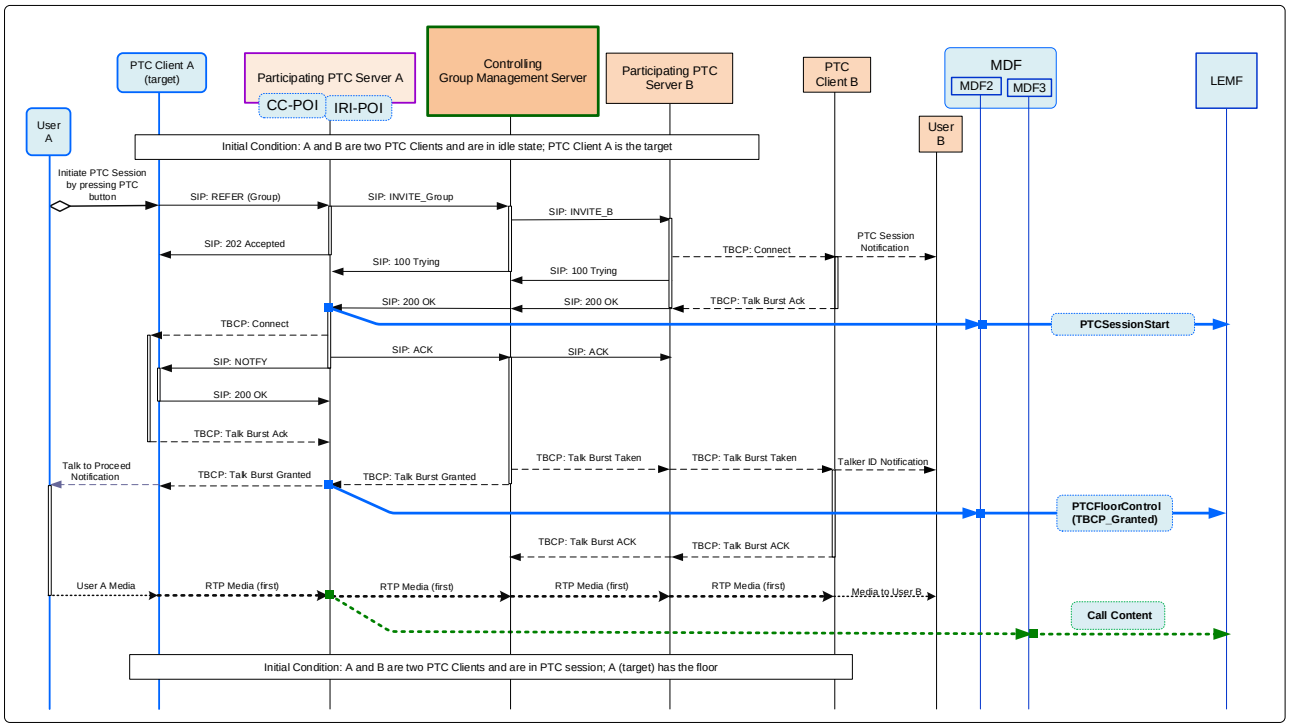


Figure 5.6.2-1: Pre-arranged PTC session from PTC client A (target) with pre-established session

As shown, the LI functions are provided by the IRI-POI present in the PTC Server A that provides the participating PTC function.

5.6.3 PTC session end

In the flow-diagram shown in figure 5.6.3 1, PTC Client A (target) ends the pre-arranged PTC session previously setup with only one destination client (PTC Client B) shown.

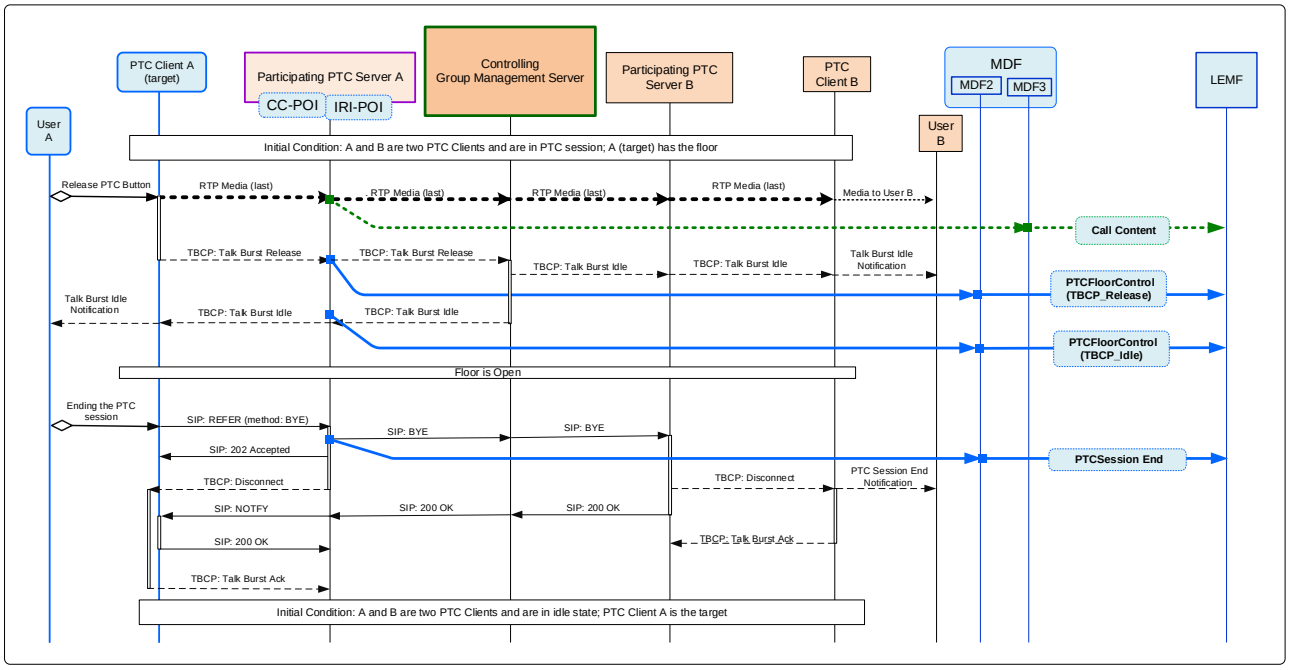


Figure 5.6.3-1: On-demand PTC session to PTC client B (target) is ended

A pre-arranged PTC session ends when the session initiator (PTC Client A in the illustration) ends the session. When any of the destination client may just self-end the session.

5.7 On-demand chat-group PTC sessions with chat-group ID as the target

5.7.1 General

The flows included in this clause show on-demand chat-group PTC sessions with chat-group ID as the target. Each PTC client joins the chat-group session. There can be multiple PTC clients joining a chat-group PTC session. For simplicity of the drawing only two PTC clients joining a chat-group PTC session are shown.

The Chatgroup Server provides the controlling PTC function for a chat-group PTC session. The Chatgroup Server may be hosted by a different provider.

Each PTC client joining a chat-group PTC session has a separate PTC Server providing the participating PTC function.

When the chat-group ID is the target, the Chatgroup Server provides the LI functions.

In all the illustrations, the TBCP message names are shown from a functional perspective.

5.7.2 Chat-group PTC session start

In the flow-diagram shown in figure 5.7.2-1, PTC Client A (first) and PTC Client B (second) join a chat-group PTC session. The chat-group ID is the target.

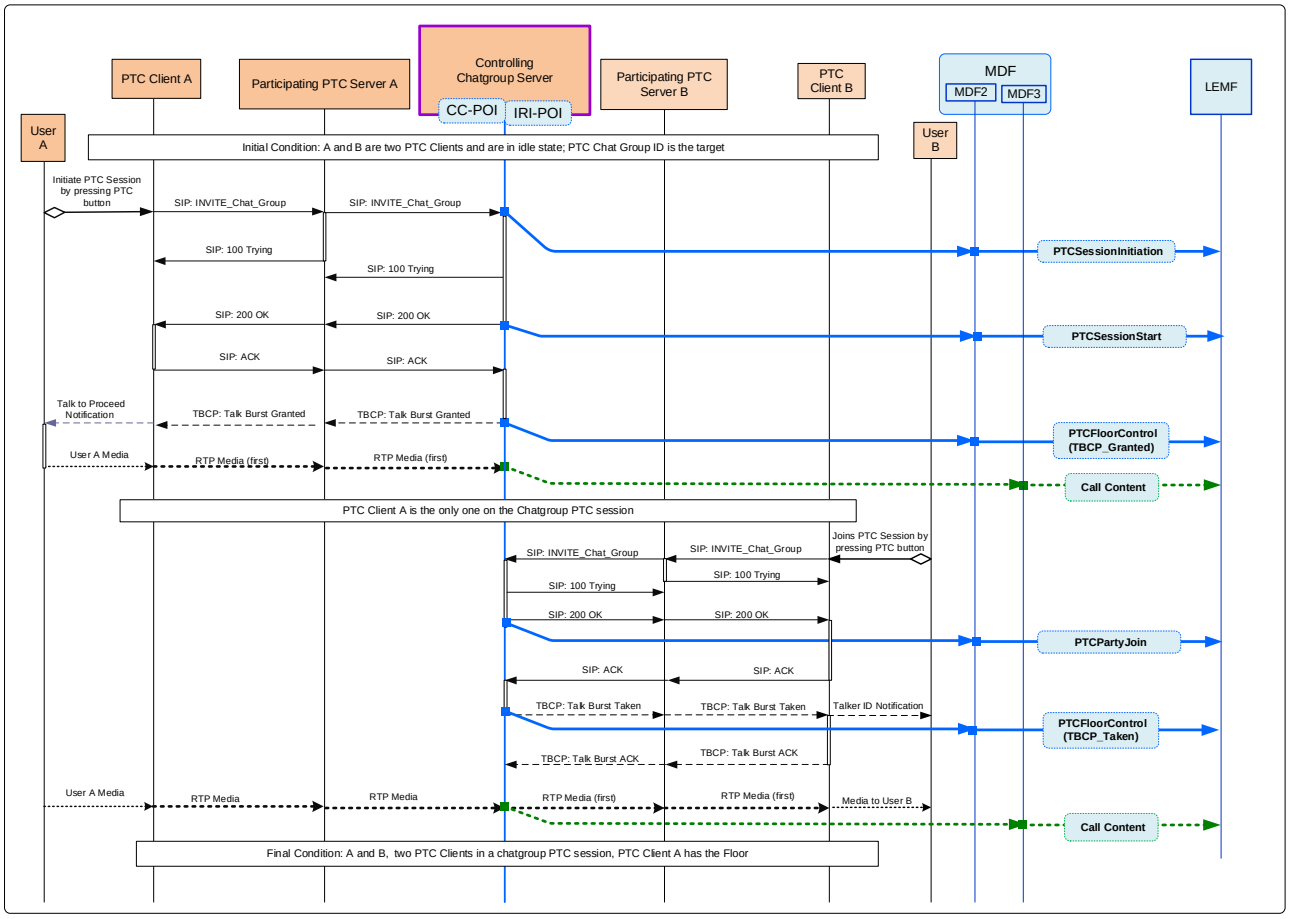


Figure 5.7.2-1: On-demand chat-group PTC session, chat-group ID is the target

As shown, the LI functions are provided by the IRI-POI and CC-POI present in the Chatgroup Server.

5.7.3 Chat-group PTC session end

In the flow-diagram shown in figure 5.7.3-1, PTC Client A (first) and then PTC Client B (last) end the chat-group PTC session.

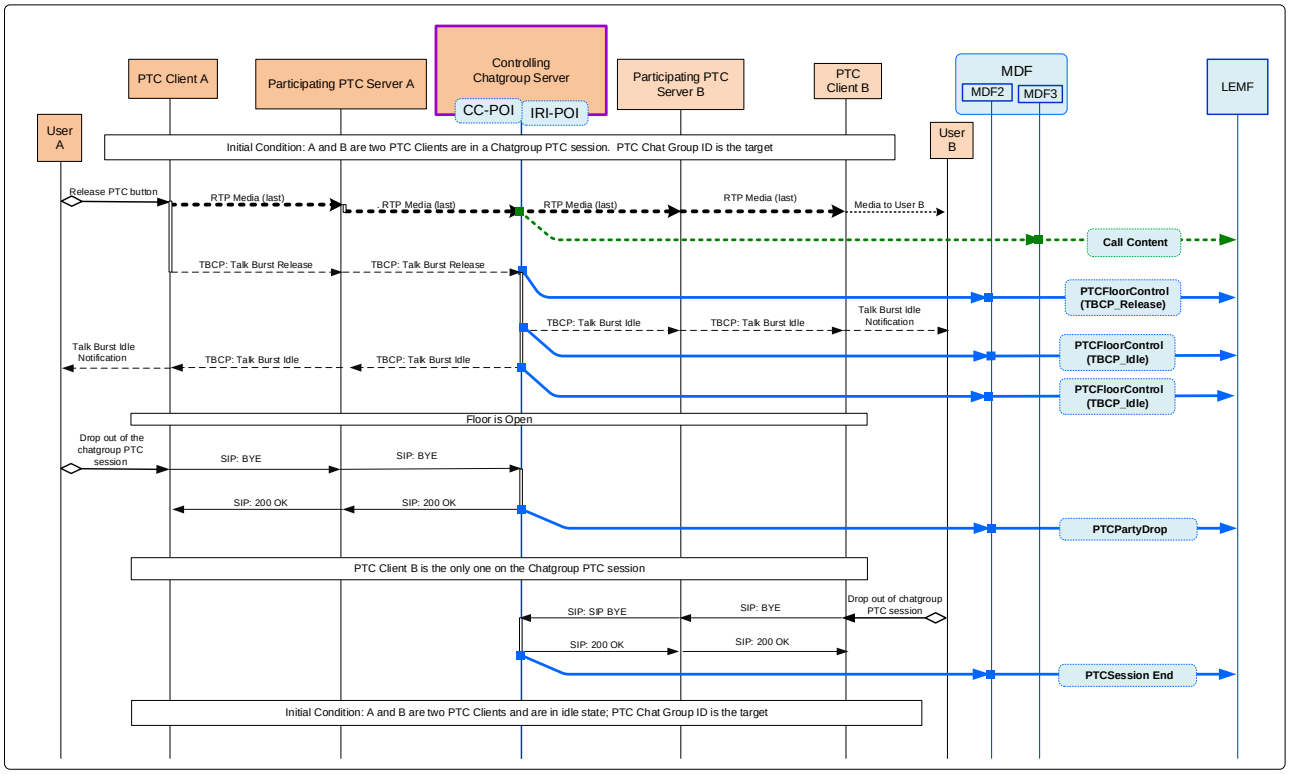


Figure 5.7.3-1: On-demand chat-group PTC session ends - chat-group ID is the target

As shown in figure 5.7.3-1, when a PTC client drops out of a chat-group PTC session, the session ends for that PTC client. A chat-group PTC session ends, when all the PTC clients end their chat-group PTC sessions.

5.7.4 Floor control procedures

5.7.4.1 General

All the floor control procedures illustrated in clause 5.3 are shown here to show the LI functions in a chat-group PTC session. The same procedures apply to all PTC clients.

5.7.4.2 Flow-control switching

The flow-diagram shown in figure 5.7.4.2-1 illustrates the floor-control switching.

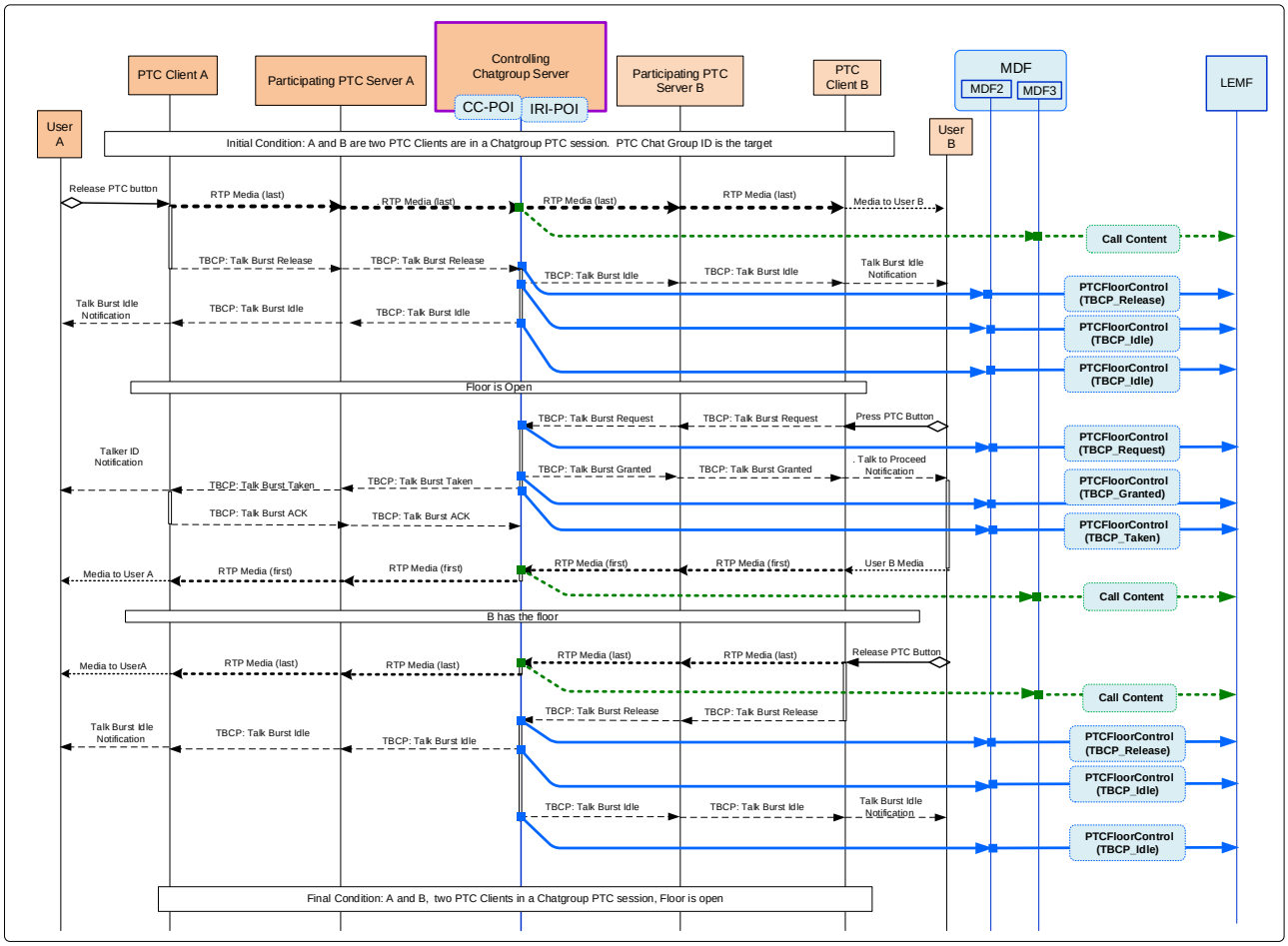


Figure 5.7.4.2-1: Chat-group PTC session – floor control switching

At the end, as shown, the floor is open and can be acquired by any of the PTC clients still on the PTC session. As shown, the floor control related xIRIs are generated from the IRI-POI present in the Chatgroup Server.

The xIRIs are generated as specified in TS 33.128 [4].

5.7.4.3 Floor control request queued

The flow-diagram shown in figure 5.7.4.3-1 illustrates an example of floor-control request is being queued.

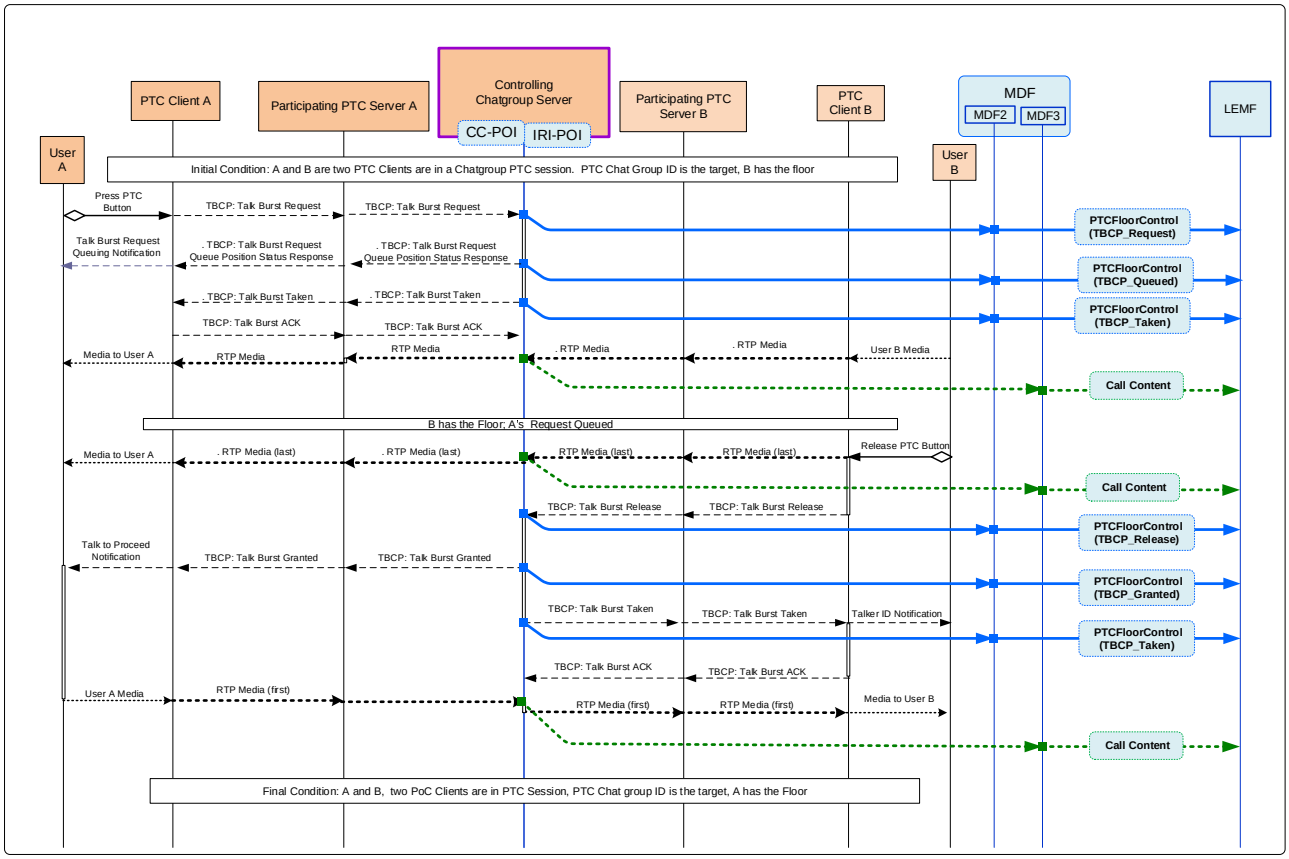


Figure 5.7.4.3-1: Chat-group PTC session –floor control request is queued

The illustration shows that a floor-control request received when the floor is not open can be queued. Such a queuing can happen for floor control request received from any of the PTC clients present on the chat-group PTC session. In the illustration, a floor control request from PTC Client A is queued.

The xIRIs are generated as specified in TS 33.128 [4].

5.7.4.4 Floor control request denied

The flow-diagram shown in figure 5.7.4.4-1 illustrates an example of floor-control request is being denied.

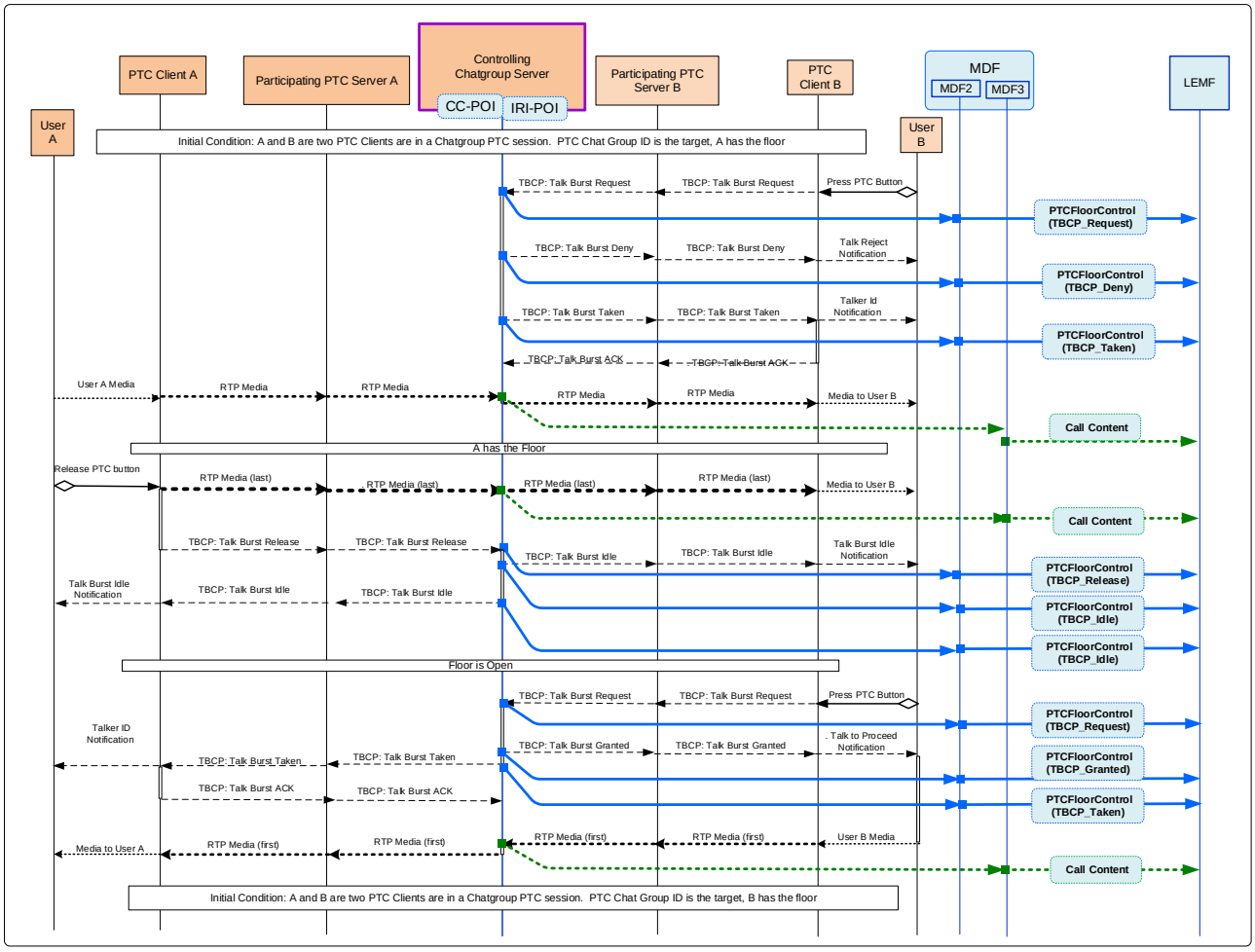


Figure 5.7.4.4-1: Chat-group PTC session – floor control request is denied

The illustration shows that a floor-control request received when the floor is not open can be denied. Such a denial can happen for floor control request received from any of the PTC clients present on the chat-group PTC session.

The xIRIs are generated as specified in TS 33.128 [4].

5.7.4.5 Floor control request revoked

The flow-diagram shown in figure 5.7.4.5-1 illustrates an example of floor-control request is being revoked.

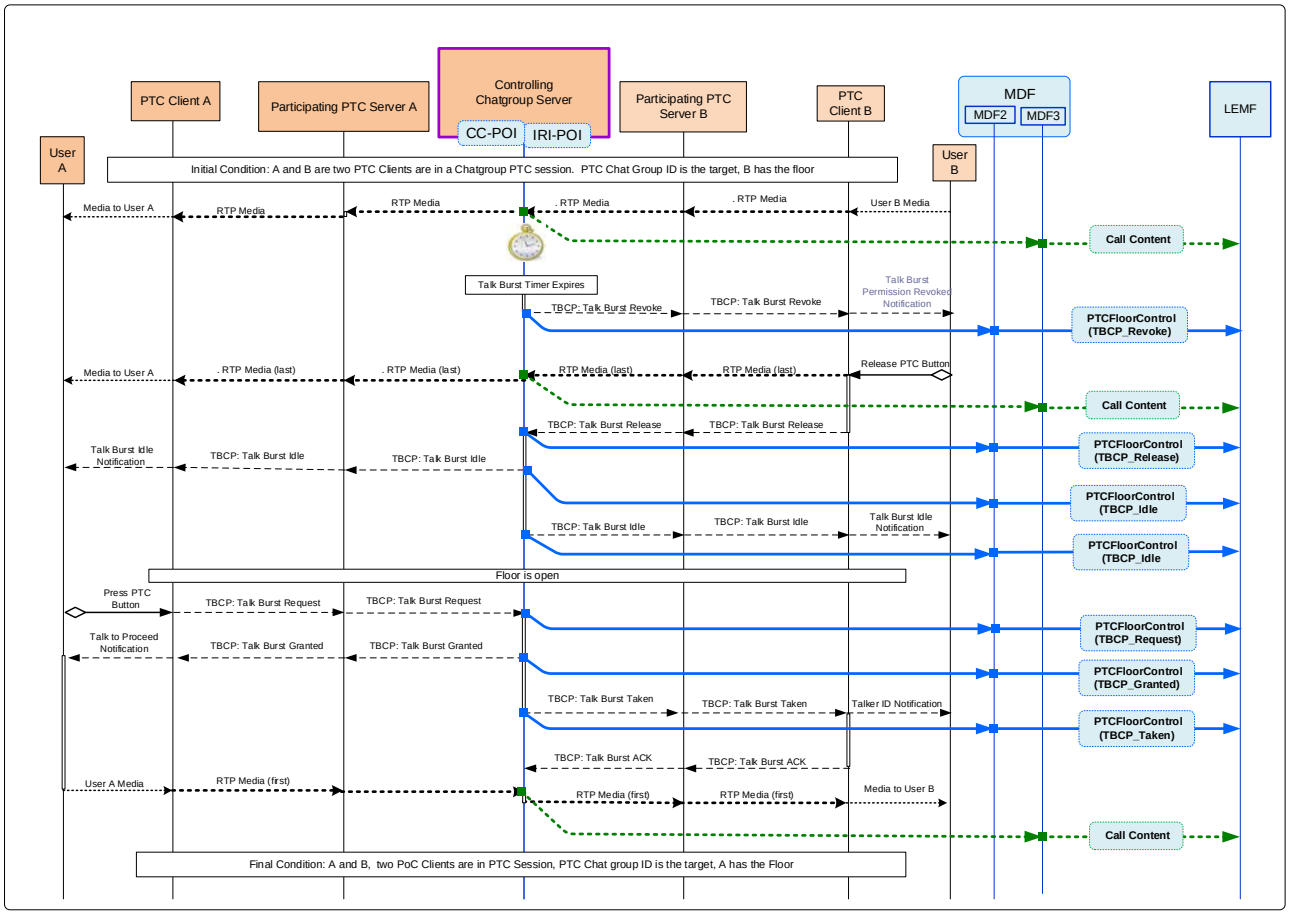


Figure 5.7.4.5-1: Chat-group PTC session – floor control request is revoked

The illustration shows that a floor-control that a PTC client has can be revoked. Such a revocation can happen when any of the PTC clients have the floor on a chat-group PTC session.

The xIRIs are generated as specified in TS 33.128 [4].

5.8 Chat-group PTC sessions with clients having pre-established sessions

5.8.1 General

The flows included in this clause show chat-group PTC session with one or more PTC clients joining the chat-group session having the pre-established sessions.

In the illustration, each PTC client joins the chat-group session. There can be multiple PTC clients joining a chat-group PTC session. For simplicity of the drawing only two PTC clients joining a chat-group PTC session are shown.

The Chatgroup Server provides the controlling PTC function for a chat-group PTC session. The Chatgroup Server may be hosted by a different provider.

Each PTC client joining a chat-group PTC session has a separate PTC Server providing the participating PTC function.

When the chat-group ID is the target, the Chatgroup Server provides the LI functions.

In all the illustrations, the TBCP message names are shown from a functional perspective.

Once the chat-group session is on, the floor control variations are same as illustrated for the on-demand chat-group PTC sessions and hence, not included here. In all the illustrations, the TBCP message names are shown from a functional perspective.

5.8.2 PTC session start

In the flow-diagram shown in figure 5.8.2-1, PTC Client A (first) and PTC Client B (second) join a chat-group PTC session. The chat-group ID is the target. In the illustration, both PTC clients have pre-established sessions.

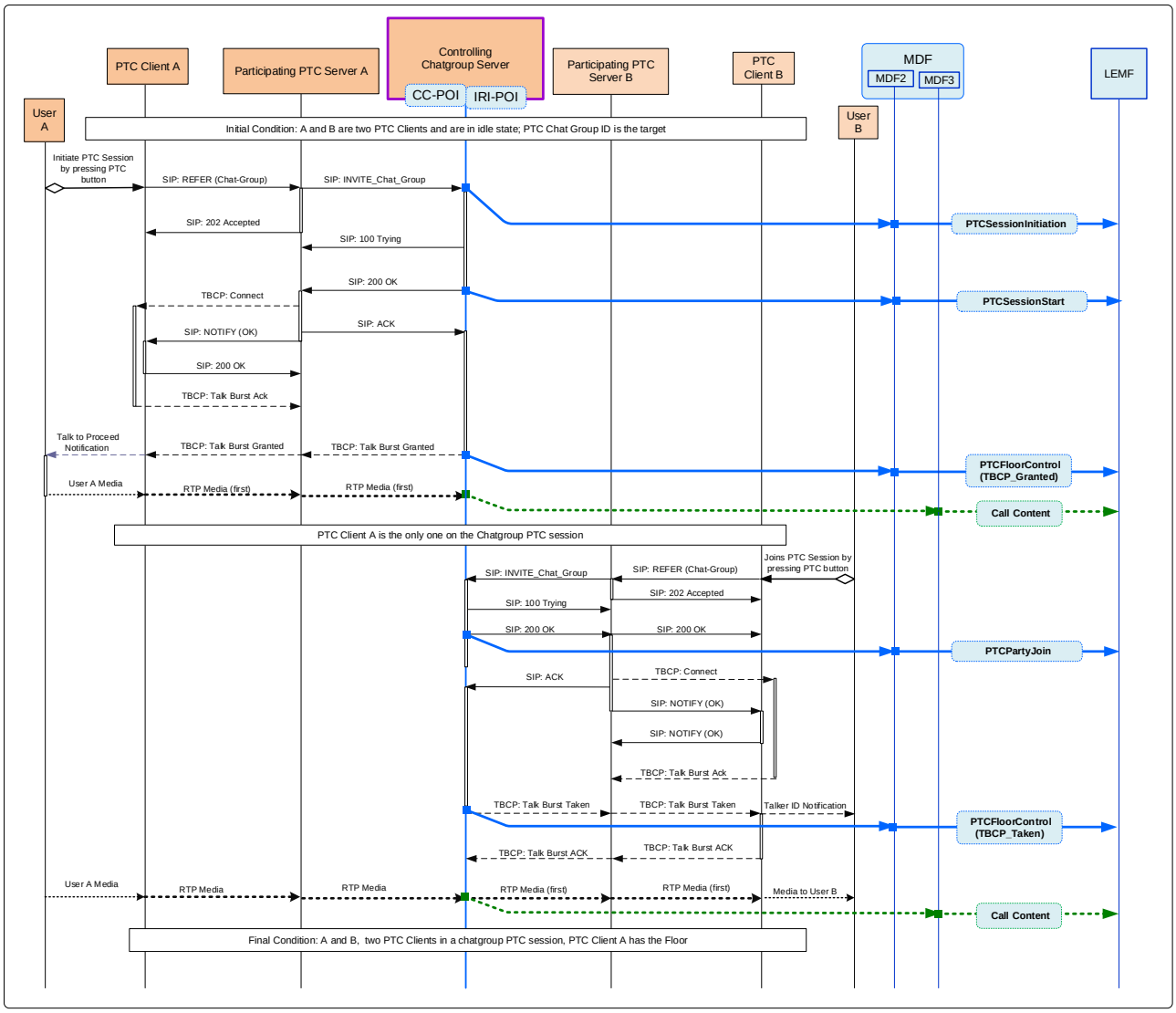


Figure 5.8.2-1: Chat-group PTC session, PTC clients have pre-established sessions

As shown, the LI functions are provided by the IRI-POI and CC-POI present in the Chatgroup Server.

5.8.3 PTC session end

In the flow-diagram shown in figure 5.8.3-1, PTC Client A (first) and then PTC Client B (last) end the chat-group PTC session.

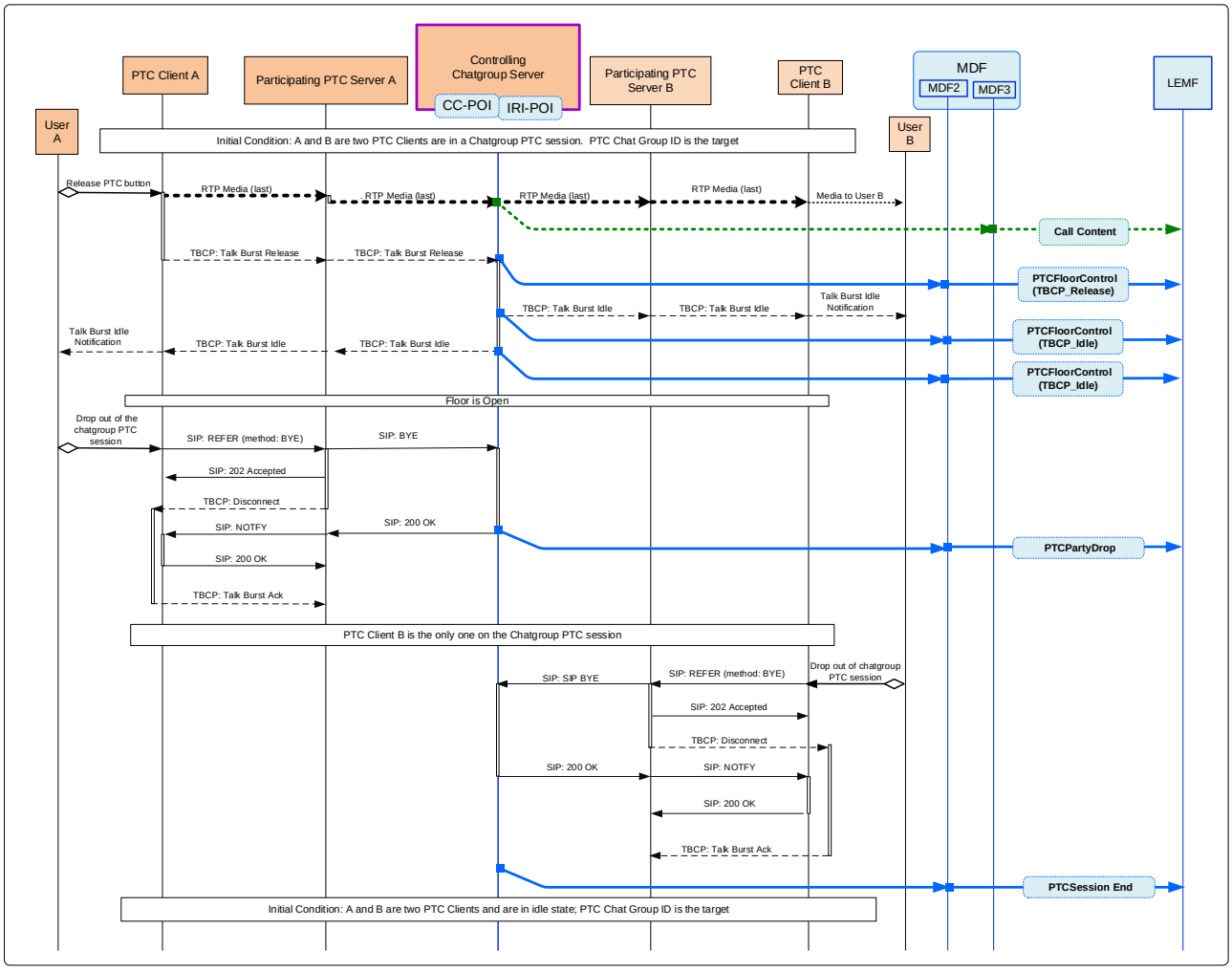


Figure 5.8.3-1: Chat-group PTC session is ended, PTC clients have pre-established sessions

As shown in figure 5.8.3-1, when a PTC Clients drops out of a chat-group PTC session, the session ends for that PTC client. A chat-group PTC session ends, when all the PTC clients end their chat-group PTC sessions.

5.9 On-demand chat-group PTC sessions with PTC client as the target

5.9.1 General

The flows included in this clause show on-demand chat-group PTC sessions with one of the PTC clients joining the chat-group PTC session as the target. In the illustrations, each PTC client joins the chat-group session. There can be multiple PTC clients joining a chat-group PTC session. For simplicity of the drawing only two PTC clients joining a chat-group PTC session are shown.

The Chatgroup Server provides the controlling PTC function for a chat-group PTC session. The Chatgroup Server may be hosted by a different provider.

Each PTC client joining a chat-group PTC session has a separate PTC Server providing the participating PTC function. The server associated with the PTC client as a target provides the LI functions.

In all the illustrations, the TBCP message names are shown from a functional perspective.

5.9.2 Chat-group PTC session start

In the flow-diagram shown in figure 5.9.2-1, PTC Client A (first) and PTC Client B (second) join a chat-group PTC session. The PTC Client A is the target.

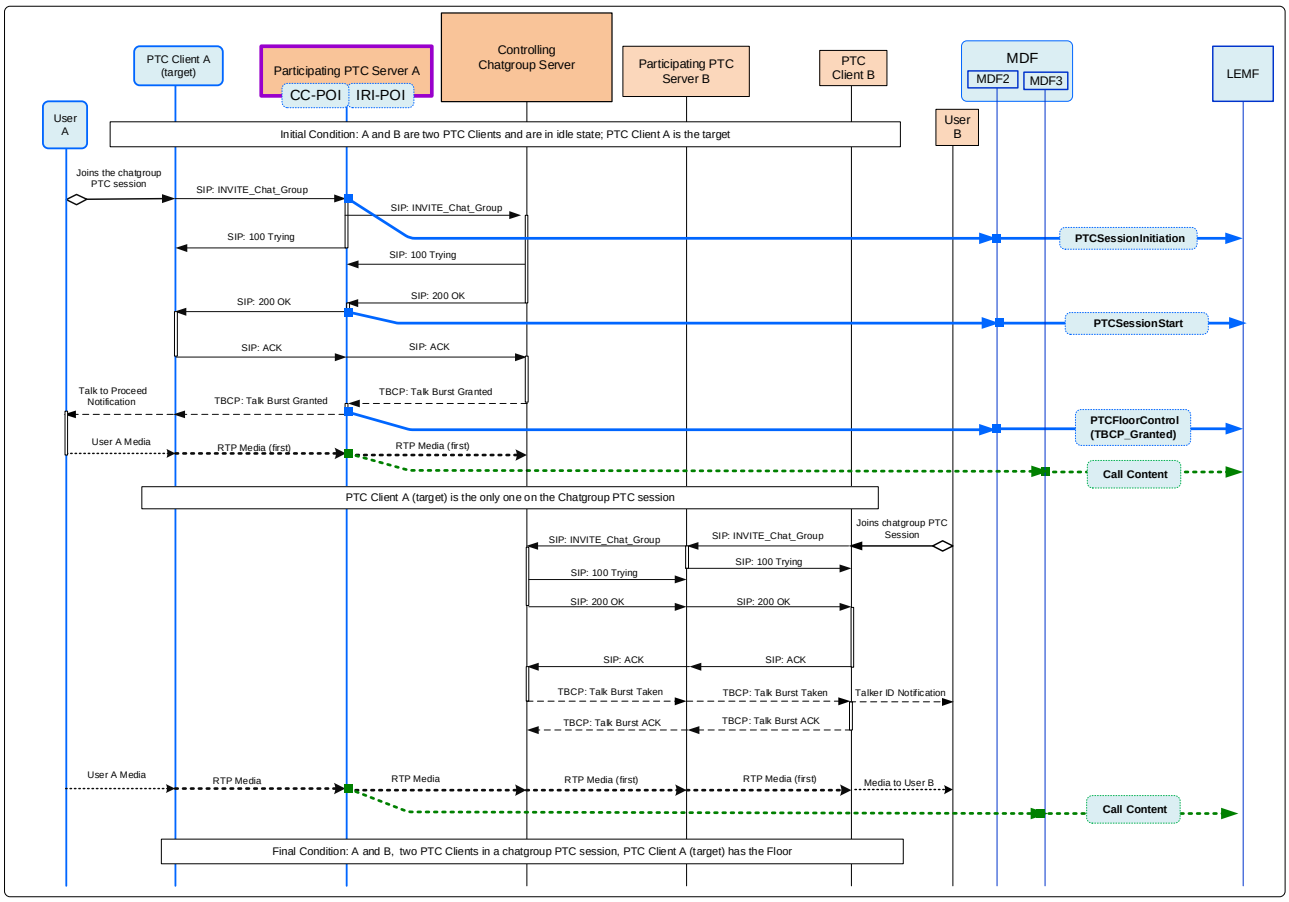


Figure 5.9.2-1: On-demand chat-group PTC session, PTC client A is the target

As shown, the LI functions are provided by the IRI-POI and CC-POI present in the PTC Server A that provides the participating PTC function to PTC Client A, the target.

5.9.3 Chat-group PTC session end – target releases first

In the flow-diagram shown in figure 5.9.3-1, PTC Client A (first) and then PTC Client B (last) end the chat-group PTC session. PTC Client A is the target and drops out of the chat-group PTC session first.

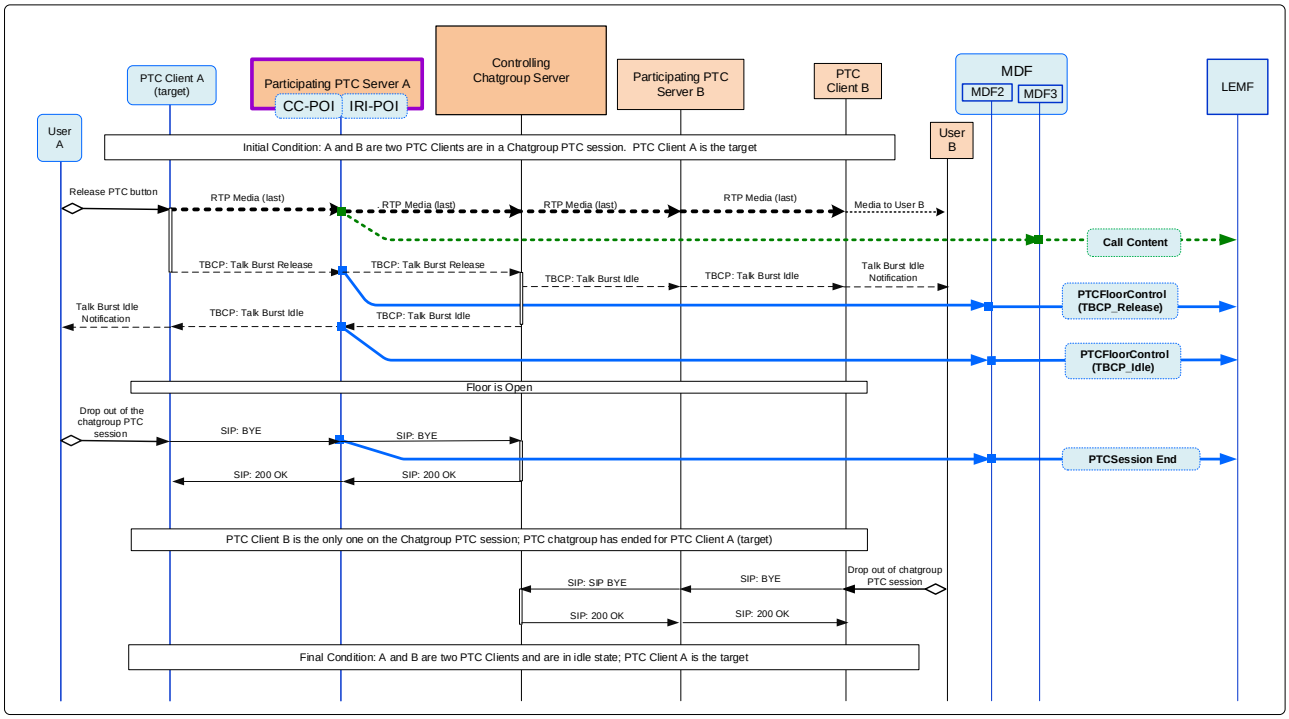


Figure 5.9.3-1: On-demand chat-group PTC session ends - PTC client A is the target

As shown in figure 5.9.3-1, when a PTC client drops out of a chat-group PTC session, the session ends for that PTC client. A chat-group PTC session ends, when all the PTC clients end their chat-group PTC sessions.

For the target PTC Client A, the chat-group PTC session ends even though the chat-group PTC session still remains on because of the other PTC clients still being on the chat-group PTC session.

5.9.4 Chat-group PTC session end – target releases last

In the flow-diagram shown in figure 5.9.4-1, PTC Client A (last) and then PTC Client B (first) end the chat-group PTC session. PTC Client A is the target and drops out of the chat-group PTC session last.

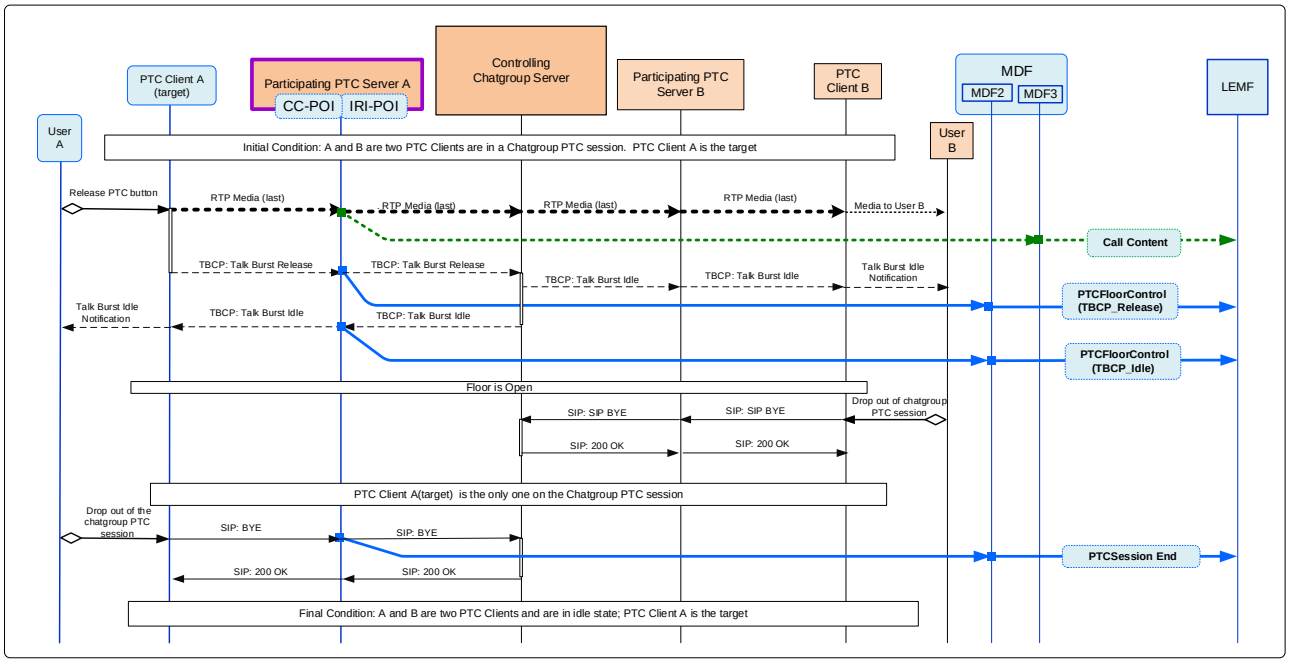


Figure 5.9.4-1: On-demand chat-group PTC session ends - PTC client A is the target

As shown in figure 5.9.4-1, when a PTC client drops out of a chat-group PTC session, the session ends for that PTC client. A chat-group PTC session ends, when all the PTC clients end their chat-group PTC sessions.

In this illustration, the target PTC Client A drops out of the chat-group PTC session last.

5.9.5 Floor control procedures

5.9.5.1 General

All the floor control procedures illustrated in clause 5.3 are shown here to show the LI functions in a chat-group PTC session. The same procedures apply to all PTC clients.

5.9.5.2 Flow-control switching

The flow-diagram shown in figure 5.9.5.2-1 illustrates the floor-control switching.

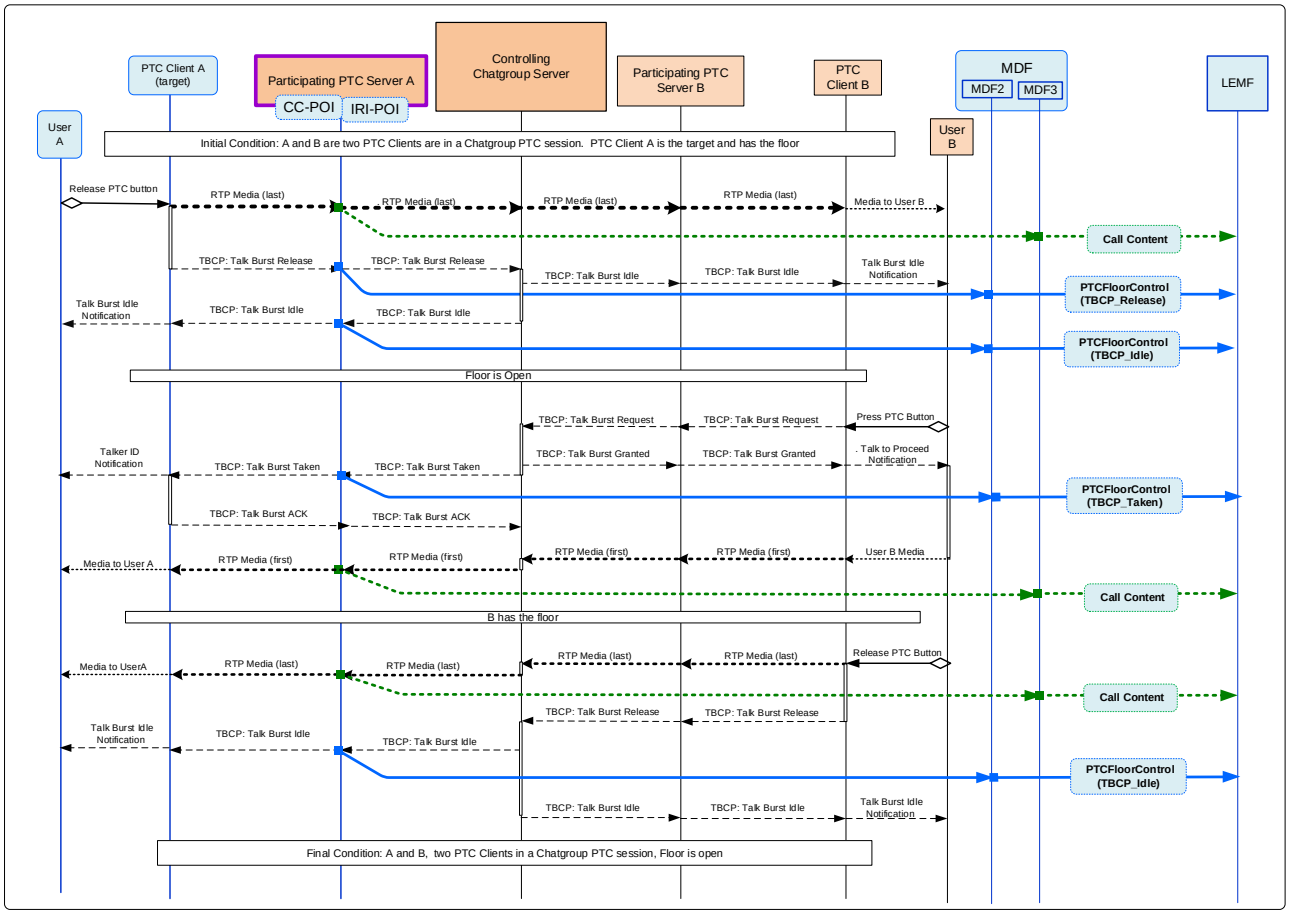


Figure 5.9.5.2-1: Chat-group PTC session – floor control switching

At the end, as shown, the floor is open and can be acquired by any of the PTC clients still on the PTC session. As shown, the floor control related xIRIs are generated from the IRI-POI present in the PTC Server A that provides the participating PTC function to the PTC Client A, the target.

The xIRIs are generated as specified in TS 33.128 [4].

5.9.5.3 Floor control request queued

The flow-diagram shown in figure 5.9.5.3-1 illustrates an example of floor-control request is being queued.

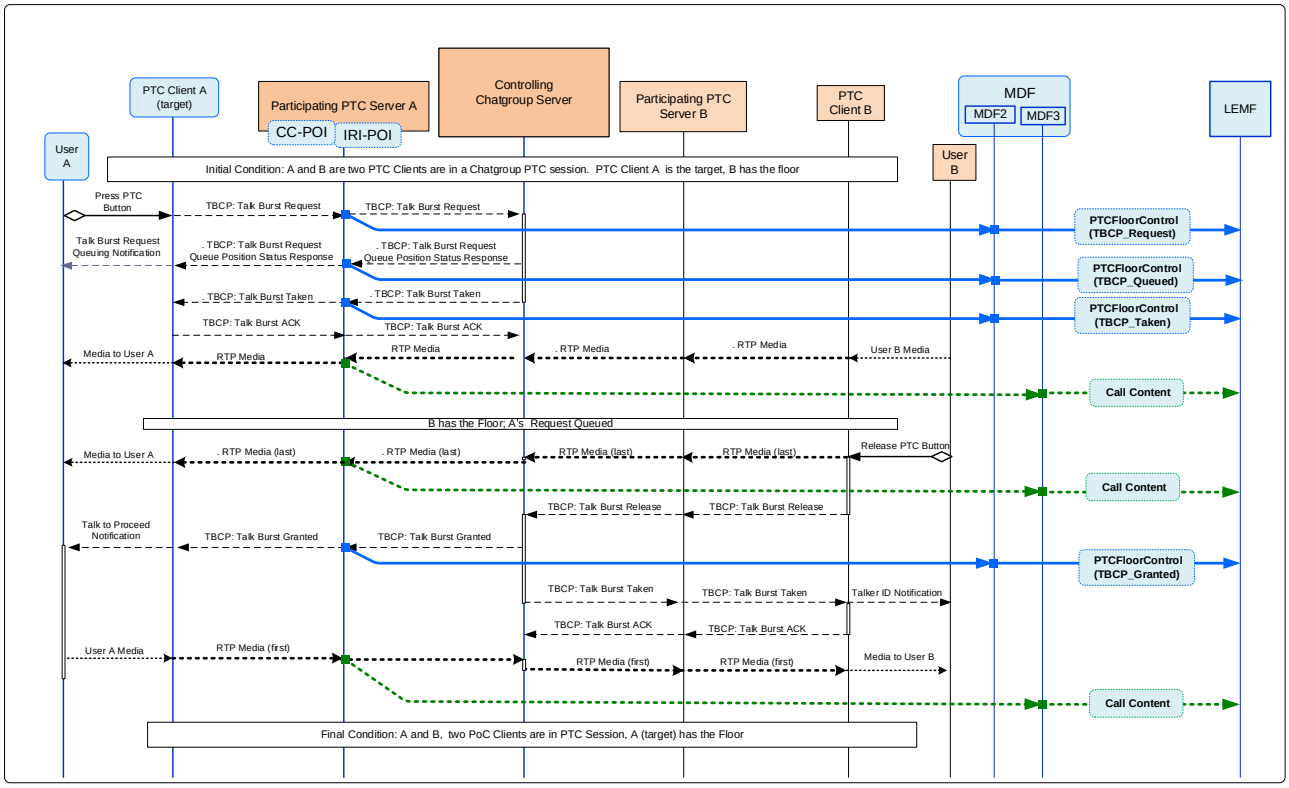


Figure 5.9.5.3-1: Chat-group PTC session –floor control request is queued

The illustration shows that a floor-control request received when the floor is not open can be queued. Such a queuing can happen for floor control request received from any of the PTC clients present on the chat-group PTC session, however, reporting to the LEMF happens if and only if the corresponding TBCP message passes the PTC server that provides the participating PTC function to the target.

In the illustration, a floor control request from PTC Client A is queued and the TBCP message related to the queuing passes through the PTC Server A and hence, an xIRI is sent.

The xIRIs are generated as specified in TS 33.128 [4].

5.9.5.4 Floor control request denied

The flow-diagram shown in figure 5.9.5.4-1 illustrates an example of floor-control request is being denied.

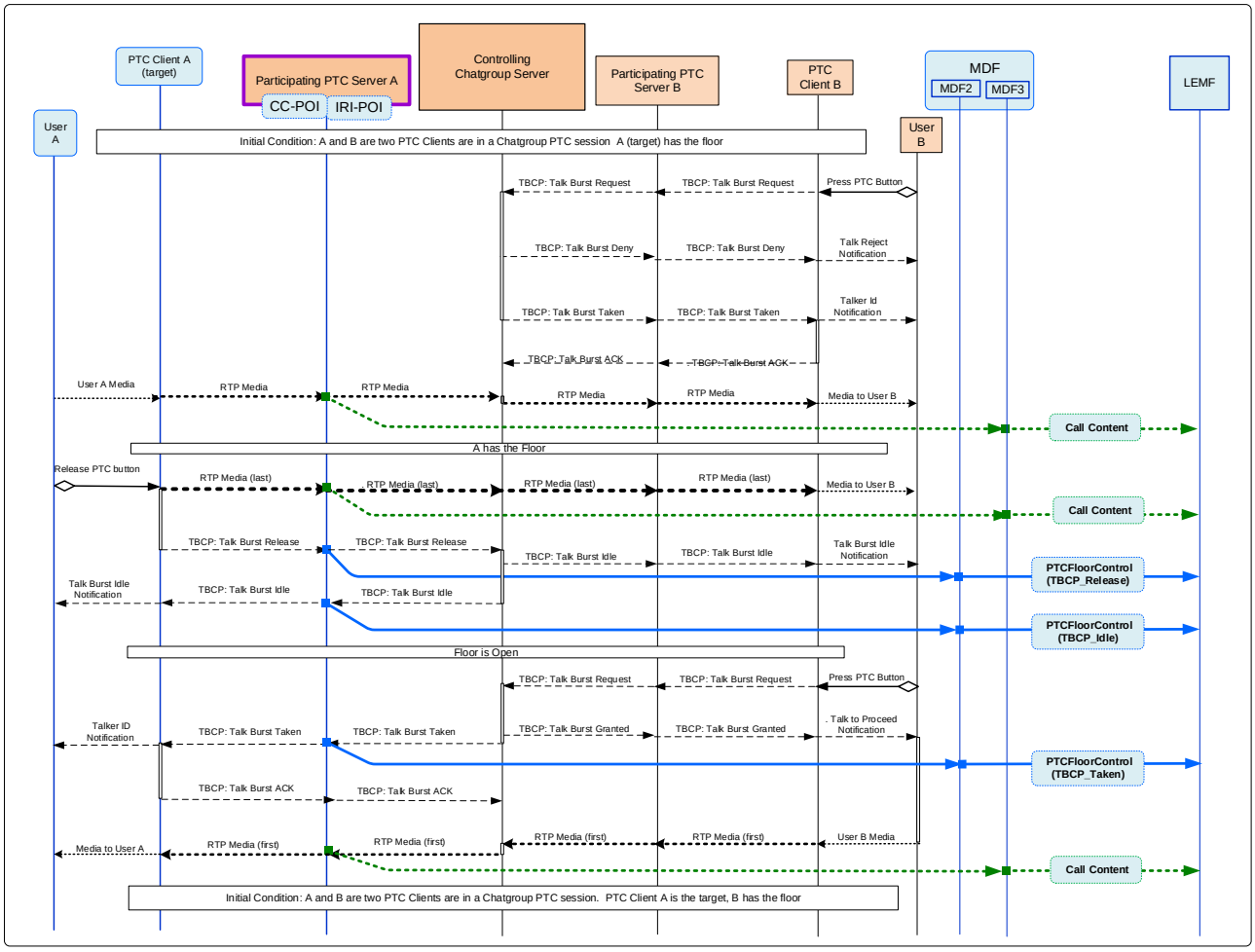


Figure 5.9.5.4-1: Chat-group PTC session – floor control request is denied

The illustration shows that a floor-control request received when the floor is not open can be denied. Such a denial can happen for floor control request received from any of the PTC clients present on the chat-group PTC session, however, reporting to the LEMF happens if and only if the corresponding TBCP message passes the PTC server that provides the participating PTC function to the target.

In the illustration, a floor control request from PTC Client B is denied and the TBCP message related to the denial does not pass through the PTC Server A that provides the participating PTC function to PTC Client A, the target, and hence, no xIRI related to the denial is generated.

The xIRIs are generated as specified in TS 33.128 [4].

5.9.5.5 Floor control request revoked

The flow-diagram shown in figure 5.9.5.5-1 illustrates an example of floor-control request is being revoked.

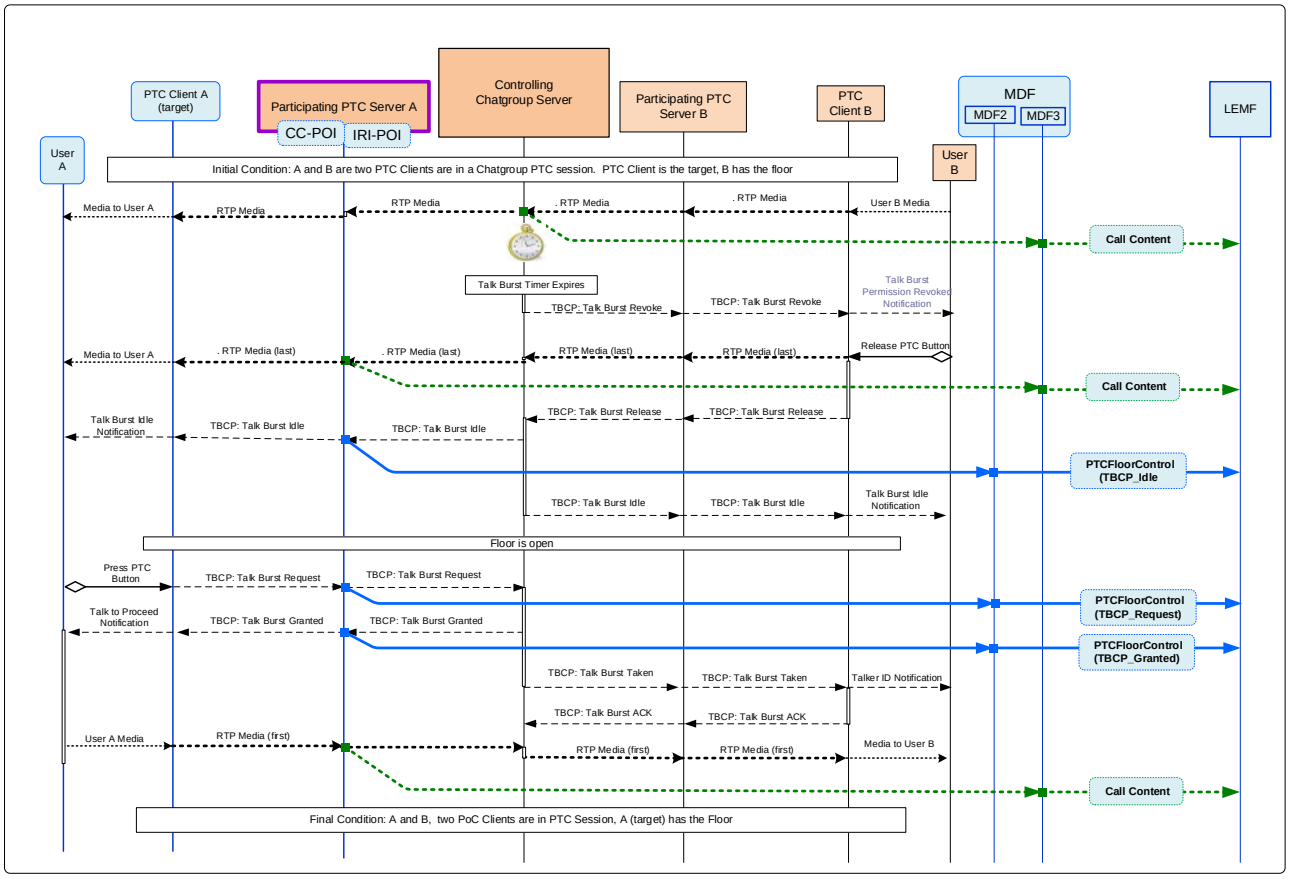


Figure 5.9.5.5-1: Chat-group PTC session – floor control request is revoked

The illustration shows that a floor-control that a PTC client has can be revoked. Such a revocation can happen when any of the PTC clients have the floor on a chat-group PTC session, however, reporting to the LEMF happens if and only if the corresponding TBCP message passes the PTC server that provides the participating PTC function to the target.

In the illustration, a floor control for PTC Client B is revoked and the TBCP message related to the revocation does not pass through the PTC Server A that provides the participating PTC function to PTC Client A (the target), and hence, no xIRI related to the revocation is generated.

The xIRIs are generated as specified in TS 33.128 [4].

5.10 Chat-group PTC sessions with clients having pre-established sessions, one of the PTC client is a target

5.10.1 General

The flows included in this clause show chat-group PTC session with one or more PTC clients joining the chat-group session having the pre-established sessions. One of those PTC clients is a target.

In the illustrations, each PTC client joins the chat-group session. There can be multiple PTC clients joining a chat-group PTC session. For simplicity of the drawing only two PTC clients joining a chat-group PTC session are shown.

The Chatgroup Server provides the controlling PTC function for a chat-group PTC session. The Chatgroup Server may be hosted by a different provider.

Each PTC client joining a chat-group PTC session has a separate PTC Server providing the participating PTC function. The server associated with the PTC client as a target provides the LI functions.

In all the illustrations, the TBCP message names are shown from a functional perspective.

Once the chat-group session is on, the floor control variations are same as illustrated for the on-demand chat-group PTC sessions and hence, not included here. In all the illustrations, the TBCP message names are shown from a functional perspective.

5.10.2 PTC session start

In the flow-diagram shown in figure 5.10.2-1, PTC Client A (first) and PTC Client B (second) join a chat-group PTC session. The PTC Client A is the target. In the illustration, both PTC clients have pre-established sessions.

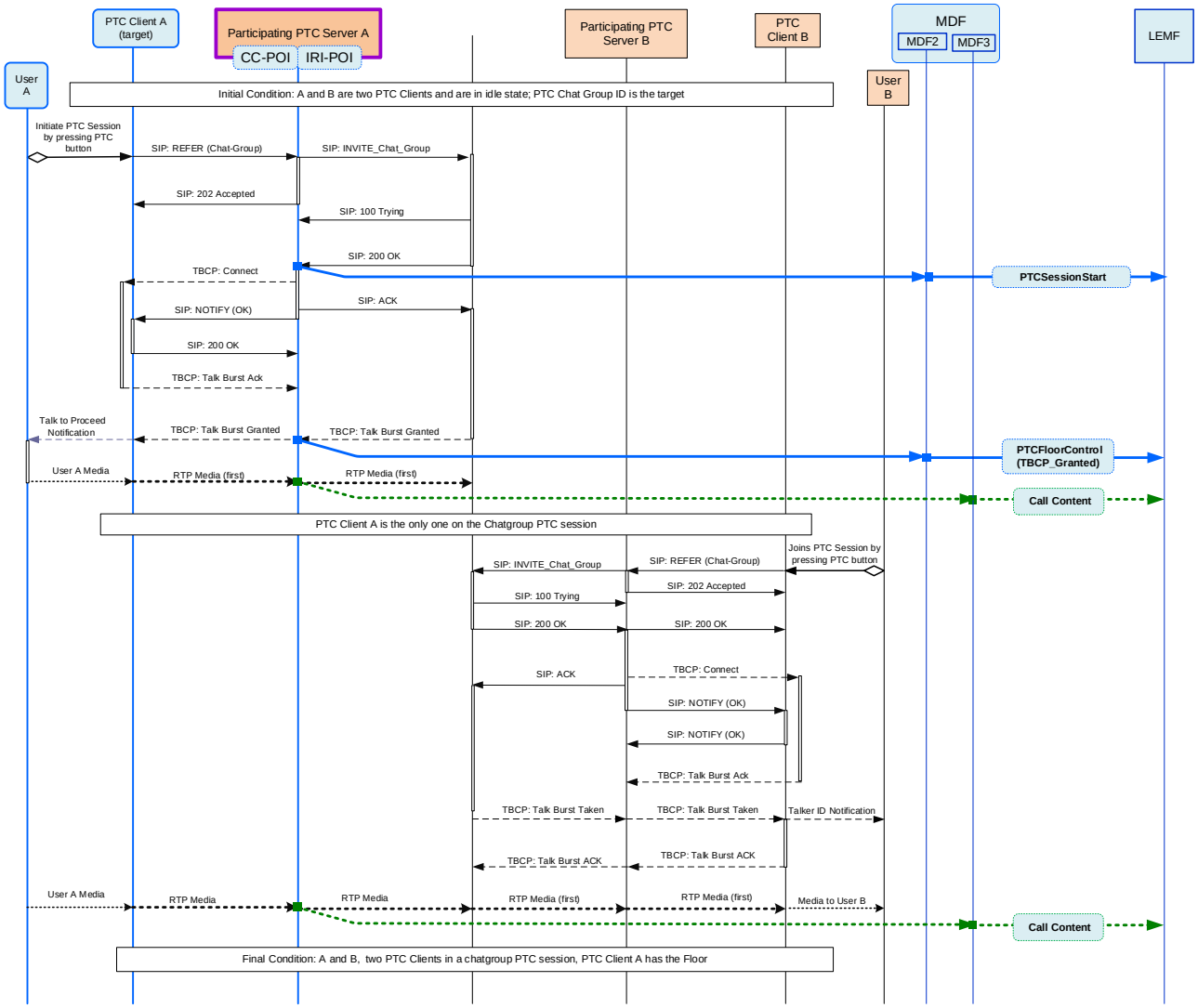


Figure 5.10.2-1: Chat-group PTC session, PTC clients have pre-established sessions

As shown, the LI functions are provided by the IRI-POI and CC-POI present in the PTC Server A that provides the participating PTC function for the PTC client A (target).

5.10.3 PTC session end – target releases first

In the flow-diagram shown in figure 5.10.3-1, PTC Client A (first) and then PTC Client B (last) end the chat-group PTC session. PTC Client A is the target and drops out of the chat-group PTC session first.

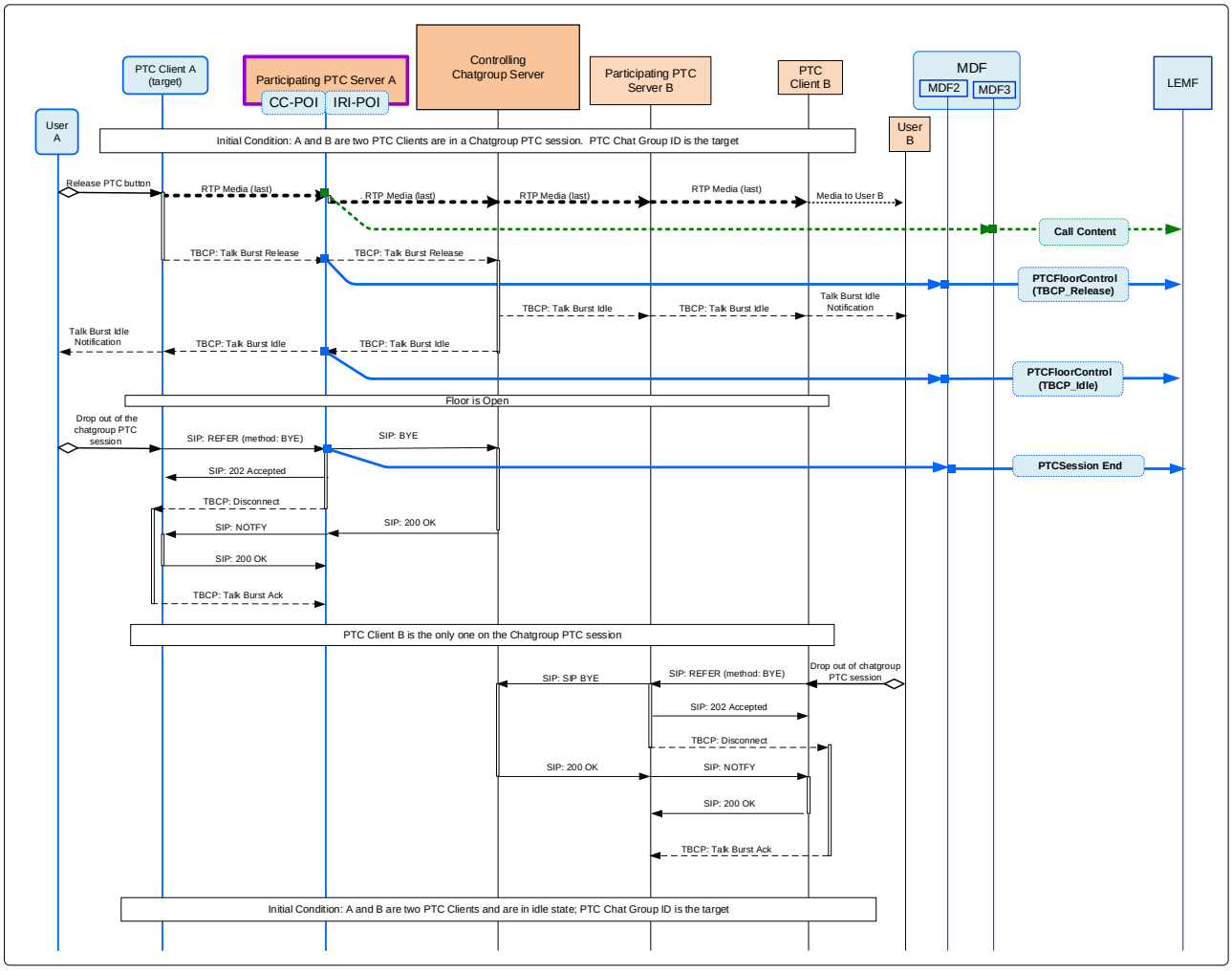


Figure 5.10.3-1: Chat-group PTC session is ended, PTC clients have pre-established sessions

As shown in figure 5.10.3-1, when a PTC client drops out of a chat-group PTC session, the session ends for that PTC client. A chat-group PTC session ends, when all the PTC clients end their chat-group PTC sessions.

For the target PTC Client A, the chat-group PTC session ends even though the chat-group PTC session still remains on because of the other PTC clients still being on the chat-group PTC session.

5.10.4 PTC session end – target releases last

In the flow-diagram shown in figure 5.10.4-1, PTC Client A (first) and then PTC Client B (last) end the chat-group PTC session. PTC Client A is the target and drops out of the chat-group PTC session last.

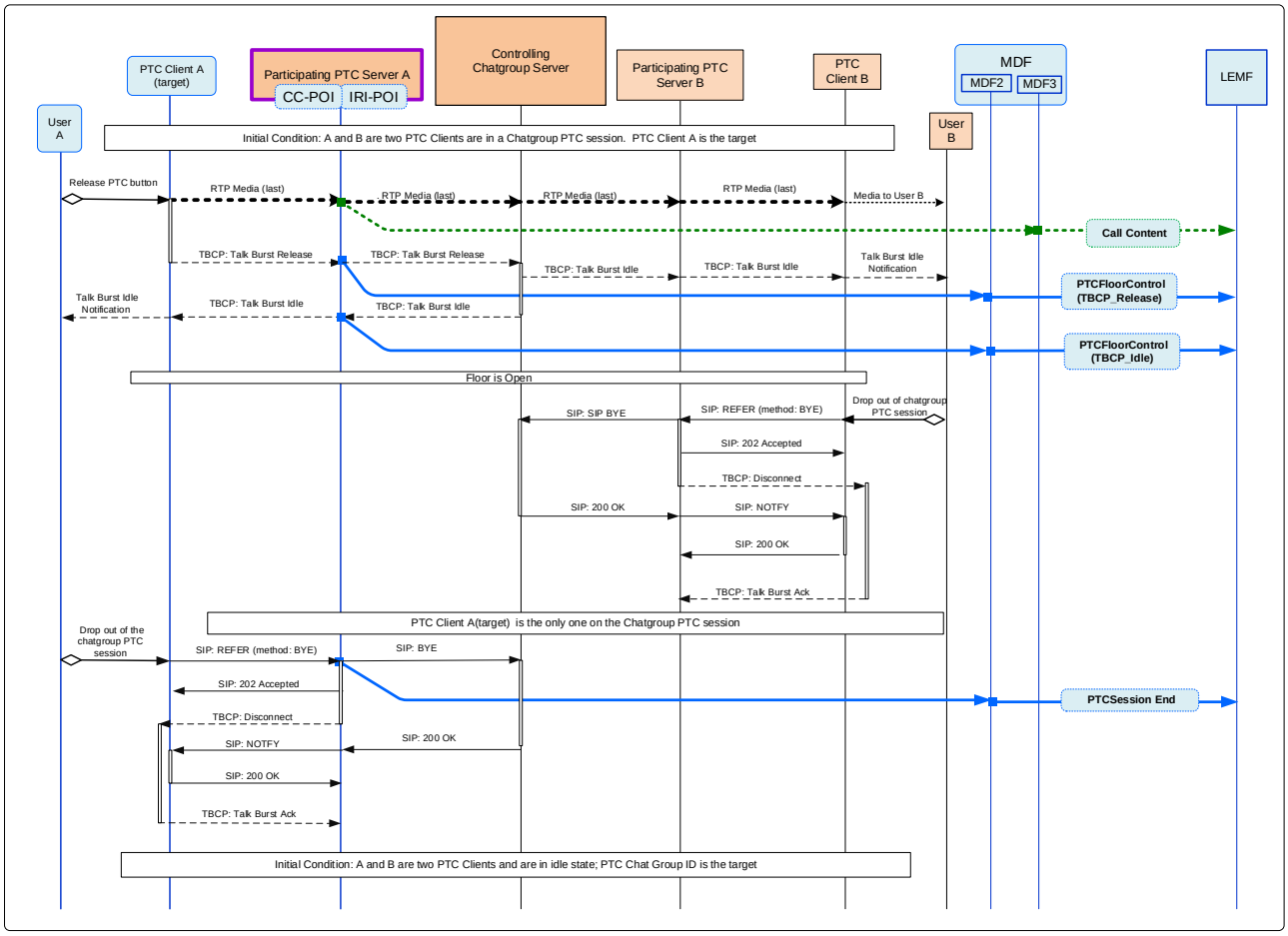


Figure 5.10.4-1: Chat-group PTC session is ended, PTC clients have pre-established sessions

As shown in figure 5.10.4-1, when a PTC client drops out of a chat-group PTC session, the session ends for that PTC client. A chat-group PTC session ends, when all the PTC clients end their chat-group PTC sessions.

In this illustration, the target PTC Client A drops out of the chat-group PTC session last.

5.11 One-to-Many-One PTC session – floor control procedures

5.11.1 General

One-to-Many-One PTC session is a variation of pre-arranged group PTC session, however, when the participant PTC client (other than the one who initiated the PTC session) has the floor, the voice can be heard by only the PTC session initiating client.

The PTC session can be started either on-demand (see clause 5.5) or with pre-established sessions (see clause 5.6).

The floor control illustrated here happens after the PTC session is started.

Since the flow is specific to illustrate the fact that that speech can only be heard by the PTC session initiator, three PTC clients are shown. The LI functions are provided by the PTC server that provides the participating PTC function to the target.

In all illustrations, the TBCP message names are shown from a functional perspective.

5.11.2 PTC session initiator is the target

Figure 5.11.2-1 illustrates when PTC Client A who initiated the pre-arranged PTC session to PTC Client B and PTC Client C is the target.

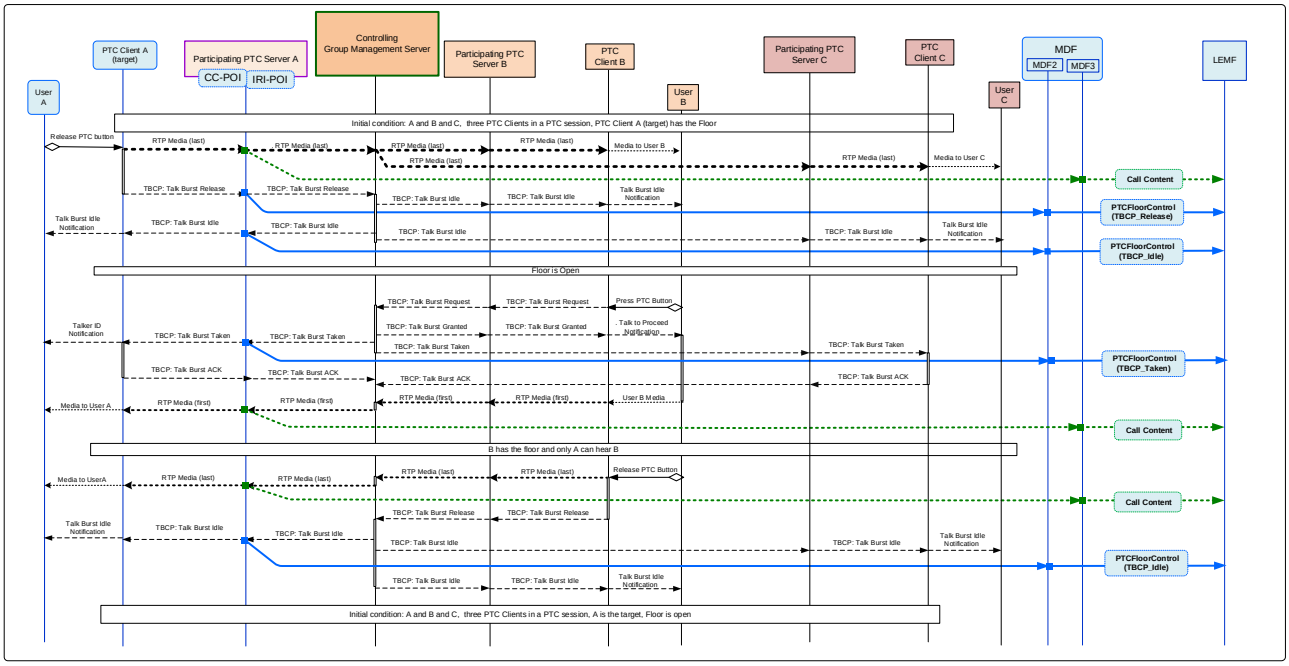


Figure 5.11.2-1: PTC client A is the target

As shown in figure 5.11.2-1, the PTC Server A that provides the participating PTC function to PTC Client A (target) provides the LI functions. When the PTC Client B has the floor, the speech is heard only by the PTC Client A. However, since the PTC Client A target always hears the speech, the LI functions in the PTC Server A are able to generate the xIRIs and xCC for the messages sent or received from the PTC Client A (target.

5.11.3 PTC session participant is the target

Figure 5.11.3-1 illustrates when PTC Client C who is one of the destination clients of a pre-arranged PTC session is a target.

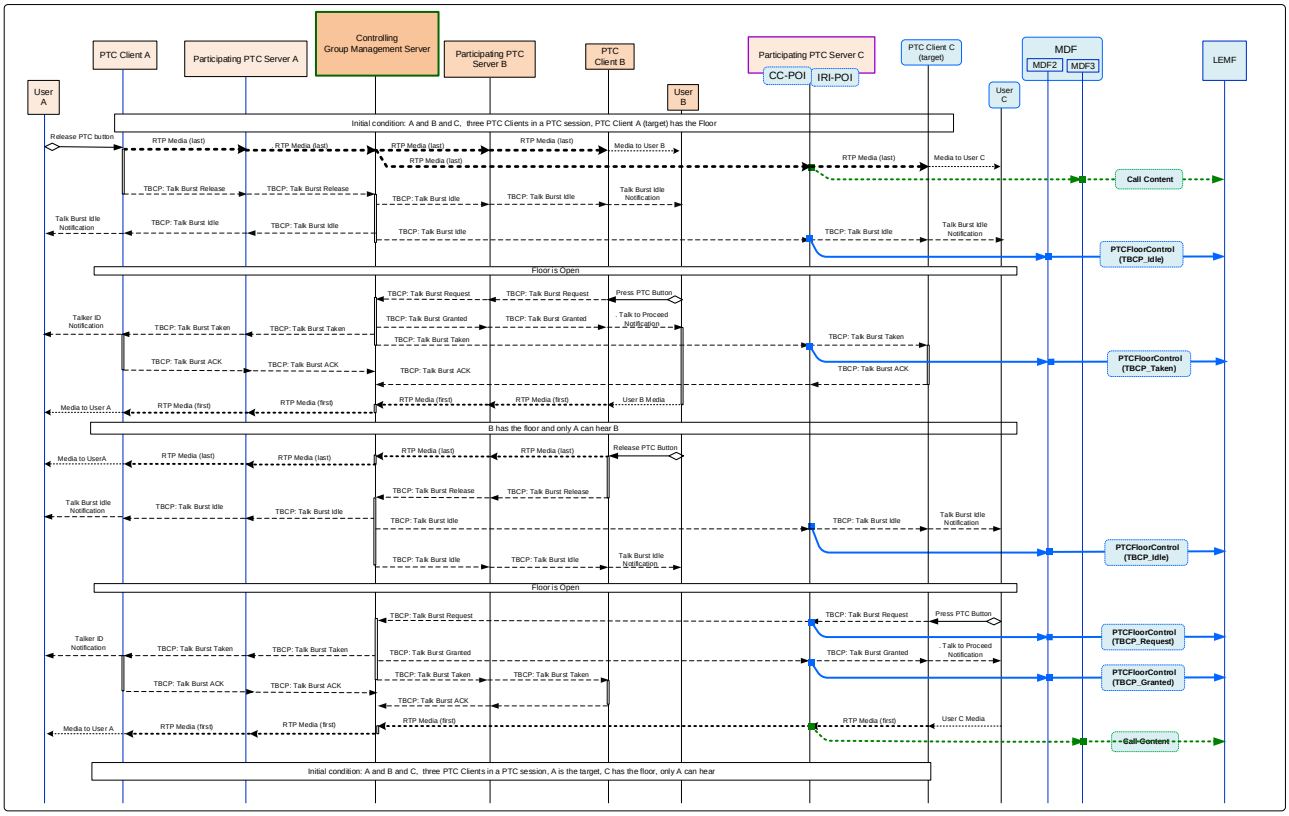


Figure 5.11.3-1: PTC client B is the target

As shown in figure 5.11.3-1, the PTC Server C that provides the participating PTC function to PTC Client C (target) provides the LI functions. When the PTC Client B has the floor, the speech is not heard only by the PTC Client C and therefore, the media is not seen at the PTC Server C. Accordingly, for the media that transfers from PTC Client B to PTC Client A, no xCC is generated by the CC-POI present in the PTC Server C. However, when the PTC Client C has the floor, the speech is heard by the PTC Client A and the media from PTC Client C passes through the PTC Server C and therefore, the CC-POI present in the PTC Server C is able to generate the xCC.

Annex A:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-07	SA3LI#94	S3i240512				Initial version	0.0.1
2025-11	SA3LI#99	S3i250681				Implemented pCRs agreed at SA3LI#99	0.0.2
2026-07	SA3LI#102	S3i260364				Agreed for Change Control	0.0.3
2026-08	SA#113	SP-260735				Submitted to SA#113 Plenary for Approval	1.0.0
2026-09	SA#113	-				Approved by SA#113 Plenary for Change Control	19.0.0

History

Version	Date	Status
V19.0.0	October 2026	Publication